A smoother way to Bernstein's theorem

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Contents

Preface **4**
Acknowledgements . . . 5
1 Introduction . . . 6
1.1 Minimal graphs and Bernstein's problem . . . 6
1.2 Philosophical orientations . . . 9
1.3 Outline of the paper . . . 10

I The smooth theory **12**
2 Minimal surfaces . . . 12
2.1 Connections and curvature . . . 12
2.2 Sectional curvature . . . 15
2.3 Riemannian submanifolds . . . 18
2.4 Variations . . . 23
2.5 First variation of area . . . 25
3 Stable minimal surfaces . . . 26
3.1 Second variation formula . . . 27
3.2 Schrödinger operators . . . 28
3.3 Stable minimal hypersurfaces . . . 30
4 Minimal graphs and calibrations . . . 32
4.1 Calibrations . . . 32
4.2 Bernstein's theorem, $n = 2$ . . . 35

II A rough detour **38**
5 Theory of Caccioppoli sets . . . 38
5.1 The classical Riesz–Markov–Kulkarni theorem . . . 39
5.2 Vector-valued Radon measures . . . 41
5.3 Functions of bounded variation . . . 41
5.4 Caccioppoli sets . . . 45
6 Cones and the monotonicity formula . . . 47
6.1 Cones . . . 47
6.2 Monotonicity formula . . . 48
7 Smooth limits of smooth hypersurfaces . . . 50
7.1 C^∞_{loc} convergence of minimal hypersurfaces . . . 51
8 Tangent cones at infinity . . . 53

8.1 Iterated tangent cones 53
8.2 Dimensional reduction 54

III A return to the smooth world **58**

9 Separation of variables 58
9.1 Proof of separation of variables 59
10 Simons equation 63
10.1 Simons identity 64
10.2 Simons inequality 66
11 Simons cone theorem and Bernstein's theorem 68

Appendix **70**

Preface

Mathematics is a collection of tautologies. A list of propositions we deem as true, each linked to one another by rigorous proof, such that the entire edifice stands tall, as strong as the axioms upon which it all rests. Nevertheless, certain tautologies appear to us as more surprising, beautiful, confusing, elegant, interesting, etc. than others. Strictly speaking, purely as logical objects it makes no sense to place these value judgments on tautologies—what more is there to say about a tautology besides that it is true?

Mathematics is also a collection of stories. Stories told by one person to another, weaving together a diverse cast of characters reaching as far as the mind can traverse. Stories with twists and turns, unexpected connections and crossovers—stories that illuminate the intersubjective world of abstract ideas that we call modern mathematics.

This honors thesis is an homage to one of these tautologies, and the story that comes with it: the solution to Bernstein's problem. Bernstein's problem, as originally formulated by Serge Bernstein, asks the following. (See [Ber17], German translation [Ber27].) Suppose I'm in 3-dimensional space $\mathbb{R}^3$ and you give me a two-variable function $u(x, y)$ and look at its graph; i.e., the points (x, y, z) at which $z = u(x, y)$. Moreover, suppose the graph behaves like a soap film, and hence is *minimal*[1] in the following sense: if I were to wiggle the graph slightly, the area can only go up. Certainly, if the graph was flat (i.e., a 2-dimensional plane), this would be true. Bernstein wondered whether or not the converse was true: if my graph behaves like a soap film, must it be flat?

One's experience with soap films might indicate that this is probably true: after all, in the absence of any boundary geometry that might curve the soap film, it feels like this should be true. Bernstein proved that indeed, this is the case: any 2-dimensional minimal graph in $\mathbb{R}^3$ is flat. However, his methods exploited deep facts about the number 2: namely, that 2-dimensional geometry and topology can utilize the rich theory of complex analysis. So, it was unclear how we could approach the general question, which came to be known as *Bernstein's problem*: Is every n-dimensional minimal graph in $\mathbb{R}^{n+1}$ flat?

If geometric intuition might be able to help us in low dimensions, it has a tendency to fail us in higher dimensions. Bernstein's problem is true for $n \leq 7$ and false for $n \geq 8$. In some sense, the reason for this fact is somewhat banal: in order for Bernstein's problem to be true, the inequality

$$(n-2) - \frac{(n-3)^2}{4} \geq 0$$

must be true.[2] The quadratic formula tells us that the positive root of this polynomial is $5 + \sqrt{8} \approx 7.82843$, hence Bernstein's problem is true for $n \leq 7$.[3] This is a deliberate simplification: Bernstein's problem for $n \leq 7$ took the

[1]This is not precise; see 1.6 and 2.40 for the definition and 3.15 for a remark on why this is a reasonable heuristic.
[2]Strictly speaking, this is either a white lie or a deliberate simplification, but every good story starts with a lie...
[3]The astute reader will notice that the negative root of this polynomial is $5 - \sqrt{8} \approx 2.17157$, so here is another lie.

concerted work of Fleming [Fle62], de Giorgi [dG65], Almgren [Alm66], and finally Simons [Sim68]. On the other hand, if you solve some ODEs and choose some constants carefully, you can construct a nonflat minimal graph with 8 variables; appending additional variables gives counterexamples to Bernstein's problem for $n \geq 8$, as proved by Bombieri–de Giorgi–Giusti [BdGG69].

We have arrived at the tautology which will be the subject of this thesis:

$$\text{Every } n\text{-dimensional minimal graph in } \mathbb{R}^{n+1} \text{ is flat} \iff n \leq 7.$$

I hope you enjoy the story.

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This thesis was a labor of love. I hope that it reflects not just the love that I have put into it, but also the love and support I have received from my friends, family, and the communities I have found myself enmeshed in.

Section 1

Introduction

§1.1 Minimal graphs and Bernstein's problem

In this section we will set up and state Bernstein's problem with more precision. Note that all statements will be in terms of C^∞ smooth objects, even though they could be phrased in terms of lower regularity classes. We will often implicitly lean on elliptic regularity to ensure that all statements which in general "should" be stated in terms of C^1 or C^2 functions are permitted to be stated in terms of C^∞ smooth functions.

Definition 1.1 (Graph of a function). Let $\Omega \subseteq \mathbb{R}^n$ be an open set, and let $u \in C^\infty(\Omega)$. Then, its *graph* is

$$\operatorname{graph} u = \{(x, y) \in \Omega \times \mathbb{R} : \ y = u(x)\} \subseteq \Omega \times \mathbb{R}.$$

In the case where Ω has smooth boundary and u extends smoothly to the boundary, the graph is an embedded n-manifold with boundary $\operatorname{graph} u|_{\partial\Omega}$.

Let $\iota : \ \Omega \hookrightarrow \mathbb{R}^{n+1}$ be the inclusion of this graph: $\iota : \ x \mapsto (x, u(x))$. Then, we can compute the area of the graph in terms of the pullback metric along ι, as follows.

Let $\overline{g}$ be the Euclidean metric on $\mathbb{R}^{n+1}$ and let $g = \iota^*\overline{g}$ be the pullback metric on Ω. Let $(x^1, \dots, x^n)$ denote the standard coordinates on Ω. In these coordinates, we can write the area[4] as

$$\operatorname{Area}(\operatorname{graph} u) = \int_\Omega \sqrt{\det g_{ij}}\, dx^1 \wedge \cdots \wedge dx^n.$$

When dealing with subsets of Euclidean space, we will often drop the volume form (or equivalently, the Lebesgue measure) from the notation. In this case, we can write the area in terms of u and its derivatives.

Proposition 1.2 (Area of a graph). Let $\Omega \subseteq \mathbb{R}^n$ be open and $u \in C^\infty(\Omega)$. Then,

$$\operatorname{Area}(\operatorname{graph} u) = \int_\Omega \sqrt{1 + |\nabla u|^2}.$$

The crux of this formula is the following linear algebra fact.

Lemma 1.3. Let $v \in \mathbb{R}^n$ and let $A = I + vv^\top$; that is, $A_{ij} = \delta_{ij} + v_i v_j$. Then,

$$\det A = 1 + |v|^2 .$$

Proof. First note that v is an eigenvector for A:

$$Av = v + vv^\top v = (1 + \langle v, v\rangle)v.$$

Moreover, for any $w \in v^\perp$,

$$Aw = w + vv^\top w = w + v\,\langle v, w\rangle = w.$$

Hence A has eigenvalues 1 and $1 + |v|^2$ with multiplicity $n - 1$ and 1, respectively. (Of course, if $v = 0$ then $A = I$ has eigenvalue 1 with multiplicity n.) □

[4] Technically, we should restrict attention to Ω bounded so that this integral is guaranteed to converge. In practice, if Ω is noncompact we will only need the area formula over compact subdomains $W \Subset \Omega$.

Proof of Proposition 1.2. We compute the metric. Let us denote $u_{;i} = \partial_i u$. Note that $d\iota_x : T_x\Omega \to T_{\iota(x)}\mathbb{R}^{n+1}$ is given by the matrix

$$d\iota_x = \begin{bmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 1 \\ u_{;1} & \cdots & u_{;n} \end{bmatrix},$$

and hence the metric tensor is given by

$$g_{ij} = \delta_{ij} + u_{;i}u_{;j}.$$

By Lemma 1.3, it follows that $\det g_{ij} = 1 + |\nabla u|^2$, and we are done. □

In minimal surface theory, we seek to find minimizers of the area within a given class of geometric objects. Experience with single-variable calculus should indicate that it is useful to consider "minimizers" in three distinct senses: (i) critical points, (ii) local minima, and (iii) global minima. It turns out to be useful to broaden our scope: minimal surfaces are surfaces which are critical points for the area, in a suitable sense.

Our first task is to make these notions precise. Given a smooth manifold M and a smooth function $f : M \to \mathbb{R}$, we say that $p \in M$ is a *critical point* if $df_p : T_pM \to T_{f(p)}\mathbb{R}$ is not surjective; in this case of target dimension 1, this is equivalent to $df_p = 0$. In practice this can be tricky to compute, but we can make use of the path definition of tangent vectors to convert this into a problem about the images of paths: $df_p = 0$ if and only if for every path $\gamma : (-\varepsilon, \varepsilon) \to M$ with $\gamma(0) = p$, $(f \circ \gamma)'(0) = 0$. Notice that this definition of a critical point does not require us to define the differential df explicitly; we use the fact that $f \circ \gamma$ is a single-variable function and hence we can use the usual tools of single-variable calculus to make sense of the derivative.

This is the strategy we employ in our case, where now we consider $\mathrm{Area} : C^\infty(\Omega) \to \mathbb{R}$. The idea is to consider 1-parameter famillies in $C^\infty(\Omega)$ which only change on a compact set: this is the notion of a *compactly supported variation.*

Suppose $\eta \in C^\infty_c(\Omega)$ is a smooth function on Ω with compact support, so that $\eta|_{\partial\Omega} \equiv 0$. Then, $u + t\eta$ is a path in $C^\infty(\Omega)$ with fixed values outside of a compact set $\operatorname{supp}\eta$ (and hence the area integral is finite, so we can make sense of these quantities).

Definition 1.4. Let $\Omega \subseteq \mathbb{R}^n$ be an open domain and $u \in C^\infty(\Omega)$. Then, graph u is a *critical point for the area functional,*[a] if for all $\eta \in C^\infty_c(\Omega)$ with $\eta|_{\partial\Omega} = 0$,

$$\left.\frac{d}{dt}\right|_{t=0} \mathrm{Area}(\mathrm{graph}\,(u + t\eta)) = 0.$$

[a] We will give a more general definition later, and show that that definition reduces to this one for graphs.

This condition yields the following equation for graphical hypersurfaces, called the *minimal surface equation* (or the MSE for short). This is the Euler-Lagrange equation for the area functional.

Proposition 1.5 (Minimal surface equation). Let $\Omega \subseteq \mathbb{R}^n$ be an open domain and $u \in C^\infty(\Omega)$. Then,

$$\text{graph } u \text{ is a critical point for the area functional} \iff \operatorname{div}\left(\frac{\nabla u}{\sqrt{1 + |\nabla u|^2}}\right) = 0.$$

Proof. Let $\eta \in C_c^\infty(\Omega)$. Computing,

$$\begin{aligned} \frac{d}{dt} \text{Area}(\text{graph}\,(u + t\eta))\Big|_{t=0} &= \int_\Omega \frac{d}{dt} \sqrt{1 + |\nabla u + t\nabla \eta|^2}\Big|_{t=0} \\ &= \int_\Omega \frac{1}{2} \left(1 + |\nabla u + t\nabla\eta|^2\right)^{-1/2} 2\,\langle \nabla u + t\nabla \eta, \nabla \eta\rangle\Big|_{t=0} \\ &= \int_\Omega \frac{\langle \nabla u, \nabla \eta\rangle}{\sqrt{1+|\nabla u|^2}} \end{aligned}$$

The first equality comes from η being compactly supported, so that we can exchange the derivative with the integral; the second equality comes from taking the chain rule and the identity $(|x(t)|^2)' = 2\langle x(t), x'(t)\rangle$. Let $\nu : \Omega \to \mathbb{R}^{n+1}$ be the upwards unit normal map, given by

$$\nu(x) = \frac{(-\partial_1 u, \dots, -\partial_n u, 1)}{\sqrt{1 + |\nabla u|^2}}.$$

Then, integrating by parts,

$$\begin{aligned} \frac{d}{dt} \text{Area}(\text{graph}\,(u + t\eta))\Big|_{t=0} &= \int_\Omega \frac{\langle \nabla u, \nabla \eta\rangle}{\sqrt{1+|\nabla u|^2}} \\ &= \int_{\partial\Omega} \eta \frac{\langle \nu, \nabla u\rangle}{\sqrt{1+|\nabla u|^2}} - \int_\Omega \eta \operatorname{div}\left(\frac{\nabla u}{\sqrt{1+|\nabla u|^2}}\right) \\ &= -\int_\Omega \eta \operatorname{div}\left(\frac{\nabla u}{\sqrt{1+|\nabla u|^2}}\right) \end{aligned}$$

because $\eta|_{\partial\Omega} = 0$. Because our choice of η was arbitrary, this formula entails our result. □

The above proposition justifies the following definition:

Definition 1.6 (Minimal graph). A graphical hypersurface graph u is *minimal* if one of the following equivalent conditions holds:

(a) graph u is a critical point for the area functional,

(b) u satisfies the minimal surface equation.

Moreover, u is an *entire* solution to the MSE if $\Omega = \mathbb{R}^n$, in which case we say that graph u is an *entire minimal graph* or *entire graphical minimal hypersurface.*

Finally, we say that u is *affine* if it is a degree 1 polynomial, in which case graph u is a hyperplane. Then, we can state Bernstein's problem, as follows:

Question (Bernstein's Problem). Let $u \in C^\infty(\mathbb{R}^n)$ be an entire solution to the minimal surface equation. Then, is u affine? Equivalently, is every entire graphical minimal hypersurface a hyperplane?

Remark 1.7 (Bernstein for $n = 1$). We can solve Bernstein's problem in $n = 1$ solely by analyzing the minimal surface equation: in this dimension the MSE is

$$\frac{d}{dx}\left(\frac{u'}{\sqrt{1+(u')^2}}\right) = \frac{u''\left(1+(u')^2\right)^{1/2} - u'(1+(u')^2)^{-1/2}u'u''}{1+(u')^2}$$

$$= \frac{u''(1 + (u')^2 - (u')^2)}{(1 + (u')^2)^{3/2}}$$
$$= \frac{u''}{(1 + (u')^2)^{3/2}}.$$

The denominator is always $\neq 0$, so u solves the MSE if and only if $u'' = 0$ identically, which entails that u is affine.

As mentioned in the introduction, the answer is yes for $n \leq 7$ and no for $n \geq 8$. For $n \geq 8$ there exists a counterexample constructed by [BdGG69], see [Giu84] for a derivation. In this paper, we will chiefly concern ourselves with the $n \leq 7$ case. Roughly speaking, as n increases we will need increasingly sophisticated tools to handle the problem.

§1.2 Philosophical orientations

Classically, minimal surfaces were studied from the point of view of differential geometry, PDE, and the calculus of variations. A classical motivating problem is the following: *Plateau's problem* (see [CM11] for more discussion).

Question (Plateau's problem). Let $\Gamma^1 \subseteq \mathbb{R}^3$ is a C^1 simple closed curve, and let

$$a_\Gamma = \inf \left\{ \text{Area}(\Sigma) : \Sigma^2 \subseteq \mathbb{R}^3 \text{ surface with boundary } \partial\Sigma = \Gamma \right\}.$$

Does there exist a smooth $\Sigma \subseteq \mathbb{R}^3$ such that $\text{Area}(\Sigma) = a_\Gamma$?

The straightforward way to solve this geometric optimization problem is to employ the *direct method.*

(1) Pick a minimizing sequence: $\Sigma_j \subseteq \mathbb{R}^3$ with $\partial\Sigma = \Gamma$ and $\text{Area}(\Sigma_j) \searrow a_\Gamma$.

(2) Equip the space of permissible geometric objects with a suitable topology, and take the completion: in this case, the space is $\left\{\Sigma^2 \subseteq \mathbb{R}^3 : \partial\Sigma = \Gamma\right\}$.

(3) Prove a *compactness theorem* to argue that the limit (up to taking a subsequence) exists.

(4) Prove a *regularity theorem* to argue that the fact that the limit satisfies the optimization problem implies that it is smooth.

Despite the fact that minimal surfaces were originally conceived of as smooth objects, it turned out that the compactification step in Plateau's problem could not be handled purely with smooth methods: inaugurating what would become geometric measure theory (GMT).

Geometric measure theory has the advantage of having good compactness properties, and hence is incredibly powerful for attacking geometric optimization problems like Plateau's problem. But, the disadvantage is that much of the machinery can get incredibly technical, and proving regularity is, in general, very difficult.

From the perspective of solving Bernstein's problem, there is another perhaps aesthetic reason why geometric measure theory methods are not entirely satisfactory. On its face, Bernstein's problem is an entirely differential geometric problem: all of the relevant objects are smooth, so one might hope that there would exist a proof of Bernstein's theorem using entirely smooth techniques. This is not the case—the conventional proof takes a detour through several mature results in geometric measure theory. This thesis is a contribution towards the following (perhaps fuzzy) question.

QUESTION. Does there exist a solution to Bernstein's problem for $n \leq 7$ utilizing only smooth techniques?

The answer (as far as we are aware) is still no: we still have to make a detour through the rough theory (i.e., geometric measure theory). But, this thesis attempts to make the detour as brief as possible, utilizing several simplifications.

The first simplification is a standard one: because minimal graphs carry a natural orientation (i.e., a global unit normal) and are codimension 1, we only need to develop geometric measure theory in the oriented codimension 1 setting. In this case, we only need to develop the theory of *Caccioppoli sets*, in which the requisite analytic tools are relatively simple, only involving so-called *functions of bounded variation*.

The second simplification has to do with one of the foundational regularity theorems in geometric measure theory: Allard's ε-regularity theorem [All72]. In full generality Allard's theorem is very technical, but in the category of objects which are limits of smooth minimal surfaces, there is an easier version of Allard's theorem due to White [Whi05], hence reducing our reliance on heavy-duty GMT machinery.

The third simplification is a new result (or rather, a new proof of a known result). One of the crucial steps in the conventional proof is *de Giorgi's cone splitting theorem* [dG65], which allows us to perform a dimension reduction step which extends the proof of Bernstein's theorem from $n \leq 6$ to $n \leq 7$. In the spirit of White's easy Allard, we present a weaker but easier version (Theorem 8.9) of de Giorgi's theorem which nonetheless allows us to perform the dimension reduction step as in the conventional proof.

This is the spirit with which we embark on our journey: a smoother way to Bernstein's theorem for $n \leq 7$.

§1.3 Outline of the paper

This paper will be organized into three chapters, each building up the theoretical tools necessary to prove Bernstein's theorem for $n \leq 7$.

In Chapter I, we will go over the basics of minimal surface theory from the smooth (i.e., differential geometric) perspective, culminating in a proof of Bernstein's theorem for $n = 2$. In a suitable sense, minimal graphs are not just critical points of the area, but also *stable* (local minimizers / satisfies a second derivative test) and *area-minimizing on compact sets* (global minimiziers). The area-minimizing ensures that the area of a minimal graph on $B^n = B_1(0) \subseteq \mathbb{R}^n$ is bounded above by the area of the unit n-sphere, giving us area growth bounds. Then, stability ensures that the curvature of the graph[5] satisfies an integral inequality called the *stability inequality*. Then, we use a trick called the *log cutoff trick*, which for dimensional reasons is specific to *quadratic* area growth, to prove Bernstein's theorem for $n = 2$.

In Chapter II, we take a detour into to rough (i.e., non-smooth) approaches to minimal surface theory. The aim of this chapter is to translate Bernstein's problem on minimal graphs into a problem about area-minimizing cones (sets which are radially invariant (i.e., $x \in C \implies rx \in C$ for any $r > 0$)). The idea is as follows: for a non-flat minimal graph graph $u \subseteq \mathbb{R}^{n+1}$, we take the *blowdown*, which intuitively is the limit as we zoom out to infinity. The area-minimizing property of the graph, as well as non-flatness, is preserved in the limit, and we get a (potentially singular) non-flat area-minimizing cone $C \subseteq \mathbb{R}^{n+1}$. Hence, if we can prove that every area-minimizing cone in $\mathbb{R}^{n+1}$ is flat, we get a resolution to Bernstein's problem in dimension n. However, the proof of such a statement

[5]More accurately, the second fundamental form, which takes the place of the Hessian in the usual second derivative test for a local minimum.

would involve delving into mature technical facts in geometric measure theory (i.e., the rough theory), so instead we steer back towards the smooth world.

For a cone $C \subseteq \mathbb{R}^{n+1}$, we define its *link* as $\Sigma = C \cap S^n$. Note that the cone is a plane if and only if Σ is an equatorial sphere. Nonetheless, if Σ is a smooth manifold we can use smooth tools to attempt to force it to be an equatorial sphere, which would resolve Bernstein's problem. With this in mind, we prove a *non-smooth blowdown theorem* (Theorem 8.9): the blowdown of a nonflat minimal graph never has smooth link. This is a new proof of a known result; this fact follows from de Giorgi's cone splitting theorem, but unlike de Giorgi we prove this result using primarily smooth methods, arguing by contradiction.

Now, we are in a position to maneuver back into the smooth world, using an *iterated tangent cone* argument. We say a point $x \in C$ is *singular* if there exists no smooth chart at x;[6] note that a cone with smooth link has $\operatorname{sing} C = \{0\}$. The idea is that if a cone C has a singular point $x \in \operatorname{sing} C \setminus \{0\}$, then we can take the blowup at that point, zooming in and taking the limit. Then, radial invariance in the x direction becomes translation invariance in the limit, and as such the limit cone splits a line: the blowup of C at x gives $C' \times \mathbb{R}$. Moreover, C' remains minimizing, so we can iterate this until we arrive at a minimizing cone with smooth link. The importance of our non-smooth blowdown theorem is that we can perform this dimension reduction *at least once*, such that if every minimizing cone with smooth link in $\mathbb{R}^{k+1}$ is flat, then Bernstein's problem is true in dimension $k + 1$.

In Chapter III we finish the proof of Bernstein's theorem for $n \leq 7$ with Simons's theorem on stable minimal cones. Every minimizing cone with smooth link is also *stable*, so to resolve Bernstein's theorem for the remaining cases $3 \leq n \leq 7$, it suffices to show that every stable minimal cone in $\mathbb{R}^{k+1}$ for $2 \leq k \leq 6$ is flat. Indeed, this is precisely Simons's theorem, so we are done.

This thesis will assume a working knowledge of Riemannian geometry, measure theory, functional analysis, and elliptic PDE, at the level of first-year graduate courses in each.[7] We have compiled important results and terminology in an appendix; we encourage the reader to refer to the appendix as needed, especially for clarification on terminology or notation.

[6] This is a good heuristic definition, see for a more workable definition.

[7] At Stanford, this would correspond to 215C, 205A, 205B, and 256A, respectively.

Chapter I

The smooth theory

In single-variable calculus, we learn that to solve optimization problems with differentiable functions, it is often beneficial to study critical points, then apply the second derivative test to find local extrema. Hence there are three kinds of optima: (i) critical points, (ii) local extrema, and (iii) global extrema. In the geometric setting, these correspond to (i) minimal surfaces, (ii) stable minimal surfaces, and (iii) area-minimizing surfaces. For Bernstein's theorem in $n = 1$ we saw in Remark 1.7 that (i) is sufficient. For $n = 2$, however, we will need all three. Luckily, as we will see, minimal graphs are optima in all three senses.

This chapter is organized according to these three themes. Section 2 develops the basic theory of minimal surfaces as such, reviewing the necessary notions in Riemannian geometry and arriving at the *first variation formula*, which is a generalization of the derivation of the minimal surface equation (Proposition 1.5). Section 3 develops the theory of stable minimal surfaces and the analysis of certain partial differential operators, *Schrödinger equations*, which arise in their study. Section 4 moves on to the area-minimizing setting, developing tools in *calibrated geometry* which allow us to establish that minimal graphs are area-minimizing, culminating in the proof of Bernstein's theorem for $n = 2$.

Section 2

Minimal surfaces

Minimal surfaces can be understood either as critical points of the area functional or as solutions to the geometric PDE $H = 0$, where H is the *mean curvature*. The goal of this section is to see that these are, in fact, equivalent characterizations.

As such, we first review some basic notions in Riemannian geometry, with an eye towards results which will be useful throughout this thesis. Our primary references are [LeeRM] and [CM11].

§2.1 Connections and curvature

Connections on arbitrary smooth manifolds

The foundational technical tool of Riemannian geometry is the notion of a *connection*, which allows us to differentiate sections of vector bundles.

Definition 2.1 (Connection). Let $\pi : E \to M$ be a smooth vector bundle over a smooth manifold M. A *connection*, or a *covariant derivative*, is a map $\nabla : \mathfrak{X}(M) \times \Gamma(E) \to \Gamma(E)$ satisfying the following properties:

(a) $\nabla_X Y$ is $C^\infty(M)$-linear in X,

(b) $\nabla_X Y$ is $\mathbb{R}$-linear in Y, and

(c) $\nabla_X Y$ satisfies the following product rule: for $f \in C^\infty(M)$, $X \in \mathfrak{X}(M)$, and $Y \in \Gamma(E)$,

$$\nabla_X(fY) = (Xf)Y + f\nabla_X Y.$$

The paradigmatic case is the bundle $TM \to M$, since sections are just vector fields. But, it turns out that having a connection on $TM \to M$ actually fixes a unique connection on any (k,l)-tensor bundle $T^{(k,l)}TM \to M$.

Proposition 2.2 (Extension of connection on TM to tensor bundles, LeeRM Prop 4.15). Let M be a smooth manifold, and let ∇ be a connection on TM. Then, there exists a unique connection on each tensor bundle $T^{(k,l)}TM$, also denoted by ∇, satisfying the following four conditions.

(a) For $T^{(1,0)}TM = TM$, ∇ agrees with the given connection.

(b) For $T^{(0,0)}TM = M \times \mathbb{R}$, ∇ is given by ordinary differentiation of functions: $\nabla_X f = Xf$.

(c) For $F \in \Gamma(T^{(k,l)}TM)$ and $G \in \Gamma(T^{(k',l')}TM)$, ∇ obeys the following product rule:

$$\nabla_X(F \otimes G) = (\nabla_X F) \otimes G + F \otimes (\nabla_X G).$$

(d) ∇ commutes with contractions: if tr denotes a trace on any pair of indices, one contravariant and one covariant, then

$$\nabla_X(\operatorname{tr} F) = \operatorname{tr}(\nabla_X F).$$

Moreover, for any $F \in \Gamma(T^{(k,l)}TM)$, $\nabla_X F$ obeys the following computational identity: for smooth 1-forms $\omega^1, \dots, \omega^k$ and smooth vector fields $Y_1, \dots, Y_l$,

$$\begin{aligned}(\nabla_X F)(\omega^1, \dots, \omega^k, Y_1, \dots, Y_l) &= X(F(\omega^1, \dots, \omega^k, Y_1, \dots, Y_l)) \\ &\quad - \sum_{i=1}^{k} F(\omega^1, \dots, \nabla_X \omega^i, \dots, \omega^k, Y_1, \dots, Y_l) \\ &\quad - \sum_{j=1}^{l} F(\omega^1, \dots, \omega^k, Y_1, \dots, \nabla_X Y_j, \dots, Y_l).\end{aligned}$$

Note that since $\nabla_X F$ is linear over $C^\infty(M)$ in X, we can think of $\nabla_{(-)} F$ as a $(k, l+1)$-tensor field. This is the *total covariant derivative*, which we will often have use for in computations.

Definition 2.3 (Total covariant derivative). Let M be a smooth manifold and let ∇ be a connection on TM. The *total covariant derivative* $\nabla : \Gamma(T^{(k,l)}M) \to \Gamma(T^{(k,l+1)}TM)$ is given by

$$\nabla F(\omega^1, \dots, \omega^k, Y_1, \dots, Y_l, X) = (\nabla_X F)(\omega^1, \dots, \omega^k, Y_1, \dots, Y_l).$$

The total covariant derivative allows us to define the *second covariant derivative*, which generalizes the usual

Hessian of a function.

DEFINITION 2.4 (Second covariant derivative). Let M be a smooth manifold and let ∇ be a connection on TM. The *second covariant derivative* of a (k, l)-tensor field $F \in \Gamma(T^{(k,l)}TM)$ is

$$\nabla^2_{X,Y} F(\dots) = (\nabla^2 F)(\dots, Y, X).$$

In practice, we often compute using the following formula.

PROPOSITION 2.5 (Formula for the second covariant derivative, LeeRM Prop 4.21). Let M be a smooth manifold and let ∇ be a connection on TM. The second covariant derivative of a (k, l)-tensor field $F \in \Gamma(T^{(k,l)}TM)$ satisfies

$$\nabla^2_{X,Y} F = \nabla_X(\nabla_Y F) - \nabla_{\nabla_X Y} F.$$

Connections on Riemannian manifolds

In order to do more with connections, we would like to be able to swap perspectives between covariant and contravariant indices at will. However, we run into two obstacles: (i) such an assignment is highly noncanonical, since a finite dimensional vector space is only noncanonically isomorphic to its dual, and (ii) an arbitrary smooth manifold admits an infinite dimensional affine space of connections (see LeeRM, Theorem 4.14). The tool which allows us to avoid these two obstacles is a choice of an inner product on each fiber of TM; i.e., a Riemannian metric. Let us denote by Sym^k the k-fold symmetric power of a vector bundle.

DEFINITION 2.6 (Smooth fiber metric). Let $\pi : E \to M$ be a smooth vector bundle. A *smooth fiber metric* on E is a section $g \in \Gamma(\mathrm{Sym}^2(E^*))$ such that $g_p \in \mathrm{Sym}^2(E^*_p)$ is an inner product on E_p for every $p \in M$.

DEFINITION 2.7 (Riemannian manifold). Let M be a smooth manifold. A *Riemannian metric* on M is a smooth fiber metric on TM, and a pair (M, g) is called a *Riemannian manifold*.

In the case of a Riemannian manifold (M, g), there is a unique choice of connection that is compatible with both the metric structure and the smooth structure.
Regarding compatibility with the metric structure, note that since $g \in \Gamma(T^{(0,2)}TM)$ is a $(0, 2)$-tensor, we can take its total covariant derivative ∇g. Recall that a tensor field $F \in \Gamma(T^{(k,l)}TM)$ is called *parallel* if $\nabla F = 0$. A connection ∇ on a Riemannian manifold is then called a *metric connection*, or *compatible with the metric*, if g is parallel with respect to ∇: $\nabla g = 0$. Equivalently, the connection satisfies the following product rule over the metric:

$$\nabla_X \langle Y, Z\rangle = \langle \nabla_X Y, Z\rangle + \langle Y, \nabla_X Z\rangle.$$

The second condition is a bit more mysterious, though it can be seen as a compatibility with the smooth structure. Recall that the *Lie bracket* of two vector fields $X, Y \in \mathfrak{X}(M)$ is given by

$$[X, Y]f = X(Yf) - Y(Xf).$$

A connection is called *torsion-free* or *symmetric* if

$$\nabla_X Y - \nabla_Y X = [X, Y].$$

It is a wonderful fact that these two conditions are enough to specify a unique connection.

Theorem 2.8 (Levi-Civita connection, LeeRM Thm 5.10). Let (M, g) be a Riemannian manifold. Then, there exists a unique connection ∇, called the *Levi-Civita connection*, that is both compatible with the metric and torsion-free.

Going forward, if (M, g) is a Riemannian manifold, we denote by ∇ the Levi-Civita connection. The connection allows us to define various notions of curvature, as a failure for second derivatives to commute.

Definition 2.9 (Riemann and Ricci curvature tensors). Let (M, g) be a Riemannian manifold. The $(1, 3)$-*curvature tensor* $R \in \Gamma(T^{(1,3)}TM)$ is defined by

$$R(X, Y)Z = [\nabla_X, \nabla_Y]Z - \nabla_{[X,Y]}Z = \nabla_X\nabla_Y Z - \nabla_Y\nabla_X Z - \nabla_{[X,Y]}Z.$$

The $(0, 4)$-*curvature tensor* (or *Riemann curvature tensor*) Rm, is given by lowering an index: $\mathrm{Rm} = R^\flat$, or

$$\mathrm{Rm}(X, Y, Z, W) = \langle R(X, Y)Z, W\rangle .$$

The *Ricci curvature tensor* $\mathrm{Ric} \in \Gamma(T^{(0,2)}TM)$ is given by taking the trace of the Riemann curvature tensor on the first and last index:

$$\mathrm{Ric}(X, Y) = \mathrm{tr}\,(Z \mapsto R(Z, X)Y) .$$

The connection, along with the metric, also allows us to define the Laplacian of an arbitrary tensor field.

Definition 2.10 (Laplacian). Let (M, g) be a Riemannian manifold and $F \in \Gamma(T^{(k,l)}TM)$ be a (k, l) tensor field. Then, the *Laplacian operator* $\Delta : \Gamma(T^{(k,l)}TM) \to \Gamma(T^{(k,l)}TM)$ is defined by

$$\Delta F = \mathrm{tr}_g \nabla^2 F,$$

where the trace is taken on the last two entries.

§2.2 Sectional curvature

We will often have use for computations for hypersurfaces in the model spaces $\mathbb{R}^{n+1}$ and S^n, particularly for the case $\Sigma^{n-1} \hookrightarrow S^n \hookrightarrow \mathbb{R}^{n+1}$. These spaces are examples of manifolds with *constant sectional curvature*, which are especially nice for computations, as their curvature tensors take on an easily computable form. To understand this theory, we first work through some linear algebra, then apply it to the manifold setting.

Algebraic curvature tensors

We say that a covariant 4-tensor on a finite dimensional vector space is an *algebraic curvature tensor* if it satisfies the same symmetries as the Riemann curvature tensor:

Definition 2.11 (Algebraic curvature tensor). Let V be an n-dimensional real vector space. A covariant 4-tensor $T \in T^4V^*$ is a *algebraic curvature tensor* if it satisfies the symmetries

(a) ((12) skew symmetry) $T(w, x, y, z) = -T(x, w, y, z)$,

(b) ((34) skew symmetry) $T(w, x, y, z) = -T(w, x, z, y)$,

(c) (interchange symmetry) $T(w, x, y, z) = T(y, x, w, z)$,

(d) (Bianchi identity) $T(w, x, y, z) + T(x, y, w, z) + T(y, w, x, z) = 0$.

We can define the sectional curvature of a 2-plane $\Pi \subseteq V$ with respect to an algebraic curvature tensor:

Definition 2.12 (Sectional curvature of an algebraic curvature tensor). Let V be a finite dimensional inner product space, let $T \in T^4 V^*$ be an algebraic curvature tensor, and let $v, w \in V$ be linearly independent vectors spanning the 2-plane $\Pi \subseteq V$. The *sectional curvature of* Π is defined to be

$$\sec_T(\Pi) = \sec_T(v, w) = \frac{T(v, w, w, v)}{|v \wedge w|^2},$$

where $|v \wedge w|^2 = |v|^2 |w|^2 - \langle v, w \rangle^2$.

It turns out that sectional curvatures uniquely determine an algebraic curvature tensor.

Proposition 2.13 (Sectional curvatures determine algebraic curvature tensor, LeeRM Prop 8.31). Let V be a finite dimensional inner product space, and let $T_1, T_2 \in T^4 V^*$ be two algebraic curvature tensors. If $\sec_{T_1}(\Pi) = \sec_{T_2}(\Pi)$ for every 2-plane $\Pi \subseteq V$, then $T_1 = T_2$.

The Kulkarni–Nomizu product

The motivating problem for the Kulkarni–Nomizu product is as follows. On a Riemannian manifold, is it possible to build the Riemann curvature tensor out of the symmetric covariant 2-tensors g and Ric? That is, is it possible to furnish an inverse to the map $\operatorname{tr}_g : \operatorname{Rm} \mapsto \operatorname{Ric}$? For dimensional reasons we cannot hope for a genuine inverse, but the Kulkarni–Nomizu product allows us to write down a right inverse for tr_g (see LeeRM Prop 7.23).

Definition 2.14 (Kulkarni–Nomizu product). Let (V, g) be a finite dimensional inner product space, and let $h, k \in \operatorname{Sym}^2(V^*)$ be two symmetric covariant 2-tensors on V. The *Kulkarni–Nomizu product*, denoted $h \owedge k$, is the covariant 4-tensor defined by

$$h \owedge k(w, x, y, z) = h(w, z)k(x, y) + h(x, y)k(w, z) - h(w, y)k(x, z) - h(x, z)k(w, y).$$

Note that by construction, the Kulkarni–Nomizu product of two symmetric covariant 2-tensors is an algebraic curvature tensor. We will also later have use for properties about the trace of the Kulkarni–Nomizu product of two symmetric covariant 2-tensors, especially when one or both are the metric g.

Lemma 2.15 (Properties of the Kulkarni–Nomizu product, LeeRM 7.22). Let V be an n-dimensional vector space with inner product g, let h be a symmetric covariant 2-tensor on V, and let tr_g denote the trace on the first and last indices. Then,

(a) $\operatorname{tr}_g(h \owedge g) = (n-2)h + (\operatorname{tr}_g h)g$,

(b) $\operatorname{tr}_g(g \owedge g) = 2(n-1)g$.

Proof. 1. Picking a basis, we can compute that

$$(\operatorname{tr}_g(h \owedge g))_{jk} = g^{il}(h_{il}g_{jk} + h_{jk}g_{il} - h_{ik}g_{jl} - h_{jl}g_{ik})$$

$$= h_i{}^i g_{jk} + g_i{}^i h_{jk} - h_{jk} - h_{jk}.$$

Since $\operatorname{tr}_g g = g_i{}^i = n$, (a) follows.
2. (b) follows simply by taking $h = g$ and again noting that $\operatorname{tr}_g g = n$. □

Manifolds of constant sectional curvature

We now return to the manifold setting. By the usual arguments, all of these operations extend to the appropriate sections of vector bundles on a Riemannian manifold (M, g).

Definition 2.16 (Sectional curvature). Let (M, g) be a Riemannian manifold, let $p \in M$, and let $v, w \in T_pM$ be linearly independent vectors spanning the 2-plane $\Pi \subseteq T_pM$. The *sectional curvature of* Π is defined to be

$$\sec(\Pi) = \sec(v, w) = \frac{\operatorname{Rm}_p(v, w, w, v)}{|v \wedge w|^2}.$$

We say that (M, g) has *constant sectional curvature* κ if $\sec(\Pi) = \kappa$ for all 2-planes $\Pi \subseteq T_pM$ at every $p \in M$.

Remark 2.17. We say that (M, g) is a *space form* if it is complete and has constant sectional curvature. The theory of space forms closely resembles the result of the uniformization theorem for surfaces: by the Killing–Hopf theorem (LeeRM Thm 12.4), every simply connected space form is isometric to either $\mathbb{R}^n$, $S^n(R)$, or $\mathbb{H}^n(R)$, which have constant sectional curvature 0, R^{-2}, or $-R^{-2}$, respectively (see LeeRM Prop 8.34). Hence, S^n and $\mathbb{R}^{n+1}$ are two of the nicest manifolds of constant sectional curvature, and in a sense (if we impose completeness) they account for two-thirds of the story.

Proposition 2.18 (Riemann curvature tensor in constant sectional curvature, LeeRM Prop 8.36). A Riemannian manifold (M^n, g) has constant sectional curvature κ if and only if its curvature tensor Rm satisfies

$$\operatorname{Rm} = \frac{1}{2}\kappa g \owedge g.$$

In this case,

$$\operatorname{Ric} = (n-1)\kappa g.$$

Proof. 1. For any linearly independent $v, w \in T_pM$,

$$(g \owedge g)(v, w, w, v) = g(v, v)g(w, w) + g(w, w)g(v, v) - g(v, w)g(w, v) - g(w, v)g(v, w) = 2\,|v \wedge w|^2.$$

Therefore, the sectional curvature of $\frac{1}{2}\kappa g \owedge g$ is

$$\frac{\kappa}{2}\frac{(g \owedge g)(v, w, w, v)}{|v \wedge w|^2} = \kappa.$$

Since sectional curvatures determine the curvature tensor (Proposition 2.12), it follows that Rm has constant sectional curvature κ if and only if $\operatorname{Rm} = \frac{1}{2}\kappa g \owedge g$.
2. Let tr_g denote taking the trace with respect to the first and last indices. Since $\operatorname{tr}_g(g \owedge g) = 2(n-1)g$ (Lemma 2.15), it follows that

$$\operatorname{Ric} = \operatorname{tr}_g \operatorname{Rm} = \frac{1}{2}\kappa \operatorname{tr}_g(g \owedge g) = (n-1)g.$$

□

§2.3 Riemannian submanifolds

We can now discuss submanifolds in an ambient Riemannian manifold (M, g). We will always assume our submanifolds are embedded, although many of the results in this thesis can be phrased in terms of immersions. Moreover, given an embedded submanifold $\iota : \Sigma^k \hookrightarrow (M^n, g)$, we will always give Σ the induced metric $\iota^* g$. We will sometimes distinguish the metrics as $g_\Sigma = \iota^* g$ and $g_M = g$, other times we will abuse notation and write both by g. We say that $\iota : (\Sigma, g_\Sigma) \hookrightarrow (M, g_M)$ is a *Riemannian submanifold* if $\iota^* g_M = g_\Sigma$.
In this section, we will always denote ∇ the connection on Σ and $\widetilde{\nabla}$ the connection on M; at other points in this thesis we also write ∇_Σ and ∇_M. We apologize in advance.

The second fundamental form

Let $\Sigma \hookrightarrow M$ be a Riemannian submanifold. Then, at each point we have $T_p\Sigma \subseteq T_pM$, and given the metric structure we can define the *normal space at* p: $N_p\Sigma = T_p\Sigma^\top$. We can then define the *normal bundle* $N\Sigma \to \Sigma$ on Σ by taking $N\Sigma = \bigsqcup_{p\in\Sigma} N_p\Sigma$; it is straightforward to show that $N\Sigma$ is a smooth manifold of dimension $\dim\Sigma + \operatorname{codim}\Sigma = \dim M$. Given any smooth vector field $X \in \mathfrak{X}(\Sigma)$ (or more generally $\Gamma(TM|_\Sigma)$), we can thus decompose it into its *normal* and *tangential* parts:

$$X = X^\perp + X^\top,$$

where $X^\perp \in \Gamma(N\Sigma)$ and $X^\top \in \Gamma(T\Sigma)$. This gives us a splitting $TM|_\Sigma = N\Sigma \oplus T\Sigma$ with

$$\pi^\perp(X) = X^\perp \in \Gamma(N\Sigma) \qquad \text{and} \qquad \pi^\top(X) = X^\top \in \Gamma(T\Sigma) = \mathfrak{X}(\Sigma).$$

We can thus decompose the Levi-Civita connection $\widetilde{\nabla}$ on M into its normal and tangential parts, for vector fields in $\mathfrak{X}(\Sigma)$.

Definition 2.19 (Vector second fundamental form). The *(vector) second fundamental form* $\mathrm{I\!I} \in \Gamma(T^{(1,2)}\Sigma)$ is the map $\mathrm{I\!I} : \mathfrak{X}(\Sigma) \times \mathfrak{X}(\Sigma) \to \Gamma(N\Sigma)$ given by

$$\mathrm{I\!I}(X, Y) = (\widetilde{\nabla}_X Y)^\perp.$$

Magically, the tangential part turns out to simply be equal to the Levi-Civita connection ∇ on Σ, by the *Gauss formula*:

Proposition 2.20 (Gauss formula, LeeRM Thm 8.2). Let $\Sigma \hookrightarrow (M, g)$ be a Riemannian submanifold, with Levi-Civita connections ∇ and $\widetilde{\nabla}$, respectively. Then, for any $X, Y \in \mathfrak{X}(\Sigma)$,

$$\widetilde{\nabla}_X Y = \nabla_X Y + \mathrm{I\!I}(X, Y).$$

We now recall several results and definitions regarding Riemannian submanifolds.

Proposition 2.21 (Gauss Equation, LeeRM Thm 8.5). Let $\Sigma \hookrightarrow (M, g)$ be a Riemannian submanifold, with Riemann curvature tensors Rm and $\widetilde{\mathrm{Rm}}$, respectively. Then, for any $W, X, Y, Z \in \mathfrak{X}(\Sigma)$,

$$\widetilde{\mathrm{Rm}}(W, X, Y, Z) = \mathrm{Rm}(W, X, Y, Z) - \langle \mathrm{I\!I}(W, Z), \mathrm{I\!I}(X, Y)\rangle + \langle \mathrm{I\!I}(W, Y), \mathrm{I\!I}(X, Z)\rangle.$$

Definition 2.22 (Mean curvature). Let $\Sigma \hookrightarrow (M, g)$ be a Riemannian submanifold. Then, the (vector) *mean curvature*, denoted $H \in \Gamma(N\Sigma)$, is defined as

$$H = \operatorname{tr}_g \mathrm{I\!I}.$$

If $(E_i)_{i=1}^k$ is an orthonormal frame for Σ, then

$$H = \sum_{i=1}^{k} \mathrm{I\!I}(E_i, E_i).$$

Definition 2.23 (Norm squared of second fundamental form). Let $\Sigma \hookrightarrow (M, g)$ be a Riemannian submanifold. Then, the *norm squared of the second fundamental form*, denoted $|\mathrm{I\!I}|^2 : \Sigma \to \mathbb{R}$, is defined as

$$|\mathrm{I\!I}|^2 = \sum_{i=1}^{k} |\,\mathrm{I\!I}(E_i, E_i)\,|^2$$

for an orthonormal frame $(E_i)_{i=1}^k$.

Hypersurfaces

Now let us suppose that $\Sigma^n \hookrightarrow (M^{n+1}, g)$ is a Riemannian hypersurface. Moreover, let us take Σ to be *two-sided*:

Definition 2.24 (Two-sided hypersurface). Let $\iota : \Sigma^n \hookrightarrow (M^{n+1}, g)$ be a Riemannian hypersurface. Then, Σ is *two-sided* if $N\Sigma$ is trivial. Equivalently, Σ admits a global smooth unit normal $\nu \in \Gamma(N\Sigma)$.

In the codimension 1 two-sided case, $N\Sigma \to \Sigma$ is a rank 1 trivial bundle, so $N\Sigma \cong \Sigma \times \mathbb{R}$. Then, for a choice of unit normal $\nu \in \Gamma(N\Sigma)$ (i.e., a choice of basis for $N_x\Sigma$ at each point $x \in \Sigma$), we get an isomorphism between their spaces of sections $\langle -, \nu \rangle : \Gamma(N\Sigma) \to C^\infty(\Sigma)$. Thus, we can define a canonical (up to sign) *scalar* second fundamental form and *scalar* mean curvature.

Definition 2.25 (Scalar second fundamental form and mean curvature). Let $\iota : \Sigma^n \hookrightarrow (M^{n+1}, g)$ be a two-sided hypersurface. Then, the *scalar second fundamental form*, denoted $A_\Sigma : \mathfrak{X}(\Sigma) \times \mathfrak{X}(\Sigma) \to C^\infty(\Sigma)$, is defined by

$$A_\Sigma(X, Y) = \langle \mathrm{I\!I}(X, Y), \nu \rangle .$$

The *scalar mean curvature*, denoted $H \in C^\infty(\Sigma)$, is

$$H = \operatorname{tr}_g A_\Sigma.$$

Definition 2.26 (Shape operator of a hypersurface). Let $\iota : \Sigma^n \hookrightarrow (M^{n+1}, g)$ be an isometric two-sided immersion. The *shape operator*, denoted $s : \mathfrak{X}(\Sigma) \to \mathfrak{X}(\Sigma)$, is the symmetric linear map corresponding to the scalar second fundamental form:

$$\langle sX, Y \rangle = \langle \mathrm{I\!I}(X, Y), \nu \rangle .$$

On a two-sided hypersurface, we have the following fundamental equations.

Theorem 2.27 (Fundamental equations for a hypersurface, LeeRM Thm 8.13). Let $\Sigma^n \hookrightarrow (M^{n+1}, g)$ be a two-sided hypersurface with unit normal ν, and let $\nabla, \widetilde{\nabla}$ and $\mathrm{Rm}, \widetilde{\mathrm{Rm}}$ denote their respective connections and Riemann curvature tensors.

(a) *Gauss formula:* If $X, Y \in \mathfrak{X}(\Sigma)$, then

$$\widetilde{\nabla}_X Y = \nabla_X Y + A_\Sigma(X, Y)\nu. \tag{2.1}$$

(b) *Weingarten equation:* If $X \in \mathfrak{X}(\Sigma)$, then

$$\widetilde{\nabla}_X \nu = -sX. \tag{2.2}$$

(c) *Gauss equation:* If $W, X, Y, Z \in \mathfrak{X}(\Sigma)$, then

$$\widetilde{\mathrm{Rm}}(W, X, Y, Z) = \widetilde{Rm}(W, X, Y, Z) - \frac{1}{2}(A_\Sigma \owedge A_\Sigma)(W, X, Y, Z). \tag{2.3}$$

(d) *Codazzi equation:* If $X, Y, Z \in \mathfrak{X}(\Sigma)$, then

$$\widetilde{Rm}(X, Y, Z, \nu) = (\nabla A_\Sigma)(X, Y, Z) - (\nabla A_\Sigma)(Z, X, Y). \tag{2.4}$$

In practice, we will be using these equations inside of an ambient manifold of constant sectional curvatures, in which computations are very nice.

Lemma 2.28 (Gauss equation). Let $\Sigma^n \hookrightarrow (M^{n+1}, g)$ be a two-sided hypersurface in a Riemannian manifold of constant sectional curvature κ. Then,

$$\mathrm{Rm} = \frac{1}{2} A_\Sigma \owedge A_\Sigma + \widetilde{\mathrm{Rm}} = \frac{1}{2} A_\Sigma \owedge A_\Sigma + \frac{1}{2}\kappa g \owedge g.$$

In coordinates,

$$\mathrm{Rm}_{ijkl} = A_{il}A_{jk} - A_{ik}A_{jl} + \kappa(g_{il}g_{jk} - g_{ik}g_{jl}).$$

Proof. This follows immediately from Proposition 2.18 and the definition of the Kulkarni–Nomizu product. □

Lemma 2.29 (∇A_Σ is symmetric). Let $\Sigma^n \hookrightarrow (M^{n+1}, g)$ be a two-sided hypersurface with unit normal ν. If M has constant sectional curvature κ, then ∇A_Σ is symmetric in all three entries.

Proof. 1. Since A_Σ is symmetric, we have that ∇A_Σ is symmetric in its first two entries. Thus, it suffices to show that it is symmetric with respect to the cyclic permutation (123). By the Codazzi equation (2.4), it suffices to show that

$$\mathrm{Rm}_M(X, Y, Z, \nu) = (\nabla A_\Sigma)(X, Y, Z) - (\nabla A_\Sigma)(Z, X, Y) = 0.$$

2. Since M has constant sectional curvature, by Proposition 2.18, $\mathrm{Rm}_M = \frac{1}{2}\kappa g \owedge g$. Computing,

$$(g \owedge g)(X, Y, Z, \nu) = g(X, \nu)g(Y, Z) + g(Y, Z)g(X, \nu) - g(X, Z)g(Y, \nu) - g(Y, \nu)g(X, Z) = 0,$$

since $g(-, \nu) = 0$ on $\mathfrak{X}(\Sigma)$. Thus the result holds. □

Computations in hypersurfaces

In concrete situations where a hypersurface is given as the level set to a smooth function (this is true, for instance, for $S^n \hookrightarrow \mathbb{R}^{n+1}$ and for graphs of functions), the scalar second fundamental form and mean curvature are easy to compute. We begin with a quick linear algebra lemma.

> Lemma 2.30. Let $(V, \langle -, - \rangle)$ be a finite dimensional inner product space and $w \in V$ has $|w| = 1$. Let $A : V \to V$ preserve $w^\perp$, and let $\Pi_{w^\perp} : V \to V$ denote the projection
>
> $$\Pi_{w^\perp} : v \mapsto v - \langle v, w \rangle w.$$
>
> Then, $\operatorname{tr}(A|_{w^\perp}) = \operatorname{tr}(A \circ \Pi_{w^\perp})$.

Proof. Assume $\dim V = n+1$ and let $e_1, \dots, e_n$ be an orthonormal basis for $w^\perp$, so that $w, e_1, \dots, e_n$ is an orthonormal basis for V which respects the decomposition $V \cong w \oplus w^\perp$. Then, writing A in terms of this basis, we get that

$$A = \begin{bmatrix} c_0 & 0 & \cdots & 0 \\ c_1 & b_{11} & \cdots & b_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ c_n & b_{n1} & \cdots & c_{nn} \end{bmatrix}$$

where $Aw = c_0 w + c_1 e_1 + \cdots + c_n e_n$ and $(A|_{w^\perp})_{ij} = b_{ij}$. Note that $\Pi_{w^\perp}|_{w^\perp} = \mathrm{id}_{w^\perp}$ and $\Pi_{w^\perp}|_w = 0$, so

$$A \circ \Pi_{w^\perp} = \begin{bmatrix} 0 & 0 & \cdots & 0 \\ c_1 & b_{11} & \cdots & b_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ c_n & b_{n1} & \cdots & c_{nn} \end{bmatrix}.$$

Hence, $\operatorname{tr}(A_{w^\perp}) = \sum_{i=1}^n b_{ii} = \operatorname{tr}(A \circ \Pi_{w^\perp})$. □

> Proposition 2.31 (Mean curvature of a regular hypersurface). Let $F : M^{n+1} \to \mathbb{R}$ be a map with $y \in \mathbb{R}$ a regular value. Then, $\nu = \frac{\nabla F}{|\nabla F|}$ is a unit normal of $\Sigma = F^{-1}(y)$, and its scalar second fundamental form is
>
> $$A_\Sigma(X, Y) = -\frac{\nabla^2 F(X, Y)}{|\nabla F|},$$
>
> and its scalar mean curvature is
>
> $$H_\Sigma = -\operatorname{div}_M \nu.$$

Proof. 1. Since y is a regular value, ∇F vanishes nowhere on Σ, so $\nu = \frac{\nabla F}{|\nabla F|}$ is a well-defined unit vector along Σ. Moreover, since $T_p\Sigma = \ker dF_p$, we have that for $v \in T_p\Sigma$, $\langle \nabla F(p), v \rangle = dF_p(v) = 0$, we have that $T_p\Sigma \subseteq \nu^\perp$ and therefore $T_p\Sigma = \nu^\perp$ since they are both dimension n. Thus ν is a global unit normal on Σ.
2. We can write the Hessian as

$$\nabla^2 F(X, Y) = X(YF) - (\nabla_X Y)F = \langle \nabla_X \nabla F, Y \rangle + \langle \nabla F, \nabla_X Y \rangle - \langle \nabla F, \nabla_X Y \rangle = \langle \nabla_X \nabla F, Y \rangle.$$

Recall that the second fundamental form satisfies

$$A_\Sigma(X, Y) = -\langle \nabla_X \nu, Y \rangle.$$

Computing explicitly,

$$\nabla_X \nu = (X|\nabla F|^{-1})\nu + |\nabla F|^{-1} \nabla_X \nabla F,$$

so since $Y \in \mathfrak{X}(\Sigma)$ and hence $\langle Y, \nu \rangle = 0$, we get

$$A_\Sigma(X,Y) = -\langle \nabla_X \nu, Y \rangle = -|\nabla F|^{-1} \langle \nabla_X \nabla F, Y \rangle = -\frac{\nabla^2 F(X,Y)}{|\nabla F|}.$$

3. Recall the shape operator $s : \mathfrak{X}(\Sigma) \to \mathfrak{X}(\Sigma)$ is defined by

$$\langle sX, Y \rangle = A_\Sigma(X,Y) = -\langle \nabla_X \nu, Y \rangle .$$

Now, note that $\nabla \nu : \mathfrak{X}(M) \to \mathfrak{X}(M)$ defined by $\nabla \nu : X \mapsto \nabla_X \nu$ restricts to $\nabla \nu|_{\mathfrak{X}(\Sigma)} = -s$ and hence preserves the subspace $\mathfrak{X}(\Sigma) = \nu^\perp$, and hence by Lemma 2.30 we have

$$H_\Sigma = \mathrm{tr}_\Sigma(s) = -\mathrm{tr}_M(\nabla \nu|_{\mathfrak{X}(\Sigma)}) = -\mathrm{tr}_M(\nabla \nu \circ \Pi_{T\Sigma}),$$

where $\Pi_{T\Sigma} : TM \to T\Sigma$ is the tangential projection. Now note that $\nabla \nu \circ \Pi_{T\Sigma} : X \mapsto \nabla_{X^\top} \nu$, and that

$$\langle \nabla_\nu \nu, \nu \rangle = \frac{1}{2} \nabla_\nu \langle \nu, \nu \rangle = 0,$$

so for an orthonormal basis $e_1, \dots, e_n \in T_p\Sigma$,

$$\mathrm{tr}_M(\nabla \nu \circ \Pi_{T\Sigma})(p) = \sum_{i=1}^n \left\langle \nabla_{e_i} \nu, e_i \right\rangle = \langle \nabla_\nu \nu, \nu \rangle + \sum_{i=1}^n \left\langle \nabla_{e_i} \nu, e_i \right\rangle = \mathrm{tr}_M(\nabla \nu)(p),$$

so it follows that

$$H_\Sigma = -\mathrm{tr}_M(\nabla \nu) = -\mathrm{div}_M \nu .$$

□

Lemma 2.32 (Second fundamental form of the sphere). The second fundamental form of $S^n \hookrightarrow \mathbb{R}^{n+1}$, equipped with the outward pointing unit normal, satisfies

$$A_{S^n} = -g_{S^n}.$$

Proof. Let $F = \frac{1}{2}|x|^2$; note that $S^n = F^{-1}(1)$. Note that $\nabla F = x$, so on S^n we have $|\nabla F| = 1$. The Hessian is

$$\nabla^2 F(X,Y) = X^i \delta_{ij} Y^j = \langle X, Y \rangle ,$$

so we conclude that

$$A_{S^n}(X,Y) = -\langle X, Y \rangle .$$

□

Proposition 2.33 (Mean curvature of a graph). Let $\Omega \subseteq \mathbb{R}^n$ open and $u : \Omega \to \mathbb{R}$. The scalar mean curvature to $\Gamma = \mathrm{graph}\, u$ with respect to the upward unit normal is

$$H_\Gamma = \mathrm{div}_\Omega \left(\frac{\nabla u}{\sqrt{1 + |\nabla u|^2}} \right).$$

Proof. 1. Denote $(x,y) \in \Omega \times \mathbb{R}$ and let $F(x,y) = y - u(x)$. Note that $dF_{(x,u(x))}(\partial_{n+1}) = 1$, so 0 is a regular value.

Then, $\Gamma = F^{-1}(0)$, and we have the unit normal

$$\nu = \frac{\nabla F}{|\nabla F|} = \frac{(-\nabla u, 1)}{\sqrt{1 + |\nabla u|^2}}.$$

2. We now compute $H_\Gamma = -\operatorname{div}_{\mathbb{R}^{n+1}} \nu$. Let $\partial_1, \dots, \partial_{n+1}$ denote the standard coordinate vector fields on $\mathbb{R}^{n+1}$, and $u_{;i} = \partial_i u$. Then,

$$\begin{aligned}
-\operatorname{div}_{\mathbb{R}^{n+1}} \nu &= \partial_i \left(\frac{u_{;i}}{\sqrt{1 + |\nabla u|^2}} \right) - \partial_{n+1} \left(\frac{1}{\sqrt{1 + |\nabla u|^2}} \right) \\
&= \partial_i \left(\frac{u_{;i}}{\sqrt{1 + |\nabla u|^2}} \right) \\
&= \operatorname{div}_\Omega \left(\frac{\nabla u}{\sqrt{1 + |\nabla u|^2}} \right).
\end{aligned}$$

□

§2.4 Variations

In this section we follow [Cho21b].

Recall that if (M^n, g) is an n-dimensional oriented[1] Riemannian manifold, there exists a unique n-form $d\mathrm{vol}_g \in \Omega^n(M)$, the *volume form*, which in local oriented coordinates $(x^1, \dots, x^n)$ can be written as

$$d\mathrm{vol}_g = \sqrt{\det g_{ij}}\, dx^1 \wedge \cdots \wedge dx^n.$$

Then, for any open subset $\Omega \subseteq M$, we define its *volume* by

$$\mathrm{vol}_g(\Omega) = \int_\Omega d\mathrm{vol}_g,$$

Note that this integral may not be finite if Ω is not compact; if the sum does not converge then we say $\mathrm{vol}_g(\Omega) = \infty$.

Now let $\iota : \Sigma^k \looparrowright (M^n, g)$ be an immersion and let $g^\Sigma := \iota^* g$ denote the induced metric on Σ. Let $d\mathrm{vol}_\Sigma := d\mathrm{vol}_{g^\Sigma}$ denote the induced volume form on Σ. Then, for $\Omega \subseteq \Sigma$ we define its *area*

$$\mathrm{Area}(\Omega) = \int_\Omega d\mathrm{vol}_\Sigma.$$

It is customary to use *area* and *volume* to distinguish between the volumes of a submanifold and its ambient manifold; we will try to closely observe this distinction.

We now seek to find submanifolds which are critical points for the area, so we introduce the notion of a variation.

Definition 2.34. Let $\iota : \Sigma^k \looparrowright (M^n, g)$ be an immersed Riemannian submanifold. A *variation* of Σ is a smooth map $F : J \times \Sigma \looparrowright M$ with $J \subseteq \mathbb{R}$ an open interval with $0 \in J$ such that:

(a) $F_0 = \iota$,

(b) $F_t := F(t, -)$ is an immersion for each $t \in J$.

[1] In the case where M is nonorientable, we can still define a *Riemannian density* and get a volume measure, but in our discussion all manifolds will be orientable so we need not work with this level of generality.

If, in addition, there exists a compact $K \subseteq \Sigma$ such that $F_t|_{\Sigma \setminus K} = \mathrm{id}$, we say that F is *compactly supported.*

Given a variation $F : J \times \Sigma \to M$ of Σ, we can define the corresponding *variational vector field* $X \in \Gamma(TM|_\Sigma)$ by

$$X_p = \frac{d}{dt}F(t,p)\bigg|_{t=0}$$

for $p \in \Sigma$. The following proposition gives us a reverse construction: given $X \in \Gamma(TM|_\Sigma)$, we can exponentiate to get a compactly supported variation whose variational vector field is X.

Proposition 2.35. Let $\iota : \Sigma \looparrowright (M, g)$ be an immersed submanifold. Let $X \in \Gamma_c(TM|_\Sigma)$. Then, there exists $\varepsilon > 0$ and a compactly supported variation $F : (-\varepsilon, \varepsilon) \times \Sigma \to M$ with variational vector field X. Moreover, if $X \in \Gamma_c(N\Sigma)$ then F is normal.

Proof. Let $K = \operatorname{supp} X \subseteq \Sigma$, and let $\mathcal{E} = \{(p, v) \in TM : \gamma_v$ is defined on an interval containing $[0, 1]\}$ be the domain of the exponential function. Take $\varepsilon > 0$ small such that $\mathrm{graph}(\varepsilon X) = \left\{(p, v) \in TM : v = \varepsilon X_p\right\} \subseteq \mathcal{E}$. Then, let

$$F_t(p) = \exp_p(tX_p).$$

By construction, we get $F : (-\varepsilon, \varepsilon) \times \Sigma \to M$. Since $X = 0$ outside of a compact set, $F \equiv \mathrm{id}$ outside of a compact set. Making $\varepsilon > 0$ smaller if necessary, since $F_0 = \iota$ is an immersion and immersions are a stable class, we get that F_t is an immersion for all $t \in (-\varepsilon, \varepsilon)$. Finally,

$$\frac{d}{dt}F(t,p)\bigg|_{t=0} = \frac{d}{dt}\exp_p(tX_p)\bigg|_{t=0} = \frac{d}{dt}tX_p\bigg|_{t=0} = X_p.$$

so F is a compactly supported variation with variational vector field X. □

Thus we can freely change perspectives between variations and vector fields. Henceforth, we will almost always focus on the vector field perspective, as it allows us to separate into tangential and normal components, according to the decomposition $TM|_\Sigma \cong N\Sigma \oplus T\Sigma$.

Let us first focus on the normal component. We say that a variation is *normal* if the corresponding variational vector field is in $\Gamma(N\Sigma) \subseteq \Gamma(TM|_\Sigma)$; that is, if $X_p \in N_p\Sigma$ for each $p \in \Sigma$. The following proposition says that all variations can be modified by a reparametrization to become a normal variation.

Proposition 2.36. Let $F : J \times \Sigma \to M$ be a compactly supported variation of $\iota : \Sigma \looparrowright (M, g)$ with variational vector field $X \in \Gamma(TM|_\Sigma)$. Then, there exists a 1-parameter family of compactly supported diffeomorphisms $\varphi : J \to \mathrm{Diff}_c(\Sigma)$ such that the variation $G : J \times \Sigma \to M$ defined by

$$G(t,p) = F(t, \varphi_t(p)),$$

where $\varphi_t := \varphi(t)$, is a compactly supported normal variation with variational vector field $Y_p = X_p^\perp \in \Gamma(N\Sigma)$.

Proof. Let $X = X^\perp + X^\top$ be the decomposition of the variational vector field according to the decomposition $TM|_\Sigma = N\Sigma \oplus T\Sigma$. Note that since F is compactly supported, so is X and hence so is $X^\perp$ and $X^\top$. By the existence and uniqueness of solutions to ODE (for compactly supported vector fields), we can define φ_t the flow associated to $-X^\top \in \Gamma_c(T\Sigma) = \mathfrak{X}_c(\Sigma)$ for all $t \in \mathbb{R}$: that is, φ solves the ODE

$$\frac{d}{dt}\varphi_t(p) = -X^\top_{\varphi_t(p)}.$$

Then,

$$\frac{d}{dt}F(t,\varphi_t(p))\Big|_{t=0} = \frac{d}{dt}F(t,\varphi_0(p))\Big|_{t=0} + \frac{d}{dt}F(0,\varphi_t(p))\Big|_{t=0} = X_p - X_p^{\top} = X_p^{\perp},$$

since $F_0 = \iota$ and $\varphi_0 = \mathrm{id}$. □

Hence, given a compactly supported variation, after reparametrizing we can assume that the variational vector field is normal.

§2.5 First variation of area

In this section, we will state the first variation formula and some of its immediate consequences. For a proof, see [CM11].

Definition 2.37 (First variation). Let $\Sigma \looparrowright (M,g)$ be an immersed submanifold, and let $X \in \Gamma_c(TM|_\Sigma)$ with $K = \operatorname{supp} X \subseteq \Sigma$. Let $F : (-\varepsilon,\varepsilon)\times\Sigma \to M$ be the corresponding variation, and let $d\mathrm{vol}_{\Sigma_t} := d\mathrm{vol}_{F_t^* g}$ denote the induced volume form on Σ with respect to the immersion $F_t : \Sigma \looparrowright M$. The *first variation of area of* Σ *with respect to* X is

$$\delta\Sigma(X) = \frac{d}{dt}\int_K d\mathrm{vol}_{\Sigma_t}\Big|_{t=0}.$$

Note that if Σ is compact, we can write the more obvious formula

$$\delta\Sigma(X) = \frac{d}{dt}\mathrm{Area}(F_t(\Sigma))\Big|_{t=0},$$

but of course if Σ is noncompact we cannot guarantee that the area of $F_t(\Sigma)$ is finite. However, since we are only dealing with compactly supported variations, we can simply integrate over the support of the variation.

To fully justify the notation $\delta\Sigma(X)$, we need to check that the first variation does not depend on the choice of variation, as long as its variational vector field is X. To check this property, we prove the following *first variation formula.*

Theorem 2.38 (First variation formula). Let $\Sigma \looparrowright (M,g)$ be an immersed Riemannian submanifold. Then, for $X \in \Gamma_c(TM|_\Sigma)$ with $X|_{\partial\Sigma} \equiv 0$,

$$\delta\Sigma(X) = \int_\Sigma \mathrm{div}_\Sigma X \; d\mathrm{vol}_\Sigma = -\int_\Sigma \langle H, X\rangle \; d\mathrm{vol}_\Sigma,$$

where $H \in \Gamma(N\Sigma)$ is the (vector) mean curvature.

As an immediate consequence, we get many useful properties of the first variation.

Corollary 2.39. Let $\Sigma \looparrowright (M,g)$ be an immersed Riemannian submanifold. Then, the first variation, considered as a map $\delta\Sigma : \Gamma_c(TM|_\Sigma) \to \mathbb{R}$, satisfies the following properties:

(a) (independence of variation): $\delta\Sigma(X)$ does not depend upon the choice of variation F, as long as F has variational vector field X.

(b) ($\mathbb{R}$-linearity): For $a, b \in \mathbb{R}$ and $X, Y \in \Gamma_c(TM_\Sigma)$, $\delta\Sigma(aX + bY) = a\delta\Sigma(X) + b\delta\Sigma(Y)$.

(c) (Vanishes on $T\Sigma$): For $X \in \mathfrak{X}_c(\Sigma)$, $\delta\Sigma(X) = 0$.

(d) (Reduction to normal variation): For $X \in \Gamma_c(TM_\Sigma)$, $\delta\Sigma(X) = \delta\Sigma(X^\perp)$.

Proof. (a) is immediate, (b) follows from the fact that integration and $\langle H, -\rangle$ are $\mathbb{R}$-linear, and (c) and (d) follow from the fact that $H \in \Gamma(N\Sigma)$: if $X \in \Gamma(TM|_\Sigma)$, then $\langle H, X\rangle = \left\langle H, X^\perp \right\rangle$. □

However, the most important consequence of the first variation formula is the correspondence between $\delta\Sigma$ and H.

Definition 2.40 (Minimal surface). Let $\Sigma \looparrowright (M, g)$ be an immersed Riemannian submanifold. We say that Σ is *minimal* if it satisfies one of the following equivalent conditions:

(a) Σ is a critical point for the area functional: $\delta\Sigma \equiv 0$.

(b) The mean curvature vanishes identically: $H \equiv 0$.

Proof. (a) $\implies$ (b): Let $\alpha = \sup_{p\in\Sigma} |H_p|$, and assume for the sake of contradiction that $\alpha > 0$. Then, there exists some $p \in \Sigma$ such that $|H_p| > \frac{\alpha}{2}$. Then, since $|H|$ is continuous, there exists an open neighborhood U with $p \in U \subseteq \Sigma$ such that $|H| > \frac{\alpha}{2}$ for all $p \in U$. Let $\rho \in C_c^\infty(\Sigma)$ be a smooth bump function with $\rho \geq 0$, $\rho \equiv 1$ on $V \subseteq U$, and $\operatorname{supp}\rho \subseteq U$, and let $X = -\rho H$. Then, by first variation,

$$\delta\Sigma(X) = -\int_\Sigma \langle H, X\rangle \; d\mathrm{vol}_\Sigma = \int_\Sigma \rho\,|H|^2 \; d\mathrm{vol}_\Sigma \geq \int_V \rho\,|H|^2 \; d\mathrm{vol}_\Sigma \geq \frac{\alpha}{2}\,\mathrm{vol}_\Sigma(V) > 0,$$

a contradiction.

(b) $\implies$ (a): This follows immediately from the first variation formula. □

Section 3

Stable minimal surfaces

Already in the case of $\Sigma^2 \hookrightarrow \mathbb{R}^3$, there are plenty of minimal surfaces (e.g., the catenoid, helicoid, Costa–Hoffman–Meeks surface, etc., see [CM11] for more discussion). So, to resolve Bernstein's theorem in $n \geq 2$ we need to analyze more than just minimality.

Recall from multivariable calculus that to find a local minimum of a function $f : \mathbb{R}^n \to \mathbb{R}$, we employ the *second derivative test*: first find a critical point $p \in \mathbb{R}^n$, then compute the *Hessian* $f_{;ij} := \frac{\partial^2 f}{\partial x^i \partial x^j}$. The Hessian is symmetric, hence diagonalizable, and we can look at its eigenvalues. For simplicity, let us assume that the Hessian is nondegenerate, so that all of its eigenvalues are nonzero. The *Morse index* of the critical point p is the number of negative eigenvalues counted with multiplicity; equivalently, the maximal dimension of subspaces of $T_p\mathbb{R}^n$ on which the quadratic form $Q_f(\xi) = \xi^i f_{;ij} \xi^j$ is negative definite. If the Morse index is zero, then the Hessian is positive definite at the point, and hence the point is a local minimum.[2]

A remarkably similar story works in the infinite dimensional setting. Using a similar procedure as in the first variation formula, we can compute the *second variation of area* of an immersion $\Sigma \looparrowright M$, and find that it has the structure of a quadratic form on $\Gamma_c(TM|_\Sigma)$ induced by a symmetric linear (partial differential) operator, called the

[2] We can similarly define the Morse index of a minimal surface; see [Cho24] for more on this story.

stability operator. In the two-sided codimension 1 case, we can reduce this to $C_c^\infty(\Sigma)$, hence giving us a foothold to begin applying PDE techniques to the study of minimal surfaces.

§3.1 Second variation formula

Definition 3.1 (Second variation). Let $\Sigma \looparrowright (M, g)$ be an immersed submanifold, and let $X \in \Gamma_c(TM|_\Sigma)$ with $K = \operatorname{supp} X \subseteq \Sigma$. Let $F : (-\varepsilon, \varepsilon) \times \Sigma \to M$ be the corresponding variation, and let $d\mathrm{vol}_{\Sigma_t} := d\mathrm{vol}_{F_t^* g}$ denote the induced volume form on Σ with respect to the immersion $F_t : \Sigma \looparrowright M$. The *second variation of area of* Σ *with respect to* X is

$$\delta^2\Sigma(X) = \frac{d^2}{dt^2} \int_K d\mathrm{vol}_{\Sigma_t}\bigg|_{t=0}.$$

As before, we can always reparametrize by a diffeomorphism to reduce our consideration to normal vector fields.

Definition 3.2. Let $\Sigma \looparrowright (M, g)$ be an immersed minimal submanifold. We say that Σ is *stable*[a] if for all compactly supported normal vector fields $X \in \Gamma_c(N\Sigma)$, $\delta^2\Sigma(X) \geq 0$.

[a]There is a slightly weaker definition which is useful in the non-smooth context: if the second variation of area is nonnegative for all compactly supported ambient diffeomorphisms. Any vector field along Σ can be locally smoothly extended to M, so this definition is weaker than ours. Nevertheless, we will only speak of stability on smooth minimal surfaces, so we can safely paper over this distinction.

Much like the first variation, we have a formula for the second variation (see [CM11] for proof). However, we will only have use for the case of a two-sided minimal hypersurface. In this case, the second variation formula becomes particularly simple. Let $\Sigma^n \hookrightarrow (M^{n+1}, g)$ be a two-sided minimal immersion. Then, note that $N\Sigma$ is rank 1, so $\Gamma(N\Sigma)$ is a rank 1 free module over $C^\infty(\Sigma)$ with generator ν. In other words, we can freely switch between normal vector fields and smooth functions by taking $X \mapsto \langle X, \nu\rangle$ and $\varphi \mapsto \varphi\nu$. Phrasing the second variation formula in terms of variations induced by $\varphi\nu$, we get the following version of the second variation formula.

Theorem 3.3 (Second variation formula for hypersurfaces). Let $\Sigma^n \looparrowright (M^{n+1}, g)$ be a two-sided minimal immersion with unit normal ν. The second variation with respect to $\varphi \in C_c^\infty(\Sigma)$ is

$$\delta^2\Sigma(\varphi) := \delta^2\Sigma(\varphi\nu) = \int_\Sigma |\nabla_\Sigma \varphi|^2 - |A_\Sigma|^2 \varphi^2 - \mathrm{Ric}_M(\nu, \nu)\varphi^2.$$

Integrating by parts, we can write second variation in terms of the *stability operator*

$$L_\Sigma \varphi = \Delta_\Sigma \varphi + |A_\Sigma|^2 \varphi + \mathrm{Ric}_M(\nu, \nu)\varphi.^3$$

That is,

$$\delta^2\Sigma(\varphi) = -\int_\Sigma \varphi L_\Sigma \varphi.$$

In the next section, we will make the connection between stability and the stability operator more explicit.

[3]This is sometimes also called the Jacobi operator, and solutions are called Jacobi fields. We will distinguish between the operator $L_\Sigma = \Delta_\Sigma + |A_\Sigma|^2 + \mathrm{Ric}_M(\nu, \nu)$ and $L_\Sigma^\circ = \Delta_\Sigma + |A_\Sigma|^2$; we call the first one the stability operator and the second the Jacobi operator. In Euclidean space they coincide, but on the sphere they will differ by a factor of $\mathrm{Ric}_{S^n}(\nu, \nu) = n - 1$.

§3.2 Schrödinger operators

The second variation formula tells us that the stability of a two-sided minimal hypersurface $\Sigma \hookrightarrow (M, g)$ is closely related to the *stability operator* $L_\Sigma = \Delta_\Sigma + |A_\Sigma|^2 + \mathrm{Ric}_M(\nu, \nu)$. Later, we will also have use for the *Jacobi operator* $L_\Sigma^\circ = \Delta_\Sigma + |A_\Sigma|^2$. It turns out that both operators are examples of a more general kind of elliptic partial differential operator of the form $L = \Delta + V$ for V a smooth potential, called a *Schrödinger operator.* In this section, we will survey several important technical results, using elliptic PDE theory, which will later help us analyze stable minimal hypersurfaces.

Recall that on a Riemannian manifold M, the $W^{1,2}$ norm is defined by

$$\|u\|_{W^{1,2}(M)} = \int_M |u|^2 + \int_M |\nabla u|^2 .$$

We denote $W^{1,2}(M)$ the completion of $C^\infty(M)$ under the $W^{1,2}$ norm, and $W_0^{1,2}(M)$ the completion of $C_c^\infty(M)$ under the $W^{1,2}$ norm.

Definition 3.4 (First eigenvalue). Let (M^n, g) be a Riemannian manifold, $\Omega \subseteq M$ open, and let $V \in C^\infty(M)$ be a potential and $L = \Delta + V$ the corresponding Schrödinger operator. The *first eigenvalue of L on Ω* is

$$\lambda_1(L, \Omega) = \inf_{\varphi \in W_0^{1,2}(\Omega)\setminus\{0\}} \frac{\int_\Omega |\nabla\varphi|^2 - V\varphi^2}{\int_\Omega \varphi^2}. \tag{3.5}$$

If M is compact, we write $\lambda_1(L) := \lambda_1(L, M \setminus \partial M)$.

In the above definition we can safely replace $W_0^{1,2}$ with C_c^∞, by the following theorem. For a reference, see [Cho21b].

Theorem 3.5 (Main theorem on the first eigenvalue). Let (Ω, g) be a compact Riemannian manifold with boundary, let $V \in C^\infty(\Omega)$ be a potential, and let $L = \Delta + V$ be the corresponding Schrödinger operator.

(a) *Existence:* There exists a smooth solution $\varphi \in C^\infty(\Omega)$ to the Dirichlet boundary value problem

$$L\varphi + \lambda_1(L)\varphi = 0 \text{ on } \Omega, \qquad \varphi \equiv 0 \text{ on } \partial\Omega. \tag{3.6}$$

(b) *Uniqueness:* The dimension of the λ_1-eigenspace is equal to 1; that is, for any other solution $\widetilde{\varphi}$ to (3.6), there exists a constant $\mu \in \mathbb{R} \setminus \{0\}$ such that $\mu\varphi = \widetilde{\varphi}$.

Remark 3.6. Note that a function is a solution to the Dirichlet boundary value problem (3.6) if and only if it satisfies the infimum of the Rayleigh quotient (3.5). In either case, we call the function a *first eigenfunction.*

From the usual Harnack inequality (see [GT13], Thm 8.20 and [CM11], Lemma 1.46), we have the following version for Schrödinger operators on compact Riemannian manifolds.

Theorem 3.7 (Harnack inequality for Schrödinger operators). Let (Ω, g) be a compact Riemannian manifold with boundary, let $V \in C^\infty(\Omega)$ be a potential, and let $L = \Delta + V$ be the corresponding Schrödinger operator. Suppose $\varphi \in C^0(\Omega) \cap W^{1,2}(\Omega)$ satisfies $\varphi \geq 0$ in Ω and $L\varphi = 0$ weakly in Ω. Then, for every $\omega \Subset \Omega$, there exists a constant $C = C(n, \omega, \Omega, V, g) > 0$ such that

$$\sup_\omega \varphi \leq C \inf_\omega \varphi.$$

Similarly, we have the following version of the maximum principle (see [GT13] Theorem 9.1).

> THEOREM 3.8 (Maximum principle for Schrödinger operators). Let (Ω, g) be a compact Riemannian manifold with boundary. If $\varphi \in C^\infty(\Omega) \cap C^0(\overline{\Omega})$, $\Delta\varphi \geq 0$, and $u \geq 0$, then
>
> $$\sup_\Omega u = \sup_{\partial\Omega} u.$$

As a consequence, we get the following powerful characterization of first eigenfunctions. We say that a function φ is a *eigenfunction* of L with eigenvalue λ if $L\varphi + \lambda\varphi = 0$ on Ω and $\varphi \equiv 0$ on $\partial\Omega$.

> PROPOSITION 3.9 (Eigenfunctions of the first eigenvalue). Let (Ω, g) be a compact Riemannian manifold with boundary, let $V \in C^\infty(\Omega)$ be a potential, and let $L = \Delta + V$ be the corresponding Schrödinger operator.
>
> (a) Let φ be an eigenfunction of L. Then, φ is a first eigenfunction if and only if φ does not change sign.
>
> (b) *Uniqueness in Theorem 3.5:* The dimension of the first eigenspace is 1.

Proof. We prove the results out of order: first we prove the forward direction of (a), then (b), then the backward direction of (a).
1. Assume φ a first eigenfunction, not identically zero. Note that since φ is smooth, $\nabla\varphi = \nabla|\varphi|$ a.e., so $|\varphi|$ also achieves the infimum of the Rayleigh quotient. Hence it is a smooth solution to $L|\varphi| + \lambda_1(L)|\varphi|$ with $|\varphi| \equiv 0$ on $\partial\Omega$, with $|\varphi| \geq 0$. Applying the Harnack inequality to the operator $\widetilde{L} = L + \lambda_1(L)$, it follows that $|\varphi| > 0$ in the interior of Ω, as desired.
2. Assume $\varphi, \widetilde{\varphi}$ are two first eigenfunctions, and assume they are both positive. Take $c_0 = \inf\{c \in \mathbb{R} : c\widetilde{\varphi} - \varphi > 0\}$. Clearly, $c_0\widetilde{\varphi} - \varphi = 0$ at some point (else we could make c_0 smaller), so by the maximum principle $c_0\widetilde{\varphi} = \varphi$, as desired.
3. Assume φ is an eigenfunction that does not change sign. Without loss of generality assume $\varphi > 0$ Let ψ be a first eigenfunction; by (1) we can assume $\psi > 0$. Then, $\langle\varphi, \psi\rangle_{L^2} > 0$. Since eigenspaces of distinct eigenvalues are orthogonal, it follows that φ is a first eigenfunction. ☐

As a result of Theorem 3.5 and Proposition 3.9, we get that on a compact Riemannian manifold Ω with a Schrödinger operator L, if $\lambda_1(L) \geq 0$, then there exists a function $\varphi \in C^\infty(\Omega)$ with $\varphi > 0$ on $\Omega \setminus \partial\Omega$ and $L\varphi = -\lambda_1\varphi \leq 0$. The following theorem says that this is an if and only if.

> THEOREM 3.10 (Barta). Let (Ω, g) be a compact Riemannian manifold with boundary, let $V \in C^\infty(\Omega)$ be a potential, and let $L = \Delta + V$ be the corresponding Schrödinger operator. Then,
>
> $$\lambda_1(L) \geq 0 \iff \text{there exists } u \in C^\infty(\Omega \setminus \partial\Omega) \text{ with } u > 0 \text{ and } Lu \leq 0.$$

Proof. 1. The forward direction follows from Theorem 3.5 and Proposition 3.9.
2. In the backward direction, let $u \in C^\infty(\Omega \setminus \partial\Omega)$ with $u > 0$ and $Lu \leq 0$. Set $w = \log u$. Then, $\nabla w = \frac{\nabla u}{u}$, so

$$\Delta w = \frac{\Delta u}{u} - \frac{|\nabla u|^2}{u^2} \leq -V - |\nabla w|^2$$

since $Lu \leq 0$, $\Delta u \leq -Vu$. Take $\varphi \in C_c^\infty(\Omega)$, multiply by φ^2, then integrate:

$$\int_\Omega V\varphi^2 \leq -\int_\Omega (\Delta w + |\nabla w|^2)\varphi^2 = \int_\Omega 2\varphi\langle\nabla w, \nabla\varphi\rangle - |\nabla w|^2\varphi^2.$$

By Cauchy–Schwarz and AM–GM, we have

$$2\varphi\langle\nabla w,\nabla\varphi\rangle \leq 2|\varphi||\nabla w||\nabla\varphi| \leq |\varphi|^2|\nabla w|^2 + |\nabla\varphi|^2,$$

and hence

$$\int_\Omega V\varphi^2 \leq \int_\Omega |\nabla\varphi|^2,$$

so $\lambda_1(L) \geq 0$. □

§3.3 Stable minimal hypersurfaces

With these tools under our belt, we can return to our study of stable minimal hypersurfaces, and minimal graphs in particular.

Recall that the *stability operator* of a two-sided minimal hypersurface $\Sigma^n \hookrightarrow (M^{n+1}, g)$ is

$$L_\Sigma = \Delta_\Sigma + |A_\Sigma|^2 + \mathrm{Ric}_M(\nu,\nu).$$

Proposition 3.11 (Stability criterion). Let $\Sigma^n \hookrightarrow (M^{n+1}, g)$ be a two-sided minimal hypersurface. Then, it is stable if and only if $\lambda_1(L_\Sigma, \Omega) \geq 0$ for all $\Omega \Subset \Sigma$.

Moreover, if Σ is compact, then it is stable if and only if $\lambda_1(L_\Sigma) := \lambda_1(L_\Sigma, \Sigma \setminus \partial\Sigma) \geq 0$.

Proof. 1. The forward direction is easy: if Σ is stable, then for any $\Omega \Subset \Sigma$, we have

$$\delta^2\Sigma(\varphi) = \int_\Omega |\nabla_\Sigma\varphi|^2 - |A_\Sigma|^2\varphi^2 - \mathrm{Ric}_M(\nu,\nu)\varphi^2 \geq 0$$

for all $\varphi \in C_c^\infty(\Omega)$, and hence $\lambda_1(L_\Sigma, \Omega) \geq 0$.

2. The backward direction is also easy: for any $\varphi \in C_c^\infty(\Sigma)$, take $\operatorname{supp}\varphi \subseteq \Omega \Subset \Sigma$. Then, since $\lambda_1(L_\Sigma, \Omega) \geq 0$, $\delta^2\Sigma(\varphi) \geq 0$.

3. For the final step, note that $\lambda_1(L_\Sigma, -)$ is monotonically decreasing with respect to set inclusion: if $\Omega \subseteq \Omega'$, then $\lambda_1(L_\Sigma, \Omega) \geq \lambda_1(L_\Sigma, \Omega')$. Therefore,

$$\lambda_1(L_\Sigma) = \inf\{\lambda_1(L_\Sigma, \Omega) : \Omega \Subset \Sigma\},$$

so $\lambda_1(L_\Sigma) \geq 0$ if and only if $\lambda_1(L_\Sigma, \Omega) \geq 0$ for all $\Omega \Subset \Sigma$. □

Remark 3.12. Note that this discussion implies that stability is equivalent to having the following inequality for any $\varphi \in C_c^\infty(\Sigma)$:

$$\int_\Sigma |A_\Sigma|^2\varphi^2 + \mathrm{Ric}_M(\nu,\nu)\varphi^2 \leq \int_\Sigma |\nabla\varphi|^2.$$

We call this inequality the *stability inequality*.

Using Barta's theorem, this gives us a way to check that minimal graphs are stable. To apply Barta's theorem, we will need a positive function u with $L_\Sigma u \leq 0$; we can in fact find one where $L_\Sigma u = 0$.

Lemma 3.13. Let $\Sigma^n \hookrightarrow \mathbb{R}^{n+1}$ be a two-sided minimal hypersurface with unit normal ν, and let $a \in \mathfrak{X}(\mathbb{R}^{n+1})$ be a parallel vector field (i.e., $\overline{\nabla} a = 0$, where $\overline{\nabla}$ is the ambient connection on $\mathbb{R}^{n+1}$). Then, $f = \langle \nu, a\rangle$ is a Jacobi field on Σ:

$$L_\Sigma f = \Delta_\Sigma f + |A_\Sigma|^2 f = 0.$$

Proof. 1. Denote ∇ the connection on Σ and $\overline{\nabla}$ the connection on $\mathbb{R}^{n+1}$. Let $p \in \Sigma$ and pick $E_1, \dots, E_n$ a local orthonormal frame at p that is ∇-parallel at p. Then, we get that

$$\Delta_\Sigma f = \sum_{i=1}^{n} \nabla_{E_i} \nabla_{E_i} \langle \nu, a \rangle$$

at p. We first compute the first derivative term $\nabla_{E_i} \langle \nu, a \rangle$. Since the connections agree with usual differentiation of functions,

$$\nabla_{E_i} \langle \nu, a \rangle = E_i \langle \nu, a \rangle = \overline{\nabla}_{E_i} \langle \nu, a \rangle .$$

Since a is parallel (and hence $\overline{\nabla}_{E_i} a = 0$), we get

$$\overline{\nabla}_{E_i} \langle \nu, a \rangle = \left\langle \overline{\nabla}_{E_i} \nu, a \right\rangle .$$

Since $\langle \nu, \nu \rangle \equiv 1$ is constant, for any vector field X we have

$$0 = X(1) = X \langle \nu, \nu \rangle = \overline{\nabla}_X \langle \nu, \nu \rangle = 2 \left\langle \nu, \overline{\nabla}_X \nu \right\rangle ,$$

so we get that

$$\left\langle \overline{\nabla}_{E_i} \nu, a^\perp \right\rangle = \left\langle \overline{\nabla}_{E_i} \nu, \langle a, \nu \rangle \nu \right\rangle = \langle a, \nu \rangle \left\langle \overline{\nabla}_{E_i} \nu, \nu \right\rangle = 0,$$

so in conclusion

$$\nabla_{E_i} \langle \nu, a \rangle = \left\langle \overline{\nabla}_{E_i} \nu, a^\top \right\rangle = A_\Sigma(E_i, a^\top). \tag{3.7}$$

2. We now compute the second derivative term $\nabla_{E_i} \nabla_{E_i} \langle \nu, a \rangle$. Expanding,

$$\nabla_{E_i} \nabla_{E_i} \langle \nu, a \rangle = \nabla_{E_i} (A_\Sigma(E_i, a^\top)) = (\nabla A_\Sigma)(E_i, a^\top, E_i) + A_\Sigma(\nabla_{E_i} E_i, a^\top) + A_\Sigma(E_i, \nabla_{E_i} a^\top).$$

The second term $A_\Sigma(\nabla_{E_i} E_i, a^\top)$ vanishes since E_i were chosen to be parallel at p. By the Codazzi equation, (Lemma 2.29), ∇A_Σ is symmetric, so we get that

$$(\nabla A_\Sigma)(E_i, a^\top, E_i) = (\nabla A_\Sigma)(E_i, E_i, a^\top) = \nabla_{a^\top} A_\Sigma(E_i, E_i) - 2A_\Sigma(\nabla_{a^\top} E_i, E_i).$$

Again, since E_i is parallel at p, the second term vanishes. Finally, taking the trace we get by minimality that

$$\sum_{i=1}^{n} (\nabla A_\Sigma)(E_i, a^\top, E_i) = \sum_{i=1}^{n} \nabla_{a^\top} A_\Sigma(E_i, E_i) = \nabla_{a^\top} H = 0.$$

Therefore, we get that

$$\sum_{i=1}^{n} \nabla_{E_i} \nabla_{E_i} \langle \nu, a \rangle = \sum_{i=1}^{n} A_\Sigma(E_i, \nabla_{E_i} a^\top). \tag{3.8}$$

3. We now compute the $\nabla_{E_i} a^\top$ term. By the Gauss formula, $\nabla_{E_i} a^\top = (\overline{\nabla}_{E_i} a^\top)^\top$. Since a is parallel,

$$(\overline{\nabla}_{E_i} a^\top)^\top = -(\overline{\nabla}_{E_i} a^\perp)^\top = -\left(\overline{\nabla}_{E_i} (\langle \nu, a \rangle \nu)\right)^\top = -\left((E_i \langle \nu, a \rangle)\nu - \langle \nu, a \rangle \overline{\nabla}_{E_i} \nu\right)^\top = -\langle \nu, a \rangle \left(\overline{\nabla}_{E_i} \nu\right)^\top$$

since we are only taking the tangential component. Finally, this gives us that

$$\nabla_{E_i} a^\top = -\langle \nu, a \rangle \sum_{j=1}^{n} \left\langle \overline{\nabla}_{E_i} \nu, E_j \right\rangle E_j = -\langle \nu, a \rangle \sum_{j=1}^{n} A_\Sigma(E_i, E_j) E_j. \tag{3.9}$$

4. Putting (3.8) and (3.9) together, we get that

$$\Delta_\Sigma \langle \nu, a \rangle = \sum_{i=1}^{n} \nabla_{E_i} \nabla_{E_i} \langle \nu, a \rangle = -\langle \nu, a \rangle \sum_{i,j=1}^{n} A_\Sigma(E_i, E_j)^2 = -\langle \nu, a \rangle \, |A_\Sigma|^2 ,$$

and therefore $L_\Sigma \langle \nu, a \rangle = 0$, as desired. □

THEOREM 3.14 (Minimal graphs are stable). Let $\Omega \subseteq \mathbb{R}^n$ and let $u : \Omega \to \mathbb{R}$ solve the MSE. Then, $\Gamma = \operatorname{graph} u$ is stable.

Proof. 1. Let ν denote the usual upward pointing unit normal

$$\nu = \frac{(-\nabla u, 1)}{\sqrt{1 + |\nabla u|^2}}.$$

Take $f = \langle \nu, e_{n+1} \rangle$, which we can compute is

$$f = \frac{1}{\sqrt{1 + |\nabla u|^2}} > 0.$$

But, by Lemma 3.13, $L_\Sigma f = 0$. By Barta's theorem (Theorem 3.10) and the stability criterion (Proposition 3.11), it follows that Γ is stable. □

REMARK 3.15. Recall that we said at the beginning that a graph is minimal if its area only goes up when wiggled. Strictly speaking this is false, since that sounds more like a definition of stability. Nonetheless, minimal graphs are indeed stable, so this is overall a relatively harmless lie.

SECTION 4

MINIMAL GRAPHS AND CALIBRATIONS

The idea of calibrated geometry goes back to an influential paper of Harvey and Lawson [HL82]. In modern geometry and topology, we often study special classes of manifolds with certain structures (e.g., smooth, oriented, Riemannian, symplectic, almost complex, complex, Kähler, etc.) defined on them. This geometric structure is often defined in terms of either (i) the structure group of the tangent bundle (e.g., orientable manifolds with structure group $GL_+(n, \mathbb{R})$), or (ii) some globally defined section of a tensor bundle (e.g., a global nonvanishing volume form). However, we might also take a more classical point of view, understanding a geometric structure in terms of (iii) a distinguished family of submanifolds (e.g., Euclidean geometry in terms of lines and circles).

It turns out that the machinery of calibrated geometry, while abstract, readily gives us a proof that minimal graphs are area minimizing on compact sets. Using this, we can get area growth estimates for minimal graphs, which is our final recipe in the proof of Bernstein's theorem for $n = 2$.

§4.1 Calibrations

Let V be an inner product space with inner product $\langle -, - \rangle$ and norm $|-|_V$, and let $\bigwedge^k V$ be the k-fold wedge product (see [Lan05], [LeeRM], and [Fed14] for references on the relevant tensor algebra). Recall that a k-vector

$\xi \in \bigwedge^k V$ is *simple* if $\xi = v_1 \wedge \cdots \wedge v_k$, and we say a simple k-vector is *unit* if $v_1, \ldots, v_k$ is orthonormal. Then, we can define the *comass norm* on $\bigwedge^k V$ by

$$|\phi|_{\bigwedge^k V} = \sup \left\{ \langle \phi, \xi \rangle \ : \ \xi \in \bigwedge^k V \text{ unit simple } k\text{-vector} \right\}.$$

Naturally, this definition extends to differential forms on a Riemannian manifold.

Definition 4.1 (Comass of a differential form). Let M be a smooth Riemannian manifold and let $\alpha \in \Omega^k(M)$ be a differential k-form. The *comass of α at a point p* is

$$|\alpha_p|_{\bigwedge^k(T_p^* M)} = \sup \left\{ \alpha_p(v_1, \ldots, v_k) : \ v_1 \wedge \cdots \wedge v_k \text{ unit simple } k\text{-vector in } \bigwedge^k T_p M \right\},$$

and the *comass of α on M* is

$$\|\alpha\|_{\Omega^k(M)} = \sup_p |\alpha_p|_{\bigwedge^k(T_p^* M)}.$$

Definition 4.2 (Calibrations). Let (M^n, g) be a Riemannian manifold. A k-form $\alpha \in \Omega^k(M)$ is a *calibration* if $d\alpha = 0$ and the comass $\|\alpha\|_{\Omega^k(M)} \leq 1$. An embedded submanifold $\iota : \ \Sigma^k \hookrightarrow M$ is *calibrated* by α if

$$\alpha|_\Sigma = \ d\mathrm{vol}_\Sigma,$$

where $\alpha|_\Sigma \ := \iota^* \alpha$ is the restriction of the calibration to Σ and $\ d\mathrm{vol}_\Sigma \ := \ d\mathrm{vol}_{\iota^* g}$ is the induced volume form.

Note that α having comass 1 means that for any k-manifold with corners $\Sigma' \hookrightarrow M$, for any (positively oriented) unit simple k-vector $v_1 \wedge \cdots \wedge v_k \in \bigwedge^k T_p M$ we have

$$\alpha_p(v_1, \ldots, v_k) \leq 1 = (\ d\mathrm{vol}_{\Sigma'})_p(v_1, \ldots, v_k)$$

and hence by linearity, $\alpha \leq \ d\mathrm{vol}_{\Sigma'}$ (on oriented k-vectors). This is the primary utility of the calibration for proving that calibrated submanifolds are homologically area minimizing.

Proposition 4.3 (Calibrated submanifolds are homologically area-minimizing). Let $\Sigma^k \hookrightarrow (M^n, g)$ be an embedded calibrated submanifold. Then, Σ is homologically area-minimizing: for any k-chain $[\Sigma'] \in H_k(M)$ represented by a k-manifold with corners Σ',

$$\mathrm{Area}(\Sigma) \leq \mathrm{Area}(\Sigma').$$

Proof. The proof is essentially a one-line application of Stokes' theorem:

$$\mathrm{Area}(\Sigma) = \int_\Sigma d\mathrm{vol}_g = \int_\Sigma \alpha = \int_{\Sigma'} \alpha \leq \int_{\Sigma'} d\mathrm{vol}_{\Sigma'} = \mathrm{Area}(\Sigma').$$

The first and last equality are by definition, the second equality is by the fact that Σ is calibrated by α, and the fourth inequality comes from the fact that α has comass at most 1. The third equality is Stokes' theorem: if $[W] \in H_{k+1}(M)$ is a $k+1$-chain represented by a manifold with corners with $\partial[W] = [\Sigma] - [\Sigma']$, then since α is closed,

$$0 = \int_W d\alpha = \int_\Sigma \alpha - \int_{\Sigma'} \alpha.$$

□

Lemma 4.4 (Minimal graphs are calibrated). Let $\Omega \subseteq \mathbb{R}^n$ be open and let $u \in C^\infty(\Omega)$ be a solution to the MSE. Let ν denote the upward unit normal

$$\nu = \frac{(-\nabla u, 1)}{\sqrt{1 + |\nabla u|^2}}.$$

Then, the n-form $\alpha = \iota_\nu \, d\mathrm{vol}_{\mathbb{R}^{n+1}}$ is a calibration of $\Gamma = \operatorname{graph} u$.

Proof. 1. We show that α is closed. By Proposition 2.31, $H_\Gamma = -\operatorname{div}_{\mathbb{R}^{n+1}} \nu$, so

$$d\alpha = d\iota_\nu \, d\mathrm{vol}_{\mathbb{R}^{n+1}} = \operatorname{div}_{\mathbb{R}^{n+1}} \nu \, d\mathrm{vol}_{\mathbb{R}^{n+1}} = -H_\Gamma \, d\mathrm{vol}_{\mathbb{R}^{n+1}} = 0.$$

2. We show that α has comass at most 1. For any orthonormal $e_1, \dots, e_n \in T_p\mathbb{R}^{n+1}$,

$$\left|\alpha_p(e_1, \dots, e_n)\right| = \left| d\mathrm{vol}_{\mathbb{R}^{n+1}}(\nu, e_1, \dots, e_n)\right| = \left|\det(\nu_p, e_1, \dots, e_n)\right| \le 1.$$

3. We show that $\alpha|_\Gamma = d\mathrm{vol}_\Gamma$. For any oriented orthonormal $e_1, \dots, e_n \in T_p\Gamma$,

$$\alpha_p(e_1, \dots, e_n) = d\mathrm{vol}_{\mathbb{R}^{n+1}}(\nu_p, e_1, \dots, e_n) = 1 = d\mathrm{vol}_\Gamma(e_1, \dots, e_n),$$

as desired. □

Corollary 4.5 (Minimal graphs are area minimizing). Let $\Omega \subseteq \mathbb{R}^n$ be open, let $u \in C^\infty(\Omega)$ be a solution to the MSE, and let $\Gamma = \operatorname{graph} u$. Then, Γ is area-minimizing on the cylinder $\Omega \times \mathbb{R}$:

$$\operatorname{Area}(\Gamma) = \inf\left\{\operatorname{Area}(\Gamma') : \Gamma' \subseteq \Omega \times \mathbb{R} \text{ smooth hypersurface with boundary } \partial\Gamma' = \partial\Gamma\right\}.$$

The upshot of minimal graphs being area-minimizing is that we can prove area growth bounds.

Theorem 4.6 (Area growth bounds for minimal graphs). Let $u \in C^\infty(\mathbb{R}^n)$ be a solution to the MSE. Then, $\Gamma = \operatorname{graph} u$ has the following extrinsic area growth estimate:

$$\mathrm{vol}_\Gamma(\Gamma \cap B_r) \le Cr^n$$

for $C = C(n)$, where $B_r = B_r(0) \subseteq \mathbb{R}^{n+1}$.

The main idea is as follows: $\Gamma \cap \partial B_r$ cuts ∂B_r into two subsets, and we can pick one to be a competitor to $\Gamma \cap B_r$. But, this competitor is a subset of the sphere ∂B_r and hence has area bounded by $\operatorname{Area}(\partial B_r) = \sigma_n r^n$ (where $\sigma_n = \operatorname{Area}(S^n)$). Then, since minimal graphs are area-minimizing on cylinders, we get the desired estimate.
The most technical part of this proof will be ensuring the existence of this competitor. The idea is to pick a generic $r > 0$ such that $\Gamma \pitchfork \partial B_r$, then show that the intersection of $E = \operatorname{epi} u$ with B_r is a domain with $\partial[E \cap B_r] = [\Gamma \cap B_r] - [E \cap \partial B_r]$.

Proof. 1. Let us denote $\iota : \mathbb{R}^n \to \mathbb{R}^{n+1}$ the embedding $\iota(x) = (x, u(x))$, such that $\iota(\mathbb{R}^n) = \Gamma$. Then, define the map $F : \mathbb{R}^n \times \mathbb{R}_{>0} \to \mathbb{R}^{n+1}$ by

$$F(x, s) = s \cdot \iota(x) = (sx, su(x)),$$

and $F_s(x) = F(x, s)$. Note that $\iota^{-1}(\Gamma \cap \partial B_r) = F_{1/r}^{-1}(S^n)$, so $\Gamma \pitchfork \partial B_r$ if and only if $\iota \pitchfork \partial B_r$ if and only if $F_{1/r} \pitchfork S^n$.
2. Now we check that $F \pitchfork S^n$. Denote $W = F^{-1}(S^n) \subseteq \mathbb{R} \times \mathbb{R}_{>0}$, and suppose $(x, s) \in W$. Then, $dF_{(x,s)} : \mathbb{R}^{n+1} \to \mathbb{R}^{n+1}$ is given by the block matrix

$$dF_{(x,s)} = \begin{bmatrix} s\,d\iota_x & \iota(x) \end{bmatrix}.$$

Hence, we have that $dF_{(x,s)}(\partial_s) = \iota(x) \in \mathbb{R}^{n+1}$, and since $(x,s) \in W = F^{-1}(S^n)$ we have $F(x,s) = s\iota(x) \in S^n$ and hence $\iota(x) \in \partial B_{1/s}$, so $\iota(x) \in \mathbb{R}^{n+1}$ spans the normal direction to S^n, so $(T_{F(x,s)}S^n)^\perp \subseteq \operatorname{im} dF_{(x,s)}$, and hence $F \pitchfork S^n$.
3. Therefore, by the parametric transversality theorem (Theorem A.2), we have that $F_s \pitchfork S^n$ for almost every $s \in \mathbb{R}_{>0}$. Now pick a generic $r > 0$ such that $F_{1/r} \pitchfork S^n$; i.e., $\Gamma \pitchfork \partial B_r$.
4. Denote by $E = \operatorname{epi} u = \{(x,y) \in \mathbb{R}^{n+1} : y \geq u(x)\}$ the epigraph, and note that $\partial E = \Gamma$. Then, by transversality we get that

$$\partial[E \cap B_r] = [\Gamma \cap B_r] - [E \cap \partial B_r].$$

By replacing $\Gamma \cap B_r$ with $E \cap \partial B_r$, we get a competitor on a cylinder $Z_R = B_R \times \mathbb{R}$, where $R > r$. Then,

$$[\Gamma \cap Z_R] - [\Gamma \cap B_r] + [E \cap \partial B_r]$$

is a competitor for $\Gamma \cap Z_R$ on the cylinder Z_R, represented by a manifold with corners (along $\Gamma \cap \partial B_r$). Then,

$$\operatorname{vol}_\Gamma(\Gamma \cap Z_R) \leq \operatorname{vol}_\Gamma(\Gamma \cap Z_R) - \operatorname{vol}_\Gamma(\Gamma \cap B_r) + \operatorname{vol}_{\partial B_r}(E \cap \partial B_r),$$

so

$$\operatorname{vol}_\Gamma(\Gamma \cap B_r) \leq \operatorname{vol}_{\partial B_r}(E \cap \partial B_r) \leq \operatorname{vol}_{\partial B_r}(\partial B_r) = \sigma_n r^n,$$

where $\sigma = \operatorname{Area}(S^n)$.

□

Another convenient consequence is that we get an easy proof that minimal graphs are stable.

Proposition 4.7 (Minimal graphs are stable). Let $u \in C^\infty(\mathbb{R}^n)$ be a solution to the MSE. Then, $\Gamma = \operatorname{graph} u$ is stable.

Proof. Let $\varphi \in C_c^\infty(\Gamma)$, and let $F : (-\varepsilon, \varepsilon) \times \Gamma \to \mathbb{R}^{n+1}$ be the corresponding variation. Let $\Omega \Subset \mathbb{R}^n$ be a large open set with $\Omega \times \mathbb{R} \supseteq \operatorname{supp} \varphi$. Then, for all $t \neq 0$, $F_t(\Gamma)$ is a competitor for Γ in the same homology class (since $H_n(\mathbb{R}^{n+1}) = 0$) and hence $\operatorname{Area}_\Gamma(\Gamma \cap \Omega \times \mathbb{R}) \leq \operatorname{Area}_\Gamma(F_t(\Gamma) \cap \Omega \times \mathbb{R})$ for all $t \neq 0$. It follows that $\delta^2\Gamma(\varphi) \geq 0$. Since φ was arbitrary, Γ is stable. □

§4.2 Bernstein's theorem, $n = 2$

In this section, we prove Bernstein's theorem for $n = 2$. Recall that a function $f : \mathbb{R}^n \to \mathbb{R}$ is called *affine* if $f(x) = \langle a, x \rangle + b$, where $a \in \mathbb{R}^n$, $b \in \mathbb{R}$, and $\langle -, - \rangle$ is the Euclidean dot product. By Taylor's theorem, a smooth function f is affine if and only if all second derivatives vanish identically.
The proof of this theorem relies upon three key ingredients: (1) minimal graphs are stable (Theorem 3.14 and Proposition 4.7), (2) area growth bounds for minimal graphs (Theorem 4.6), and (3) the log cutoff trick (Theorem 4.9).

Definition 4.8 (Volume growth). Let $\iota : \Sigma \hookrightarrow (M, g)$ be a Riemannian submanifold. We say that Σ has *extrinsic volume growth of order k* if

$$\sup_{B_r(p) \subseteq M} \frac{\operatorname{vol}_\Sigma(\Sigma \cap B_r(p))}{r^k} < \infty,$$

where $B_r(p) \subseteq M$ is an extrinsic ball.

If M is connected, then we can simply consider the volume growth with respect to one point: this is because for any two points $p, q \in M$, we have that any geodesic ball about p is contained in a bigger geodesic ball about q: by the triangle inequality, $B_r(p) \subseteq B_{r+d_g(p,q)}(q)$. Hence, it suffices to say that for some point p, there exists a constant $C > 0$ such that

$$\mathrm{vol}_g(\Sigma \cap B_r(p)) \leq Cr^k.$$

Note that an embedded submanifold need not have volume growth on the same order as its dimension: $\Sigma = \bigcup_{n\in\mathbb{Z}} \mathbb{R} \times \{n\} \subseteq \mathbb{R}^2$ is a 1-dimensional submanifold with quadratic area growth (i.e., area growth of order 2).

Theorem 4.9 (Log cutoff trick). Let $\Sigma^2 \hookrightarrow \mathbb{R}^n$ be a smooth submanifold with quadratic area growth. Then, there exist cutoff functions $\eta_N \in C_c^{0,1}(\Sigma)$ such that $\eta_N \nearrow \mathbb{1}_\Sigma$ pointwise and $\|\nabla_\Sigma \eta_N\|_{L^2(\Sigma)} \searrow 0$ as $N \to \infty$.

Proof. For $N \in \mathbb{N}^*$, define $\eta_N \in C_c^{0,1}(\Sigma)$ by

$$\eta_N(x) = \begin{cases} 1 & |x| \in [0, e^N] \\ 2 - \frac{\log|x|}{N} & |x| \in [e^N, e^{2N}] \\ 0 & |x| \in [e^{2N}, \infty). \end{cases}$$

Clearly, we have $\eta_N \nearrow \mathbb{1}_\Sigma$ as $N \to \infty$.
Then, note that since $|\nabla_\Sigma |x|| \leq |\nabla |x|| = \left|\frac{x}{|x|}\right| = 1$, we have that

$$|\nabla_\Sigma \eta_N(x)| \leq \frac{1}{N|x|} \tag{4.10}$$

Now, note that $\nabla_\Sigma \eta_N$ is only nonzero on the annulus $\Sigma \cap (B_{e^{2N}} \setminus B_{e^N})$. At this point we would like to integrate with respect to spherical coordinates, but we don't have any control on the measure of the slices $\Sigma \cap \partial B_r$, so instead we use the quadratic area growth bound and sum over concentric annular regions $A_k = \Sigma \cap (B_{e^{k+1}} \setminus B_{e^k})$, constructed such that $\Sigma \cap (B_{e^{2N}} \setminus B_{e^N}) = \bigcup_{k=N}^{2N-1} A_k$. Note, then, that by the quadratic volume growth assumption, there exists $C > 0$ such that

$$\mathrm{vol}_g(A_k) \leq \mathrm{vol}_g(\Sigma \cap B_{e^{k+1}}) \leq Ce^{2k+2}.$$

Moreover, by (4.10) we have that

$$\sup_{A_k} |\nabla_\Sigma \eta_N|^2 \leq \frac{1}{N^2 e^{2k}},$$

and hence we get

$$\int_\Sigma |\nabla_\Sigma \eta_N|^2 \, d\mathrm{vol}_g \leq \sum_{k=N}^{2N-1} \int_{A_k} |\nabla_\Sigma \eta_N|^2 \, d\mathrm{vol}_g \leq \sum_{k=N}^{2N-1} \mathrm{vol}_g(A_k) \sup_{A_k} |\nabla_\Sigma \eta_N|^2 \leq \sum_{k=N}^{2N-1} \frac{Ce^2}{N^2} = \frac{Ce^2}{N}.$$

Taking $N \to \infty$, we get the desired result. □

We can thus finally prove Bernstein's theorem for $n = 2$:

Theorem 4.10 (Bernstein's theorem, $n = 2$). Let $u \in C^2(\mathbb{R}^2)$ solve the minimal surface equation

$$\mathcal{M}u = \mathrm{div}\left(\frac{\nabla u}{\sqrt{1 + |\nabla u|^2}}\right) = 0.$$

Then, u is affine.

Proof. Let $\Gamma = \operatorname{graph} u$. By Theorem 3.14, Γ is stable, and by Theorem 4.6, Γ has quadratic area growth. Thus, applying the log cutoff functions η_N (Theorem 4.9) to the stability inequality, we get

$$\int_{\Gamma \cap B_{e^N}} |A_\Gamma|^2 \leq \int_\Gamma |A_\Gamma|^2 |\eta_N|^2 \leq \int_\Gamma |\nabla_\Gamma \eta_N|^2 ,$$

and hence taking the limit as $N \to \infty$ we get $\int_\Gamma |A_\Sigma|^2 = 0$ and hence $A_\Sigma \equiv 0$. Since the only totally geodesic complete surfaces in $\mathbb{R}^3$ are planes, Γ is a plane and hence u is affine. $\square$

Chapter II

A rough detour

The core insight in the proof of Bernstein's theorem is that we can pass questions about minimal graphs to their *blowdowns*, or their *tangent cones at infinity*. The idea is as follows: suppose u is a nonflat solution to the minimal surface equation, and denote $\Sigma = \operatorname{graph} u$. For any $\lambda > 0$ we can consider the rescaled graph $\lambda\Sigma = \lambda \operatorname{graph}(u(\lambda^{-1}-))$. If we take a sequence $\lambda \to \infty$, we are "zooming in," so we should expect the graphs to converge to the tangent plane at 0. On the other hand, if we take $\lambda \to 0$, we are "zooming out," and given a suitable notion of compactness of hypersurfaces, we should expect that after choosing an adequate sequence $\lambda_j \to 0$, the rescaled graphs $\lambda_j\Sigma$ converge to a cone which captures the information of graph u "at infinity." As a concrete example, the graph of $u(x, y) = x^2 - y^2$ blows down to the cone

$$\left\{(x, y, z) \in \mathbb{R}^3 : |x| = |y|\right\} = \left\{(x, y) \in \mathbb{R}^2 : |x| = |y|\right\} \times \mathbb{R}.$$

Here, the limit is not smooth, so we can only make sense of this convergence as Caccioppoli sets; nevertheless, this cone preserves some useful properties of the original graph. The obvious example of this is that if u is affine (i.e., graph u is flat), then the blowdown is a plane. However, we can prove that this is an if and only if: if the blowdown is flat, the original graph must have been flat. Another example is that the blowdown of a minimal graph (which is minimizing as a smooth hypersurface and hence as Caccioppoli sets) must remain minimizing as a Caccioppoli set. So, if we can prove that all minimizing cones in $\mathbb{R}^{n+1}$ are flat, we get an affirmative solution to Bernstein's problem in dimension n.

Section 5

Theory of Caccioppoli sets

In order to prove Bernstein's theorem in higher dimensions, we seek to reduce minimal graphs to their *blowdowns*: that is, the limit of λ^{-1} graph u as $\lambda \to \infty$. However, generally these blowdowns are not smooth; in fact, any nonflat minimal graph blows down to a nonflat cone, hence not a smooth manifold. To handle this, we define a slightly broader class of potentially singular hypersurfaces, via the theory of *Caccioppoli sets*.

The outline of this section is as follows. First, we will review the basic features of Radon measures. Second, we discuss functions of bounded variation, which leads us into the theory of Caccioppoli sets, or sets of bounded perimeter.

§5.1 The classical Riesz–Markov–Kulkarni theorem

We review some basic facts about Radon measures on general σ-compact locally compact Hausdorff (LCH) spaces. Recall that a space is *σ-compact* if it is the union of countably many compact subspaces. In particular, every smooth manifold is LCH and second countable, and hence σ-compact.

Definition 5.1. Let X be an LCH space, and denote by $\mathcal{B}_X$ its Borel σ-algebra. Then, a map $\mu : \mathcal{B}_X \to [0, \infty]$ is called a *Radon measure* if it satisfies the following three properties:

(a) (inner regular) For $U \subseteq X$ open, $\mu(U) = \sup_{K \subseteq U \text{ compact}} \mu(K)$,

(b) (outer regular) For $A \subseteq X$ Borel, $\mu(A) = \inf_{U \supseteq A \text{ open}} \mu(U)$,

(c) (locally finite) For all $x \in X$ there exists a neighborhood $U \ni x$ with $\mu(U) < \infty$.

Let us denote by $\mathcal{R}_+(X)$ the space of Radon measures on X.

Hence, Radon measures are Borel measures (i.e., measures on the Borel σ-algebra) which are regular and locally finite, hence "behaving nicely" with respect to the topological structure on X. In fact, we can also interpret Radon measures in a distributional sense, as follows.

For any compact $K \subseteq X$, define $C(K)$ to be the Banach space of continuous functions $\varphi : K \to \mathbb{R}$ equipped with the sup norm

$$\|\varphi\|_{C(K)} = \|\varphi\|_{L^\infty} = \sup_K |\varphi|.$$

Let $C_c(X)$ denote the space of compactly supported continuous functions on X, equipped with the locally convex topology induced by the inclusions $C(K) \hookrightarrow C_c(X)$. That is, $\varphi_i \to \varphi$ in $C_c(X)$ if and only if $\varphi_i|_K \to \varphi|_K$ in $C(K)$ (that is, uniformly) for all $K \Subset X$ compact.

Definition 5.2 (Positive linear functionals on $C_c(X)$). Let X be an LCH space. Let $(C_c(X))^*$ denote the *continuous* dual: that is, $L \in (C_c(X))^*$ are linear maps $L : C_c(X) \to \mathbb{R}$ such that for every compact $K \Subset X$ there exists a constant $C_K > 0$ such that $|L(\varphi)| \leq C_K \|\varphi\|_{C(K)}$for all $\varphi \in C(K)$.

A functional $L \in (C_c(X))^*$ is called *positive* if $L(\varphi) \geq 0$ for all $\varphi \geq 0$; we denote the space of positive continuous linear functionals by

$$(C_c(X))^*_+ = \{L \in (C_c(X))^* : L \text{ positive}\}.$$

Every Radon measure gives rise to a positive linear functional on $C_c(X)$ by integration: if $\mu \in \mathcal{R}_+(X)$, then we can define

$$L_\mu(\varphi) := \int_X \varphi \, d\mu,$$

which is clearly a positive linear functional. Indeed, the map $\mathcal{R}_+(X) \to (C_c(X))^*_+$ defined by $\mu \mapsto L_\mu$ is injective: if $\mu, \nu \in \mathcal{R}_+(X)$ are Radon measures with $L_\mu = L_\nu$, then for any open set U we can find a sequence of compactly supported continuous bump functions $\varphi_i \nearrow \mathbb{1}_U$ pointwise. Then, by the monotone convergence theorem,

$$\mu(U) = \int_U d\mu = \lim_{i\to\infty} \int_X \varphi_i \, d\mu = \lim_{i\to\infty} \int_X \varphi_i \, d\nu = \int_U d\nu = \nu(U).$$

The following theorem (see [Coh13] for a proof) says that this map is also surjective, and hence we can think of Radon measures and positive linear functionals on $C_c(X)$ interchangeably.

Theorem 5.3 (Riesz–Markov–Kulkarni representation theorem). Let X be an LCH space. Then, the map $\mathcal{R}_+(X) \to (C_c(X))^*_+$ is a bijection. That is, every positive linear functional on $C_c(X)$ is represented by a unique Radon measure: for every $L \in (C_c(X))^*_+$ there exists a unique Radon measure μ such that

$$L(\varphi) = \int_X \varphi\, d\mu.$$

We can now define convergence of Radon measures in terms of weak* convergence on compact sets. Note that for a Radon measure μ, we can define its *restriction* $\mu|_K$ by $\mu|_K(A) = \mu(A \cap K)$. By uniqueness in the Riesz representation theorem, we can see that the linear functional $L_{\mu|_K}$ is precisely L_μ restricted to $C(K)$. Hence, we can define a norm on the space of Radon measures using the dual norm on $(C(K))^*$:

$$\|\mu|_K\|_{\mathcal{R}_+(K)} = \|L_{\mu|_K}\|_{(C(K))^*} = \sup\left\{\int_X \varphi\, d\mu :\ \varphi \in C(K), \|\varphi\|_{C(K)} = 1\right\}.$$

In other words, we force the bijection in the Riesz–Markov–Kulkarni representation to be an isometry. Then, all of the functional analytic facts about $(C_c(X))^*_+ \subseteq (C_c(X))^*$ transfer over to $\mathcal{R}_+(X)$.

Definition 5.4 (Weak convergence of Radon measures). Let X be an LCH space and $\mu_i, \mu \in \mathcal{R}_+(X)$ Radon measures on X. If $X = K$ is compact, then μ_i *converges weakly to* μ (denoted $\mu_i \rightharpoonup \mu$) if L_{μ_i} converges to L_μ in the weak* topology; i.e., for every $\varphi \in C(K)$,

$$\int_K \varphi\, d\mu_i \to \int_K \varphi\, d\mu$$

in $\mathbb{R}$. In general, we say that μ_i *converges weakly to* μ if $\mu_i|_K \rightharpoonup \mu|_K$ for every compact $K \Subset X$.

By characterizing convergence of Radon measures in terms of weak* convergence on $(C_c(X))^*$, we can use the Banach–Alaoglu theorem to prove a compactness result for Radon measures, provided that X is nice enough to run a diagonalization argument.

Theorem 5.5 (Radon compactness). Let X be a σ-compact LCH space, and let μ_i be Radon measures on X. If μ_i are uniformly bounded on compact sets (i.e., $\sup_i \|\mu_i|_K\| < \infty$ for all $K \Subset X$), then there exists a Radon measure μ and a weakly convergent subsequence $\mu_{i_j} \rightharpoonup \mu$.

Proof. 1. Since X is σ-compact, there exist compact $K_i \Subset X$ such that $X = \bigcup_{i=1}^\infty K_i$. Take $V_0 = \emptyset$. Since X is LCH we can inductively choose V_i open such that $V_i \supseteq K_i \cup \overline{V_{i-1}}$ and $\overline{V_i}$ is compact. Hence we get a compact exhaustion $X = \bigcup_{i=1}^\infty \overline{V_i}$.
2. For each i, j, define $L_{i,j} \in (C(\overline{V_j}))^*$ by

$$L_{i,j}(\varphi) = \int_{\overline{V_j}} \varphi\, d\mu_i.$$

By assumption, we have that $\sup_i \|L_{i,j}\|_{(C(\overline{V_j}))^*} = \sup_i \left\|\mu_i|_{\overline{V_j}}\right\| < \infty$, so by the Banach–Alaoglu theorem we have a weak* convergent subsequence $L_{i',j} \rightharpoonup^* L_j$. Choosing successive subsequences, we can take a diagonal sequence and get a weak* convergent subsequence $L_{\mu_{i'}} \rightharpoonup L$, where $L(\varphi) = L_j(\varphi)$ whenever $\operatorname{supp}\varphi \subseteq \overline{V_j}$.
3. If $\varphi \geq 0$, then $L_{i,j}(\varphi) \geq 0$ and hence $L_j(\varphi) \geq 0$, so $L(\varphi) \geq 0$, so L is positive. Finally, by the Riesz representation theorem, there exists a Radon measure μ representing $L = L_\mu$, so that $L_{\mu_{i'}}(\varphi) \to L_\mu(\varphi)$ for all $\varphi \in C_c(X)$ and hence $\mu_{i'} \rightharpoonup \mu$. $\square$

§5.2 Vector-valued Radon measures

In general, we would like a notion of Radon measures which are valued in $\mathbb{R}^n$. This is difficult to acomplish solely thorugh a measure-theoretic definition, so instead we will define vector-valued Radon measures to simply satisfy the Riesz representation theorem by fiat.

Denote $\langle -, - \rangle$ and $|-|$ the standard dot product and norm on $\mathbb{R}^n$. Just like above, let $C_c(X, \mathbb{R}^n)$ denote the set of continuous compactly supported maps $X \to \mathbb{R}^n$, equipped with the locally convex topology induced by the inclusions from the Banach spaces $C(K, \mathbb{R}^n)$ of continuous maps $K \to \mathbb{R}^n$ with sup norm $\|\varphi\|_{C(K,\mathbb{R}^n)} = \sup_{x\in K} |\varphi(x)|$.

Definition 5.6. Let X be an LCH space. A *vector-valued Radon measure* μ is an element of $(C_c(X, \mathbb{R}^n))^*$.

Then, we have the following version of the Riesz representation theorem (see [Sim14] Theorem 5.14)

Theorem 5.7 (Riesz–Markov–Kulkarni representation theorem for vector-valued Radon measures). Let X be an LCH space, and μ a vector-valued Radon measure on X. Then, there exists a Radon measure $|\mu|$ on $(X, \mathcal{B}_X)$ and a Borel measurable map $v : X \to \mathbb{R}^n$ with $|v| = 1$ $|\mu|$-a.e. on X, such that for any $\varphi \in C_c(X, \mathbb{R}^n)$,

$$\mu(\varphi) = \int_X \langle \varphi, v \rangle \, d|\mu|.$$

Note that we have an identical compactness result as that for Radon measures if X is σ-compact.

§5.3 Functions of bounded variation

Recall that for $\Omega \subseteq \mathbb{R}^n$ open, we define $W^{1,1}(\Omega)$ to be the closure of $C^\infty(\Omega)$ under the norm

$$\|u\|_{W^{1,1}(\Omega)} = \int_\Omega |u| + \int_\Omega |\nabla u|,$$

or equivalently those functions $u \in L^1(\Omega)$ which have weak distributional derivatives in L^1. Then, for any $u \in W^{1,1}(\Omega)$ and $\psi \in C_0^\infty(\Omega, \mathbb{R}^n)$ we have

$$-\int_\Omega \langle Du, \psi \rangle \, dx = \int_\Omega u \operatorname{div} \psi \, dx.$$

Now, note that the RHS gives us a functional on $C_0^\infty(\Omega, \mathbb{R}^n)$ which is continuous with respect to the sup norm on $C_0^\infty(\Omega, \mathbb{R}^n)$:

$$\left| \int_\Omega u \operatorname{div} \psi \, dx \right| \leq \|Du\|_{L^1} \|\psi\|_{L^\infty}.$$

Thus, the functional extends continuously to $C_0^1(\Omega, \mathbb{R}^n)$. Hence, using the Riesz representation theorem (Theorem 5.7) we can think of Du as a vector-valued measure with $|Du| = |Du| \, dx$ (the former is the Radon measure associated to the vector-valued measure, and the latter is the absolute value of the weak derivative multiplied by the Lebesgue measure). Nevertheless, we can insist on the functional $\int_\Omega u \operatorname{div} \psi \, dx$ to be continuous (and hence by the Riesz representation theorem, representable by a vector-valued Radon measure) without necessarily assuming that $u \in W^{1,1}$; this is precisely the space of BV functions.

Definition 5.8 (BV functions). Let $\Omega \subseteq \mathbb{R}^n$ be open. A function $u \in L^1(\Omega)$ has *bounded variation* (BV) if its distributional derivatives are representable by a finite vector-valued Radon measure:

$$\int_\Omega u \operatorname{div} \psi = -\int_\Omega \psi \, dDu$$

for Du a vector-valued Radon measure.

Similarly, a function $u \in L^1_{\mathrm{loc}}(\Omega)$ is *locally of bounded variation* ($\mathrm{BV}_{\mathrm{loc}}$) if its distributional derivatives are representable by a vector-valued Radon measure, or equivalently if for every $K \Subset \Omega$ there exists a constant $c_K > 0$ such that

$$\left| \int_K u \operatorname{div} \psi \, dx \right| \le c_K \, \|\psi\|_{L^\infty(K)}$$

for all $\psi \in C^\infty_c(K, \mathbb{R}^n)$.

The space of BV (resp. $\mathrm{BV}_{\mathrm{loc}}$) functions is denoted $\mathrm{BV}(\Omega)$ (resp. $\mathrm{BV}_{\mathrm{loc}}(\Omega)$).

To see why this nomenclature makes sense, we introduce the notion of the *variation* of a function.

Definition 5.9. Let $\Omega \subseteq \mathbb{R}^n$ open. The *variation* of a function $u \in L^1_{\mathrm{loc}}(\Omega)$ on an open set $W \subseteq \Omega$ is the norm of the functional $\int_W u \operatorname{div} \psi \, dx$; that is,

$$|Du|(W) = \sup \left\{ \int_W u \operatorname{div} \psi : \ \psi \in C^\infty_c(W) : \ \|\psi\|_{L^\infty} \le 1 \right\}.$$

Proposition 5.10 (Characterization of BV). Let $\Omega \subseteq \mathbb{R}^n$ be open. Then,

(a) $u \in L^1(\Omega)$ is BV if and only if $|Du|(\Omega) < \infty$,

(b) $u \in L^1_{\mathrm{loc}}(\Omega)$ is $\mathrm{BV}_{\mathrm{loc}}$ if and only if $|Du|(K) < \infty$ for all $K \Subset \Omega$.

Remark 5.11 ($W^{1,1}(\Omega) \subseteq \mathrm{BV}(\Omega)$). By the Riesz representation theorem, there exists a measurable map $v : \Omega \to \mathbb{R}^n$ with $|v| = 1$ almost everywhere, such that

$$\int_\Omega u \operatorname{div} \psi = -\int_\Omega \langle \psi, v \rangle \, d|Du|.$$

If in particular $u \in W^{1,1}(\Omega)$, we can see that

$$\int_\Omega u \operatorname{div} (\psi^1, \dots, \psi^n) = \sum_{i=1}^n \int_\Omega u \partial_i \psi^i = -\sum_{i=1}^n \int_\Omega \partial_i u \psi^i = -\int_\Omega \langle \nabla u, \psi \rangle$$

so in this case we can take $v = \frac{\nabla u}{|\nabla u|}$ (with $v = 0$ where $\nabla u = 0$) and $Du = |\nabla u| \, dx$. Hence, $W^{1,1} \subseteq \mathrm{BV}$.

Remark 5.12 ($W^{1,1}(\Omega) \subsetneq \mathrm{BV}(\Omega)$). However, the above inclusion does not hold, and it is instructive to consider a counterexample. Let $E \Subset \mathbb{R}^n$ be a bounded open subset with smooth boundary, and let $\mathbb{1}_E$ denote its characteristic function. Then, $\|\mathbb{1}_E\|_{L^1} = \mathrm{vol}(E) < \infty$, so $\mathbb{1}_E \in L^1(\mathbb{R}^n)$. To check that $\mathbb{1}_E \in \mathrm{BV}(\mathbb{R}^n)$, consider ν its outward pointing normal. Then, by the divergence theorem, for $\psi \in C^\infty_c(\mathbb{R}^n)$,

$$\int_{\mathbb{R}^n} \mathbb{1}_E \operatorname{div} \psi = \int_E \operatorname{div} \psi = \int_{\partial E} \langle \psi, \nu \rangle \le \int_{\partial E} |\psi| |\nu| \le \|\psi\|_{L^\infty} \mathrm{vol}(\partial E),$$

where vol is the volume with respect to the $(n-1)$-dimensional volume form on ∂E. Conversely, taking ψ to be a smooth extension of ν, the inequality above becomes an equality, so $|D\mathbb{1}_E|(\mathbb{R}^n) = \mathrm{vol}(\partial E)$. Hence $\mathbb{1}_E \in \mathrm{BV}(\mathbb{R}^n)$. However, $\mathbb{1}_E \notin W^{1,1}$; note that $\mathbb{1}_E$ is locally constant away from ∂E (which has Lebesgue measure 0), and hence any candidate for a weak derivative would be zero almost everywhere. But then $\mathbb{1}_E$ would have to be constant a.e., which is not true since E has positive measure.

Remark 5.13. Note that $\mathbb{1}_{[0,\infty)} \in \mathrm{BV}(\mathbb{R})$ with $\left|D\mathbb{1}_{[0,\infty)}\right| = \delta_0$. Also note that $[0,\infty)$ is a 1-manifold with boundary $\{0\}$. Combining this with the above remark, we can intuitively interpret $|D\mathbb{1}_E|$ as a sum of delta-measures along the boundary of E, so that taking the total variation (i.e., integrating $|D\mathbb{1}_E|$ on $\mathbb{R}^n$) amounts to adding up the volume of ∂E. We can make this more precise by looking at the support: $\operatorname{supp}|D\mathbb{1}_E| = \partial E$.

We will chiefly be interested in BV functions which are not $W^{1,1}$, like those in Remark 5.12. Nevertheless, some of the properties of the Sobolev space $W^{1,1}$ have analogues for BV; we will explore these now.

Proposition 5.14 (Semicontinuity of variation). Let $\Omega \subseteq \mathbb{R}^n$ be open, let $u, u_i \in \mathrm{BV}(\Omega)$ or $\mathrm{BV}_{\mathrm{loc}}(\Omega)$ with $u_i \to u$ in $L^1_{\mathrm{loc}}(\Omega)$. Then,

$$\int_\Omega |Du| \le \liminf_{i\to\infty} \int_\Omega |Du_i|.$$

Proof. We prove for $u \in \mathrm{BV}(\Omega)$; the $\mathrm{BV}_{\mathrm{loc}}$ case is similar by replacing Ω with an arbitrary $W \Subset \Omega$.
Let $\psi \in C^1_0(\Omega, \mathbb{R}^n)$ with $\|\psi\|_{L^\infty} \le 1$. Then, since $u_i \to u$ in L^1_{loc},

$$\int_\Omega u \operatorname{div}\psi = \lim_{i\to\infty} \int_\Omega u_i \operatorname{div}\psi \le \liminf_{i\to\infty} |Du_i|(\Omega).$$

Taking the sup on both sides gives the result. □

Proposition 5.15 (BV is a Banach space). The space $\mathrm{BV}(\Omega)$ is a Banach space with norm

$$\|u\|_{\mathrm{BV}(\Omega)} = \|u\|_{L^1(\Omega)} + \int_\Omega |Du|.$$

Proof. 1. Since $\|-\|_{L^1(\Omega)}$ and $|Du|(\Omega)$ are norms (recall that the variation is by definition the norm of the functional $\int_\Omega u \operatorname{div}\psi$, considered as an element of the dual space $(C^1_0(\Omega))^*$), $\|-\|_{\mathrm{BV}}$ is a norm.
2. It remains to prove completeness. Let $u_i \in \mathrm{BV}(\Omega)$ be a Cauchy sequence. Then, it is also a Cauchy sequence in L^1, and hence has a subsequential limit (without relabeling) $u_i \to u$ in L^1. Since u_i is Cauchy in BV, $\|u_i\|_{\mathrm{BV}}$ is bounded, so $|Du_i|(\Omega)$ is bounded. By semicontinuity (Proposition 5.14), $|Du|(\Omega) \le \liminf_{i\to\infty} |Du_i|(\Omega) < \infty$, so $u \in \mathrm{BV}(\Omega)$.
3. It remains to show that $u_i \to u$ in BV. We already have convergence in L^1, so it suffices to show that $|D(u-u_i)|(\Omega) \to 0$. Since u_i is Cauchy in BV, for $\varepsilon > 0$ we have $N > 0$ such that $i, j \ge N$ implies $\left\|u_i - u_j\right\|_{\mathrm{BV}} < \varepsilon$ and hence $\left|D(u_i - u_j)\right|(\Omega) < \varepsilon$. Since $u_j \to u$ in L^1, it follows that $u_i - u_j \to u_i - u$ in L^1 as $j \to \infty$. Thus by semicontinuity,

$$|D(u_i - u)|(\Omega) \le \liminf_{j\to\infty} \left|D(u_i - u_j)\right|(\Omega) \le \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, $|D(u_i - u)| \to 0$ as $i \to \infty$. □

REMARK 5.16. By Remark 5.11, the $W^{1,1}$ norm and the BV norm coincide on the subspace $W^{1,1} \subseteq \mathrm{BV}$. Indeed, as such the inclusion is continuous. However, unlike $W^{1,1}$ we should not expect BV to be the closure of C^∞ under the BV norm; since the BV norm coincides with the $W^{1,1}$ norm on C^∞, the closure under the BV norm would still be $W^{1,1}$. Nonetheless, we still have analogues of smooth approximation and compactness (á la Rellich compactness) results.

Let $\eta \in C_c^\infty(\mathbb{R}^n)$ be a symmetric mollifier; i.e., $0 \leq \eta \leq 1$, $\operatorname{supp} \eta \subseteq B_1(0)$, $\int_{\mathbb{R}^n} \eta = 1$, and $\eta(x) = \eta(-x)$. Let us define $\eta_\sigma(x) = \sigma^{-n}\eta(\sigma^{-1}x)$ the rescaled mollifier, and for $u \in L^1_{\mathrm{loc}}(\Omega)$ define $u^{(\sigma)} = \widetilde{u} * \eta_\sigma$, where we say $\Omega_\sigma = \{x \in \Omega : \operatorname{dist}(x, \partial\Omega) > \sigma\}$ and $\widetilde{u} = u$ on Ω_σ and $\widetilde{u} = 0$ otherwise. Recall [cf. Evans 5.3.1 Theorem 1] that if $u \in W^{1,1}$, this procedure gives a sequence $u^{(\sigma)} \in C^\infty(\Omega_\sigma)$ with $u^{(\sigma)} \to u$ in $W^{1,1}_{\mathrm{loc}}$. The following gives us a slightly weaker result for BV functions.

> PROPOSITION 5.17 (Smooth approximation). Let $\Omega \subseteq \mathbb{R}^n$ be open, and let $u \in \mathrm{BV}(\Omega)$ (resp. $u \in \mathrm{BV}_{\mathrm{loc}}(\Omega)$). Then, $u^{(\sigma)} \to u$ in $L^1(\Omega)$ (resp. $L^1_{\mathrm{loc}}(\Omega)$) and $\left|Du^{(\sigma)}\right| \rightharpoonup |Du|$ as Radon measures.

Proof. 1. The convergence $u^{(\sigma)} \to u$ in L^1_{loc} is standard (see Evans, C.5 Theorem 7(iv)). By semicontinuity (Proposition 5.14), it suffices to prove that for any $\varphi \in C_c(\Omega)$ with $\varphi \geq 0$,

$$\limsup_{\sigma \to 0} \int_\Omega \varphi \, d\left|Du^{(\sigma)}\right| \leq \int_\Omega \varphi \, d|Du|.$$

2. Note that

$$\int_\Omega \varphi \, d\left|Du^{(\sigma)}\right| = \sup\left\{\int_\Omega u^{(\sigma)} \operatorname{div} \psi : \psi \in C_c^\infty(\Omega), |\psi| \leq \varphi\right\}.$$

Computing,

$$\int_\Omega u^{(\sigma)} \operatorname{div} \psi = \int_\Omega (u * \eta_\sigma) \operatorname{div} \psi = \int_\Omega u\,(\eta_\sigma * \operatorname{div} \psi) = \int_\Omega u \operatorname{div}(\eta_\sigma * \psi),$$

where $\eta_\sigma * \psi = (\eta_\sigma * \psi^1, \dots, \eta_\sigma * \psi^n)$. Note that $|\eta_\sigma * \psi| \leq \eta_\sigma * |\psi| \leq \eta_\sigma * \varphi$. Now, pick $\sigma_0 < \operatorname{dist}(\operatorname{supp} \varphi, \partial\Omega)$, and note that on $\Omega_{\sigma_0} = \{x \in \Omega : \operatorname{dist}(x, \partial\Omega) > \sigma_0\}$, $\eta_\sigma * \varphi \to \varphi$ in C^∞_{loc} and hence uniformly. Therefore, for each $\sigma < \sigma_0$ there exists $\varepsilon > 0$ such that $|\eta_\sigma * \psi| \leq \varphi + \varepsilon$. Thus

$$\int_\Omega u^{(\sigma)} \operatorname{div} \psi \leq \int_\Omega (\varphi + \varepsilon) \, d|Du|.$$

Taking the supremum among all $\psi \in C_c^\infty(\Omega)$ with $|\psi| \leq \varphi$, we get

$$\int_\Omega \varphi \, d\left|Du^{(\sigma)}\right| \leq \int_\Omega (\varphi + \varepsilon) \, d|Du|.$$

Since for any $\sigma < \sigma_0$ we can find an $\varepsilon > 0$ making the above inequality hold, taking the limit we get the desired result. □

As a consequence of Rellich compactness (Theorem A.3), we get that $W^{1,1} \Subset L^1$; the idea is to use smooth approximation so that we can apply Rellich, then take a diagonal sequence, then use semicontinuity of variation to ensure the limit is in BV (see [Giu84] Theorem 1.19 or [Sim14] Theorem 2.6).

> THEOREM 5.18 (BV compactness). Let $\Omega \subseteq \mathbb{R}^n$ be open. Then, $\mathrm{BV}_{\mathrm{loc}}(\Omega) \Subset L^1_{\mathrm{loc}}(\Omega)$. If, in addition, Ω is bounded with C^1 boundary, then $\mathrm{BV}(\Omega) \Subset L^1(\Omega)$.

§5.4 Caccioppoli sets

Recall that in Remark 5.12 we considered the characteristic function of an open subset E of $\mathbb{R}^n$ with smooth boundary, showing that the area of the boundary is related to the Radon measure associated to the characteristic function's distributional derivative. It turns out that E need not actually have smooth boundary; it can be permitted to be only smooth almost everywhere. Indeed, note that in Remark 5.12 we actually only needed to have $\nu \in C^1(\partial E, \mathbb{R}^n)$ in order to extend it to a C^1 vector field to use it as a test function—that is, it suffices to have E have C^2 boundary. Pushing further, we might ask that E have C^1 boundary (i.e., smooth enough to make the divergence theorem work) and ask for $\nu \in W^{1,1}(\partial E, \mathbb{R}^n)$. Past this, the divergence theorem ceases to hold, but we can nevertheless force the divergence theorem to hold in a distributional sense, which allows us to build a theory of oriented hypersurfaces which are smooth almost everywhere. To make this notion precise, we define the notion of a *Caccioppoli set*.

Definition 5.19. Let $E \subseteq \mathbb{R}^n$ be Borel. We say that E is a *Caccioppoli set* if $\mathbb{1}_E \in \mathrm{BV}_{\mathrm{loc}}(\mathbb{R}^n)$.

Remark 5.20. Caccioppoli sets are also called *sets of locally finite perimeter*. We define the *perimeter of E in* $\Omega \subseteq \mathbb{R}^n$ to be

$$P(E,\Omega) := |D\mathbb{1}_E|(\Omega) = \sup\left\{ \int_E \operatorname{div}\psi : \ \psi \in C_c^\infty(\Omega,\mathbb{R}^n), \|\psi\|_{L^\infty} \le 1 \right\}.$$

Then, for E to have locally finite perimeter (i.e., $P(E,\Omega) < \infty$ for all $\Omega \Subset \mathbb{R}^n$) is equivalent to saying that $|D\mathbb{1}_E|(\Omega)$ is finite on compact sets, which by Proposition 5.10 is equivalent to saying that $\mathbb{1}_E \in \mathrm{BV}_{\mathrm{loc}}(\mathbb{R}^n)$.

Definition 5.21 (Support of a measure). Let $(X, \mathcal{B}_X)$ be a space with its Borel σ-algebra, and μ a positive Radon measure. Then, the *support* is the complement of the union of all open sets of measure zero:

$$\operatorname{supp}\mu = X \setminus \bigcup \{U \subseteq X : \ U \text{ open}, \mu(U) = 0\}.$$

If μ is vector-valued, then we define $\operatorname{supp}\mu = \operatorname{supp}|\mu|$.

Remark 5.22. For a vector-valued Radon measure, note that $|\mu|(U) = 0$ if and only if $\int_U \psi \, d\mu = 0$ for all test functions ψ.

Remark 5.23 (Support of $|D\mathbb{1}_E|$). Let E be a Caccioppoli set, such that for every precompact $\Omega \Subset \mathbb{R}^n$,

$$\int_\Omega \mathbb{1}_E \operatorname{div}\psi = -\int_\Omega \psi \, dD\mathbb{1}_E = -\int_\Omega \langle \psi, v\rangle \; d|D\mathbb{1}_E|$$

for some v measurable. We show that $\operatorname{supp} D\mathbb{1}_E \subseteq \partial E$. Note that we can characterize the support by saying that for $U \subseteq \mathbb{R}^n$ open, $U \subseteq \mathbb{R}^n \setminus \operatorname{supp} D\mathbb{1}_E$ if and only if $\int_U \psi D\mathbb{1}_E = 0$ for all $\psi \in C_0^1(U)$.

Suppose $x \notin \partial E$; then, there exists an open neighborhood $U \ni x$ such that either (i) $U \subseteq E$ or (ii) $U \subseteq \mathbb{R}^n \setminus E$. In case (i), $\mathbb{1}_E \equiv 0$, so $\int_U \psi \, dD\mathbb{1}_E = -\int_U \mathbb{1}_E \operatorname{div}\psi = 0$. In case (ii), $\mathbb{1}_E \equiv 1$, so $\int_U \psi \, dD\mathbb{1}_E = -\int_U \mathbb{1}_E \operatorname{div}\psi = 0$ since $E \cap U = \emptyset$. In either case, it follows that $U \subseteq \mathbb{R}^n \setminus \operatorname{supp} D\mathbb{1}_E$ and hence $x \notin \operatorname{supp} D\mathbb{1}_E$.

Therefore, for any compactly supported test function we can write

$$\int_E \operatorname{div}\psi = \int_{\partial E} \psi \, dD\mathbb{1}_E.$$

This is the sense in which Caccioppoli sets are engineered for the divergence theorem to hold in a generalized sense.

REMARK 5.24. We now expand on the connection between hypersurfaces in Euclidean space and Caccioppoli sets we made in Remark 5.12.
Let $M^{n-1} \hookrightarrow \mathbb{R}^n$ be a hypersurface without boundary, such that there exists a Borel set $E \subseteq \mathbb{R}^n$ with $\partial E = M$. This holds in particular if M is compact (by Jordan–Brouwer separation) or if M is a graph (take E the epigraph). Then, we claim that $\mathbb{1}_E$ is a Caccioppoli set. Clearly, for $\Omega \Subset \mathbb{R}^n$, $\|\mathbb{1}_E\|_{L^1(\Omega)} = |E \cap \Omega| < \infty$.
Let ν be a unit normal to M. By the divergence theorem, if $\psi \in C_0^\infty(\Omega)$ with $\Omega \Subset \mathbb{R}^n$,

$$\int_\Omega \mathbb{1}_E \operatorname{div}\psi = \int_{\mathbb{R}^n} \mathbb{1}_E \operatorname{div}\psi = \int_E \operatorname{div}\psi = \int_M \langle \psi, \nu\rangle \, d\mathrm{vol}_M \leq \|\psi\|_{L^\infty(\Omega)} \mathrm{vol}_M(M \cap \Omega) < \infty.$$

Hence, we say that E is a *smooth Caccioppoli set* if its topological boundary ∂E is a smooth hypersurface.

The first major upshot is that we can now make sense of limits of smooth hypersurfaces: if $M_i, M \subseteq \mathbb{R}^n$ are hypersurfaces, we say that $M_i \to M$ *weakly as Caccioppoli sets* if there exist $E_i, E \subseteq \mathbb{R}^n$ with $\partial E_i = M_i$ and $\partial E = M$ such that $\mathbb{1}_{E_i} \to \mathbb{1}_E$ in L^1_{loc} and $|D\mathbb{1}_{E_i}| \to |D\mathbb{1}_E|$ as Radon measures. Note that we could have asked for $\mathbb{1}_{E_i} \to \mathbb{1}_E$ in the BV norm, but this is too strong to extract compactness results.

The second major upshot is that we can use the theory of BV functions to derive compactness results for smooth hypersurfaces given uniform bounds in the BV norm. Note that the L^1 bounds are more or less automatic: if E is a Caccioppoli set and $\Omega \Subset \mathbb{R}^n$ is bounded open, then

$$\|\mathbb{1}_E\|_{L^1(\Omega)} = |E \cap \Omega| \leq |\Omega| < \infty.$$

Thus the substance of uniform BV bounds is a uniform bound on

$$|D\mathbb{1}_{E_i}|(\Omega) = \mathrm{vol}_{M_i}(M_i \cap \Omega),$$

which amount to uniform area bounds in the sequence.

THEOREM 5.25 (Compactness of Caccioppoli sets). Let $\Omega \subseteq \mathbb{R}^n$ be open and let $E_i \subseteq \Omega$ be Caccioppoli sets (i.e., $\mathbb{1}_{E_i} \in \mathrm{BV}_{\mathrm{loc}}(\Omega)$). Then, if E_i have uniform perimeter bounds on compact sets; that is,

$$\sup_i P(E_i, W) = \sup_i |D\mathbb{1}_{E_i}|(W) < \infty,$$

for all $W \Subset \Omega$, then there exists a Caccioppoli set E (i.e., $\mathbb{1}_E \in \mathrm{BV}_{\mathrm{loc}}(\Omega)$) such that up to a subsequence, $\mathbb{1}_{E_i} \to \mathbb{1}_E$ in $L^1_{\mathrm{loc}}(\Omega)$ and $|D\mathbb{1}_{E_i}| \rightharpoonup |D\mathbb{1}_E|$ as Radon measures.

Proof. The theorem follows immediately from BV compactness (Theorem 5.18). □

We can usually say more about Caccioppoli sets by proving regularity results about them: it turns out that up to a set of $\mathcal{H}^{n-1}$ measure zero, the boundary of a Caccioppoli set can be described as the union of at most countably many C^1 hypersurfaces. For more on this topic, see [Giu84].

Although the regularity of Caccioppoli sets is a rich and interesting topic, we will not treat it here; in fact, part of the novelty of this thesis will be to prove Bernstein's theorem without recourse to mature facts about Caccioppoli sets.

Section 6

Cones and the monotonicity formula

§6.1 Cones

We begin by discussing cones, which are (somewhat surprisingly at first glance) incredibly central objects in the story of Bernstein's problem. Although many of these definitions can be carried out in any $\mathbb{R}$-vector space (and even more generally in any vector space over an ordered field), we will state everything in Euclidean space.

Definition 6.1 (Cones). Let $x_0 \in \mathbb{R}^{n+1}$ and $C \subseteq \mathbb{R}^{n+1}$. We say that $C \subseteq \mathbb{R}^{n+1}$ is a *cone* about x_0 if for all real numbers $r > 0$, $x \in C$ implies $r(x - x_0) \in C - x_0$; that is, $r(C - x_0) \subseteq C - x_0$, where

$$r(C - x_0) = \left\{ r(x - x_0) \in \mathbb{R}^{n+1} : \ x \in C \right\}.$$

Given a subset $A \subseteq \mathbb{R}^{n+1}$, we say that $C(A) := \bigcup_{r>0} r(A - x_0) + x_0$ is the *cone over A about* x_0, and we say that $\hat{C}(A) := \bigcup_{r\geq 0} r(A - x_0) + x_0 = C(A) \cup \{x_0\}$ is the *closed cone over A about* x_0.
Given a cone $C \subseteq \mathbb{R}^{n+1}$ about x_0, we say that $N(C) := (S^n + x_0) \cap C$ is its *link*.

Note that since $C - x_0 \subseteq r^{-1}(C - x_0)$ implies $r(C - x_0) \subseteq C - x_0$, we can equivalently state the definition of a cone as $C - x_0 = r(C - x_0)$ for all $r > 0$. Indeed, a set A is a cone about x_0 if and only if $A = C(A)$. We also caution that the closed cone above need not be closed (e.g. consider the open upper half plane together with 0 as a closed cone over 0), but if A is closed then $\hat{C}(A)$ is closed. (The projection map $\mathbb{R}^{n+1} \setminus \{0\} \to S^n$ is a closed map, so the non-closed cone fails to be closed only for sequences converging to 0.) Note that in our definition, a cone may or may not include the origin. In the case where we have a cone including the origin, imposing smoothness turns out to be a very strong condition.

Proposition 6.2 (Smooth manifold and cone implies flat). Let $C = \hat{C}(C) \subseteq \mathbb{R}^{n+1}$ be a closed cone about x_0 which is also a $(k+1)$-dimensional smooth submanifold. Then it is flat; i.e., it is equal to a $(k+1)$-dimensional plane.

Proof. 1. We first set up the problem using the implicit function theorem. Translating, assume $x_0 = 0$. Rotating if necessary, assume that $T_0 C$ is the copy of $\mathbb{R}^{k+1} \subseteq \mathbb{R}^{n+1}$ given by the inclusion on the first $k+1$ coordinates: $(x^1, \dots, x^{k+1}) \mapsto (x^1, \dots, x^{k+1}, 0, \dots, 0)$. Denote by $D_r = \left\{ x \in \mathbb{R}^{k+1} : \ |x| < r \right\}$ the ball of radius r in $\mathbb{R}^{k+1}$, and $Z_r = D_R \times \mathbb{R}^{n-k}$ the cylinder over the disk $D_r \subseteq \mathbb{R}^{k+1}$. By the implicit function theorem, for sufficiently small $\varepsilon > 0$ we can write $C \cap Z_\varepsilon$ as a graph over D_ε: that is, there exists a smooth function $u : \ D_\varepsilon \to \mathbb{R}^{n-k}$ such that

$$C \cap Z_\varepsilon = \text{graph}\, u = \left\{ (x, u(x)) \in \mathbb{R}^{k+1} \times \mathbb{R}^{n-k} : \ x \in D_\varepsilon \right\}.$$

2. We now aim to prove that $u \equiv 0$. Since we assumed that $\mathbb{R}^{k+1} \subseteq \mathbb{R}^{n+1}$ is tangent to C at 0, we have that $u(0) = 0$ and $du_0 = 0$. Moreover, since $r(x, u(x)) = (rx, ru(x)) = (rx, u(rx))$, u is 1-homogenous: $u(rx) = ru(x)$. Hence, each partial derivative $\partial_i u$ is 0-homogenous:

$$\partial_i u(rx) = \lim_{h \to 0} \frac{u(rx + rh) - u(rx)}{rh} = \lim_{h \to 0} \frac{r\left(u(x + h) - u(x)\right)}{rh} = \partial_i u(x).$$

Taking $r \to 0$ and using that $\partial_i u$ is continuous (u is smooth, so in particular C^1), we get that $\partial_i u \equiv 0$. Hence u is

constant; in particular, since $u(0) = 0$, $u \equiv 0$. Therefore, $C \cap Z_\varepsilon = \mathbb{R}^{k+1} \cap Z_\varepsilon$.
3. Finally, we extend this globally. Note that for any $R > 0$,

$$\mathbb{R}^{k+1} \cap Z_{R\varepsilon} = R(\mathbb{R}^{k+1} \cap Z_\varepsilon) = R(C \cap Z_\varepsilon) = C \cap Z_{R\varepsilon},$$

so taking a union over all $R > 0$ we get $\bigcup_{R>0} Z_{R\varepsilon} = \mathbb{R}^{n+1}$ and hence $C = \mathbb{R}^{k+1}$. □

In view of this proposition, we usually reserve the term $(k+1)$-*dimensional smooth cone* for a cone $C^{k+1} \subseteq \mathbb{R}^{n+1}$ whose link is a smooth k-dimensional submanifold $N^k \hookrightarrow S^n$. Note that if we consider the punctured cone $C(N) = \bigcup_{r>0} rN$ then this set is not closed (consider a sequence of points converging to the origin), but it is a bona fide $(k+1)$-smooth manifold, diffeomorphic to the product $N \times (0, \infty)$ and hence the cylinder $N \times \mathbb{R}$. However, if we consider the closed cone $\hat{C}(N) = C(N) \cup \{0\} = \bigcup_{r\geq 0} rN$, then we get closedness at the cost of a singularity at 0. We will, for the most part, take the former option (in order to remain within the smooth category), but note that when we say that a not-necessarily-closed cone (which a priori does not contain the origin) is flat, we mean to say that $\hat{C}(C) = C \cup \{0\}$ is a linear subspace.

§6.2 Monotonicity formula

In this section, we will prove the seemingly arcane but surprisingly useful *monotonicity formula*, which states that the volume density of a smooth embedded minimal surface is nondecreasing. Then, we will discuss some corollaries, which allow us to use knowledge of the area density of a minimal submanifold to discern geometric properties (in particular, conicality).
We will need the *co-area formula* (see [Giu84] Theorem 1.23 or [Sim14] Theorem 7.3).

Monotonicity formula

Let $\iota : \Sigma^k \hookrightarrow \mathbb{R}^n$ be a properly embedded minimal submanifold, and fix a point $x_0 \in \mathbb{R}^n$ and $r > 0$. Let us write $\Sigma(x_0, r) := B_r(x_0) \cap \Sigma$ and $\Sigma(x_0, r, s) := (B_s(x_0) \setminus B_r(x_0)) \cap \Sigma$. Then, define the *density* as

$$\Theta(\Sigma, x_0, r) = \frac{\mathrm{Area}(\Sigma(x_0, r))}{\omega_k r^k}.$$

The monotonicity formula tells us that the density is nondecreasing in r. In fact, it gives us a way to quantify the rate of increase.

Theorem 6.3. Let $\iota : \Sigma^k \hookrightarrow \mathbb{R}^n$ be an embedded minimal submanifold, and fix $x_0 \in \mathbb{R}^n$. Define $X \in \mathfrak{X}(\Sigma)$ by $X_x = x - x_0$, under the standard identification $T_x\mathbb{R}^n \cong \mathbb{R}^n$. Then, for $r < s$,

$$\Theta(\Sigma, x_0, s) - \Theta(\Sigma, x_0, r) = \frac{1}{\omega_k} \int_{\Sigma(x_0,r,s)} \frac{|X^\perp|^2}{|X|^{k+2}} \,\mathrm{dvol}_\Sigma.$$

Proof. 1. We compute $\mathrm{div}_\Sigma X$. Take $x \in \Sigma$ and let $e_1, \dots, e_k \in T_p\Sigma$ be an orthonormal basis. Writing $x_0 = (x_0^1, \dots, x_0^n)$, we have that $\nabla_{e_i} X = e_i(x^1 - x_0^1, \dots, x^n - x_0^n) = e_i$. Then,

$$\mathrm{div}_\Sigma X = \sum_{i=1}^k \left\langle \nabla_{e_i} X, e_i \right\rangle = \sum_{i=1}^k \langle e_i, e_i \rangle = k.$$

2. We compute the area of $\Sigma(x_0, r)$ using the divergence theorem. Pick $r > 0$ generically such that $\partial B_r(x_0) \pitchfork \Sigma$, so that $\Sigma(x_0, r)$ is a smooth submanifold with boundary $\partial\Sigma(x_0, r)$.

Note that $\nu = \frac{X^\top}{|X^\top|}$ is an outward unit normal for $\partial\Sigma(x_0, r)$, so $\langle X^\top, \nu\rangle = |X^\top|$. By minimality, $\int_{\Sigma(x_0,r)} \operatorname{div}_\Sigma X^\perp = 0$, so by the divergence theorem,

$$\begin{aligned} \operatorname{Area}(\Sigma(x_0, r)) &= \frac{1}{k}\int_{\Sigma(x_0,r)} \operatorname{div}_\Sigma X \, d\mathrm{vol}_\Sigma \\ &= \frac{1}{k}\int_{\Sigma(x_0,r)} \operatorname{div}_\Sigma X^\top \, d\mathrm{vol}_\Sigma \\ &= \frac{1}{k}\int_{\partial\Sigma(x_0,r)} \left\langle X^\top, \nu \right\rangle \, d\mathrm{vol}_{\partial\Sigma(x_0,r)} \\ &= \frac{1}{k}\int_{\partial\Sigma(x_0,r)} \left|X^\top\right| \, d\mathrm{vol}_{\partial\Sigma(x_0,r)}. \end{aligned} \tag{6.1}$$

3. We now compute a second expression for the area of $\Sigma(x_0, r)$ using the coarea formula with the function $|X| : \Sigma \to \mathbb{R}$. By the chain rule, $\nabla|X| = \frac{X}{|X|}$, so $\nabla_\Sigma|X| = \frac{X^\top}{|X|}$.

$$\operatorname{Area}(\Sigma(x_0, r)) = \int_{\Sigma(x_0,r)} |\nabla_\Sigma|X||^{-1} |\nabla_\Sigma|X|| \, d\mathcal{H}^k = \int_0^r \int_{\partial\Sigma(x_0,\tau)} \frac{|X|}{|X^\top|} d\mathcal{H}^{k-1} \, d\tau \tag{6.2}$$

In particular, this entails that

$$\frac{d}{dr}\operatorname{Area}(\Sigma(x_0, r)) = \int_{\partial\Sigma(x_0,r)} \frac{|X|}{|X^\top|} d\mathcal{H}^{k-1},$$

and hence $\operatorname{Area}(\Sigma(x_0, r))$ is differentiable. Thus, $\Theta(\Sigma, x_0, r)$ is differentiable in r for $r > 0$.

4. We now compute $\frac{d}{dr}\Theta(\Sigma, x_0, r)$. By the product rule,

$$\frac{d}{dr}\Theta(\Sigma, x_0, r) = \frac{1}{\omega_k}\frac{d}{dr}\left(r^{-k}\operatorname{Area}(\Sigma(x_0, r))\right) = \frac{1}{\omega_k}\left(-kr^{-k-1}\operatorname{Area}(\Sigma(x_0, r)) + r^{-k}\frac{d}{dr}\operatorname{Area}(\Sigma(x_0, r))\right).$$

From (6.1) and (6.2) and noting that $r = |X|$ on $\partial\Sigma(x_0, r)$ and $d\mathcal{H}^{k-1} = d\mathrm{vol}_{\partial\Sigma(x_0,r)}$ for a.e. $r > 0$,

$$\frac{d}{dr}\Theta(\Sigma, x_0, r) = \frac{1}{\omega_k}\left(\frac{1}{r^{k+1}}\int_{\partial\Sigma(x_0,r)} \frac{|X|^2}{|X^\top|} - \left|X^\top\right|\right) d\mathcal{H}^{k-1} = \frac{1}{\omega_k r^{k+1}}\int_{\partial\Sigma(x_0,r)} \frac{\left|X^\perp\right|^2}{|X^\top|} d\mathcal{H}^{k-1} \tag{6.3}$$

by the Pythagorean theorem (i.e., $|X|^2 = |X^\top|^2 + \left|X^\perp\right|^2$).

5. We now integrate (6.3) and apply the coarea formula to get the final result. Since $r = |X|$ and recalling that $|\nabla_\Sigma|X|| = \frac{|X^\top|}{|X|}$,

$$\frac{d}{dr}\Theta(\Sigma, x_0, r) = \frac{1}{\omega_k}\int_{\partial\Sigma(x_0,r)} \frac{\left|X^\perp\right|^2}{|X|^{k+2}} |\nabla_\Sigma|X|| \, d\mathcal{H}^{k-1}.$$

Integrating and applying the coarea formula, we get

$$\Theta(\Sigma, x_0, s) - \Theta(\Sigma, x_0, r) = \frac{1}{\omega_k}\int_r^s \int_{\partial\Sigma(x_0,\tau)} \frac{\left|X^\perp\right|^2}{|X|^{k+2}} d\mathcal{H}^{k-1} \, d\tau = \int_{\Sigma(x_0,r,s)} \frac{\left|X^\perp\right|^2}{|X|^{k+2}} d\mathcal{H}^k = \int_{\Sigma(x_0,r,s)} \frac{\left|X^\perp\right|^2}{|X|^{k+2}} d\mathrm{vol}_\Sigma,$$

and we are done. □

Applications of the monotonicity formula

As immediate corollaries, we get more information about the density.

> **Corollary 6.4.** Let $\Sigma^k \hookrightarrow \mathbb{R}^n$ be an embedded minimal submanifold. Then,
>
> (a) For every $x_0 \in \mathbb{R}^n$, $\Theta(\Sigma, x_0, r)$ is nondecreasing in r.

(b) The limit

$$\Theta(\Sigma, x_0) := \lim_{r \searrow 0} \Theta(\Sigma, x_0, r)$$

exists; we call this the *density at* x_0.

Proof. 1. The nondecreasing property is an obvious corollary of the monotonicity formula: for $r \leq s$,

$$\Theta(\Sigma, x_0, s) - \Theta(\Sigma, x_0, r) = \int_{\Sigma(x_0,r,s)} \frac{|X^\perp|^2}{|X|^{k+2}}\, d\mathrm{vol}_\Sigma \geq 0.$$

2. Obviously $\Theta(\Sigma, x_0, r) \geq 0$ for all r; since any nonincreasing sequence bounded below converges, the limit $\lim_{r\searrow 0} \Theta(\Sigma, x_0, r)$ exists (and is nonnegative). □

Corollary 6.5. Let $\Sigma^k \hookrightarrow \mathbb{R}^n$ be an embedded minimal submanifold. Then, for any $x_0 \in \mathbb{R}^n$,

(a) $\Theta(\Sigma, x_0, r)$ is constant in r if and only if Σ is conical about x_0.

(b) If $x_0 \in \Sigma$, then $\Theta(\Sigma, x_0) \geq 1$.

(c) If the *density at infinity* $\Theta(\Sigma, \infty) := \lim_{r\nearrow\infty} \Theta(\Sigma, x_0, r) = 1$, then Σ is flat.

Note that (a) does not require that $x_0 \in \Sigma$, so Σ need not be flat.

Proof. Without loss of generality, let us translate to $x_0 = 0$.
1. By the monotonicity formula, if $\Theta(\Sigma, 0, r)$ is constant in r, then $|X^\perp| \equiv 0$, which implies that $X_x \in T_x\Sigma$ and hence the flow of X is a diffeomorphism of Σ. But, this flow is precisely dilation in the radial direction, so Σ is a cone.
2. By the monotonicity formula, the density is decreasing as $r \searrow 0$. But, if $0 \in \Sigma$ then by the implicit function theorem, on small scales we can write Σ as the graph of a function u over its tangent plane. Then, the density is bounded below by

$$\frac{\mathrm{Area}(\mathrm{graph}\, u \cap B_r)}{\omega_k r^k} = \frac{1}{\omega_k r^k} \int_{B_r} \sqrt{1 + |\nabla u|^2} \geq \frac{1}{\omega_k r^k} \int_{B_r} 1 = 1.$$

3. Assume $0 \in \Sigma$. By (b), $\Theta(\Sigma, 0) \geq 1$. By the monotonicity formula, however, for any $r > 0$,

$$1 \leq \Theta(\Sigma, 0) \leq \Theta(\Sigma, 0, r) \leq \Theta(\Sigma, \infty) = 1,$$

so all of the inequalities are equalities and therefore by (a) Σ is a cone. But, this means that Σ is flat. □

Remark 6.6. Note that if Σ satisfies area growth bounds of order equal to its dimension, then the density ratio is bounded above for all r and hence the density at infinity exists and is finite.

Section 7

Smooth limits of smooth hypersurfaces

In this section, we will discuss smooth limits of smooth hypersurfaces. By utilizing the theory of Caccioppoli sets we can make sense of limits of smooth hypersurfaces, though in general the limit will only be a Caccioppoli set. Nevertheless, if we have area bounds (as we do, for instance, in the case of minimal graphs), we can use the compactness theorem (Theorem 5.25) to extract a (subsequential) limit.

In the case where the limit happens to be smooth (i.e., the limit of the Caccioppoli sets corresponding to a sequence of smooth hypersurfaces is a Caccioppoli set whose topological boundary is a smooth hypersurface), we can take advantage of the implicit function theorem to exhibit our hypersurfaces as graphs over some number of tangent planes, so we can define a kind of C^∞_{loc} convergence of smooth hypersurfaces by passing to these graphs. All of these results hold for arbitrary codimension, but we restrict to the (oriented) codimension 1 case so that we can apply the theory of Caccioppoli sets; for general codimension we need to develop the theory of integral currents.

§7.1 C^∞_{loc} convergence of minimal hypersurfaces

Denote by $B^n = B(0,1)$ the unit ball in $\mathbb{R}^n$.

> Lemma 7.1 (Compactness for minimal graphs with uniform C^2 bounds). Let $u_j : B^n \to \mathbb{R}$ be functions with uniformly bounded C^2 norm such that $\Sigma_i = \operatorname{graph} u_j \subseteq B^n \times \mathbb{R}$ are minimal hypersurfaces. Then, there exists $u : B^n \to \mathbb{R}$ with $\Sigma = \operatorname{graph} u$ minimal such that up to taking a subsequence, $u_j \to u$ in $C^\infty_{\mathrm{loc}}(B^n)$.

Proof. By Arzelà–Ascoli, picking any $\alpha \in (0,1)$ we can find a subsequence (without relabeling) such that $u_j \to u$ in $C^1_{\mathrm{loc}}(B^n)$. By quasilinear PDE estimates [see GiTr], we can bootstrap to gain that $u \in C^\infty$ with $u_j \to u$ in C^∞_{loc}. Finally, since the minimal surface equation is continuous with respect to u, we get that u satisfies the minimal surface equation as well, so Σ is minimal. □

Remark 7.2. Note that in particular this implies that the unit normals converge (in C^∞_{loc} as maps in $C^\infty(B^n, \mathbb{R}^{n+1})$), since

$$\nu_j = \frac{(-\nabla u_j, 1)}{\sqrt{1 + |\nabla u_j|^2}}.$$

For general minimal surfaces, we don't have an obvious way to get uniform C^2 bounds. In place of this, we look for uniform second fundamental form bounds; by the following lemma, these are equivalent as long as the gradient is controlled.

> Lemma 7.3. Let $\Omega \subseteq \mathbb{R}^n$ be open with $u : \Omega \to \mathbb{R}$ solving the minimal surface equation, such that $\Sigma = \operatorname{graph} u$ is a minimal hypersurface. Then,
>
> $$\frac{|\nabla^2 u|}{(1 + |\nabla u|^2)^{3/2}} \le |A_\Sigma| \le \frac{|\nabla^2 u|}{(1 + |\nabla u|^2)^{1/2}}.$$

Now, we can generalize the compactness result Lemma 7.1 to minimal hypersurfaces in Euclidean space. For a two-sided hypersurface $\Sigma^n \hookrightarrow \mathbb{R}^{n+1}$ and a function $\varphi \in C^\infty(\Sigma)$, define the *graph over* Σ to be

$$\operatorname{graph}_\Sigma(\varphi) = \{p + \varphi(p)\nu(p) : p \in \Sigma\}.[1]$$

[1] In general, if we replace $\mathbb{R}^{n+1}$ with an ambient Riemannian manifold, we can exponentiate along ν as long as φ is small enough for the graph to fit in the domain of the exponential map. In a sense, this is using the identification of $C^\infty(\Sigma) \cong \Gamma(N\Sigma)$ from the unit normal ν and the tubular neighborhood theorem for $\Sigma \hookrightarrow \mathbb{R}^{n+1}$.

Now, we can make sense of smooth convergence of minimal hypersurfaces, as graphs over the limit hypersurface, potentially with sheeting behavior (for instance, consider the sequence $\lambda_j\Sigma$ for $\Sigma \subseteq \mathbb{R}^3$ the catenoid, which converges to a plane with multiplicity 2).

Definition 7.4. Let $\Omega \subseteq \mathbb{R}^{n+1}$ be open and let $\Sigma_\infty^n, \Sigma_j^n \hookrightarrow \Omega$ be smooth two-sided hypersurfaces with $\partial M_j \subseteq \partial\Omega$. We say that $\Sigma_j \to \Sigma_\infty$ *in* C^∞_{loc} *with finite multiplicity* if the following properties hold:

(a) There exists a locally constant function $m : \Sigma_\infty \to \mathbb{Z}_{\geq 1}$ called the *multiplicity*,

(b) There exists an exhaustion $W_1 \Subset W_2 \Subset \cdots \Subset \Sigma_\infty$ by precompact open sets,

(c) On each connected component of W_j, there exist functions $w_{1,j} < \cdots < w_{M,j}$, in C^∞, such that $w_{l,j} \to 0$ in C^∞_{loc} as $j \to \infty$, and

(d) There exists an exhaustion $\mathcal{G}_1 \Subset \mathcal{G}_2 \Subset \cdots \Subset \Omega$ by precompact open sets such that
$$\Sigma_j \cap \mathcal{G}_j = \bigcup_l \operatorname{graph}_{W_j}(w_{l,j}).$$

If $m \equiv 1$, we simply say $\Sigma_j \to \Sigma$ in C^∞_{loc}.

For $\Sigma^n \hookrightarrow \mathbb{R}^{n+1}$ and $p \in \Sigma$, let us denote $|A_\Sigma|(p)$ the norm of the second fundamental form of Σ at p.

Theorem 7.5 (Compactness for minimal hypersurfaces). Let $\Omega \subseteq \mathbb{R}^{n+1}$ be open and let $\Sigma_j^n \hookrightarrow \Omega$ be a sequence of minimal surfaces with $\partial\Sigma_j \subseteq \partial\Omega$, with uniform area and second fundamental form bounds on compact sets: for every $K \Subset \Omega$ there exist $\alpha(K), \beta(K) > 0$ such that
$$\sup_j \operatorname{Area}(\Sigma_j \cap K) \leq \alpha(K) \qquad \text{and} \qquad \sup_j \sup_{p \in K} \left|A_{\Sigma_j}\right|(p) \leq \beta(K).$$
Then, up to a subsequence there exists a smooth hypersurface $\Sigma_\infty^n \hookrightarrow \Omega$ such that $\Sigma_j \to \Sigma_\infty$ in C^∞_{loc} with finite multiplicity.

Proof. See [Cho25] Theorem 28.5. □

However, in the case of the blowdown we should not expect uniform second fundamental form bounds, since the curvature blows up at 0. Instead, we have the following result (see [Cho25] Lemma 29.1).

Proposition 7.6. Let $\Omega \subseteq \mathbb{R}^{n+1}$ be open and let $\Sigma_j^n \hookrightarrow \Omega$ be a sequence of minimal surfaces with $\partial\Sigma_j \subseteq \partial\Omega$. Let $p_j \in \Sigma_j$ with $p_j \to p_\infty \in \Omega$ such that the curvature blows up along this sequence: $\lambda_j := \left|A_{\Sigma_j}\right|(p_j) \nearrow \infty$. Then, we can pick another sequence $q_j \in \Sigma_j$ with $q_j \to p_\infty$ such that the rescaled sequences $\widetilde{\Sigma}_j = \lambda_j(\Sigma_j - q_j)$ has uniform second fundamental form bounds on compact sets and $\left|A_{\widetilde{\Sigma}_j}\right|(0) = 1$.

As a result, if Σ_j have uniform area bounds, we can take the limit (see [Cho25] Lemma 29.2).

Proposition 7.7. Let $\Omega \subseteq \mathbb{R}^{n+1}$ be open and let $\Sigma_j^n \hookrightarrow \Omega$ be a sequence of minimal surfaces with $\partial\Sigma_j \subseteq \partial\Omega$ and uniform area bounds on compact sets. Let μ_∞ denote the limit area measure, as a Caccioppoli set.
Let $p_j \in \Sigma_j$ with $p_j \to p_\infty \in \Omega$ such that the curvature blows up along this sequence: $\lambda_j := \left|A_{\Sigma_j}\right|(p_j) \nearrow \infty$.

Then, we can pick another sequence $q_j \in \Sigma_j$ with $q_j \to p_\infty$ such that the rescaled sequences $\widetilde{\Sigma}_j = \lambda_j(\Sigma_j - q_j)$ has uniform second fundamental form bounds on compact sets and hence we can take a subsequential limit in C^∞_{loc} to a minimal hypersurface $\widetilde{\Sigma}_\infty$ without boundary.
Moreover, $\Theta_{\widetilde{\Sigma}_\infty}(\infty) \leq \Theta_{\mu_\infty}(\infty)$.

Thus, we arrive at White's easy version of Allard's theorem.

Theorem 7.8 (Easy version of Allard's theorem). Let $\Sigma^n \hookrightarrow \mathbb{R}^{n+1}$ be a smooth minimal hypersurface with $|A_\Sigma| \leq |A_\Sigma|(0) = 1$. Then there exists $\varepsilon = \varepsilon(n)$ such that $\Theta(\Sigma, \infty) \geq 1 + \varepsilon$.

As a corollary, we get that in a suitable sense, singular points don't limit to smooth points.

Corollary 7.9. Suppose $\mu_k \rightharpoonup \mu$ and $x_k \in \operatorname{supp} \mu_k$ with $x_k \to x \in \operatorname{reg} \mu$. Then, there exists some k such that $x_k \in \operatorname{reg} \mu_k$.

Section 8

Tangent cones at infinity

In this section, we will discuss the notion of a tangent cone at infinity, and work up to a translation of the Bernstein problem from minimal graphs to minimal cones.
Recall that minimal graphs are minimizing on compact sets.

Definition 8.1 (Tangent cone at infinity). Let $\Gamma^n \hookrightarrow \mathbb{R}^{n+1}$ be a smooth connected hypersurface with $E \subseteq \mathbb{R}^{n+1}$ open with $\partial E = \Gamma$. A *tangent cone at infinity to* Γ is a set $C \subseteq \mathbb{R}^{n+1}$ corresponding to a Caccioppoli set $E_\infty \subseteq \mathbb{R}^{n+1}$ with $\partial E_\infty = C$, such that there exists a sequence $\lambda_j \searrow 0$ with $\lambda_j E \to E_\infty$.

Proposition 8.2 (Existence of tangent cones at infinity). Let $\Gamma^n \hookrightarrow \mathbb{R}^{n+1}$ be a smooth connected hypersurface with $E \subseteq \mathbb{R}^{n+1}$ with $\partial E = \Gamma$. If Γ minimizes area on compact subsets, then any sequence $\lambda_j \searrow 0$ has a subsequence such that $\lambda_j E$ converges to a tangent cone at infinity which is minimizing on compact sets.

Remark 8.3. A priori, the tangent cone at infinity may depend on the specific choice of sequence λ_j, and in general uniqueness of tangent cones at infinity is a difficult open problem.

§8.1 Iterated tangent cones

For a reference, see [Whi12] sections 8.3 and 9.1.

Theorem 8.4. Let $C \subseteq \mathbb{R}^{n+1}$ be a codimension 1 minimizing cone, with corresponding Caccioppoli set E with $\partial E = C$.

(a) The *spine*

$$\operatorname{spine} C = \{x \in C : \Theta(C, x) = \Theta(C, \infty)\}$$

is a linear subspace.

(b) The cone C splits as $C = C' \times \operatorname{spine} C$, where C' is a minimizing cone contained in $\operatorname{spine} C^{\perp}$.

Remark 8.5. If the cone is smooth but nonflat, then $\Theta(C, \infty) > 1$ but $\Theta(C, x) = 1$ for all $x \in C$. So, $\operatorname{spine} C = \{0\}$. However, the converse is not true: a cone may have a 0-dimensional spine and yet not be smooth. However, the following dimension reduction theorem solves this problem, by taking a *iterated tangent cone*.

Theorem 8.6 (Iterated tangent cones). Let $C \subseteq \mathbb{R}^{n+1}$ be a nonsmooth minimizing codimension 1 cone. Let $x \in \operatorname{sing} C \setminus \{0\}$. Then, either

(a) C is singular and it is regular except on its spine (i.e., $\operatorname{sing} C = \operatorname{spine} C$ and $\dim \operatorname{spine} C > 0$, or

(b) there exists a singular codimension 1 minimizing cone $C' \subseteq \mathbb{R}^{n+1}$ with $\Theta(C', \infty) < \Theta(C, \infty)$ and C' is regular except on its spine.

§8.2 Dimensional reduction

By the iterated tangent cone argument, if C is not smooth then we can blow up at singular points until we have a cone where $\operatorname{sing} C' = \operatorname{spine} C'$; i.e, the cone splits its spine. If we can always ensure that after taking an iterated tangent cone, we have a minimizing cone times a spine of positive dimension, then we can reduce the dimensions in which we need to prove that minimizing cones are flat in order to prove Bernstein's theorem.

In [dG65], de Giorgi proved the following *cone splitting theorem*:

Theorem 8.7 (De Giorgi cone splitting). Let $u : \mathbb{R}^n \to \mathbb{R}$ be a nonflat solution to the MSE and suppose $C \subseteq \mathbb{R}^{n+1}$ is a tangent cone at infinity. Then, C splits a line: there exists a cone $C' \subseteq \mathbb{R}^n$ such that $C = C' \times \mathbb{R}$.

As a result, the spine is at least 1-dimensional. Hence, we get a dimensional reduction for proving Bernstein's theorem:

Corollary 8.8. Suppose every minimizing cone in $\mathbb{R}^{n+1}$ is flat. Then, Bernstein's theorem is true for dimension $n + 1$: every $u : \mathbb{R}^{n+1} \to \mathbb{R}$ solving the MSE is affine.

Thus, if we can show that every minimizing cone in $\mathbb{R}^{n+1}$ for $n \leq 6$ is flat, then we can resolve Bernstein's problem for $n \geq 7$. However, note that in the proof, we actually don't need the full strength of the de Giorgi cone splitting theorem; in fact, we only need that the tangent cone at ∞ to a nonflat minimal graph is not smooth (i.e., its link is not a smooth submanifold) in order to run the iterated tangent cone argument to use Simons's cone theorem to resolve Bernstein's theorem in $n = 7$.

In this section, we prove an easier, weaker version of de Giorgi's cone splitting theorem.

Theorem 8.9 (Non-smooth blowdown). Let $u : \mathbb{R}^n \to \mathbb{R}$ be a nonflat solution to the MSE, and let C be a tangent cone at infinity to $\Gamma = \operatorname{graph} u$. Then, C is not smooth; i.e., the link $\Sigma = C \cap S^n$ is not a smooth submanifold.

Proposition 8.10 (Nonflat graph blows down to nonflat cone). Let $u : \mathbb{R}^{n+1} \to \mathbb{R}$ be a nonflat solution to the MSE. Then, any tangent cone at infinity is not flat.

Proof. This follows immediately from monotonicity: the contrapositive of this statement is that if the tangent cone at infinity is flat, then the graph must be flat. □

Proposition 8.11 (Regularity of smooth limits). Let $\Omega \subseteq \mathbb{R}^{n+1}$ be open and $\Gamma_\infty, \Gamma_j^n \hookrightarrow \Omega$ be smooth two-sided hypersurfaces with $\partial\Gamma_j \subseteq \partial\Omega$. Let E_j, E_∞ be the Caccioppoli sets corresponding to these hypersurfaces, and suppose $E_j \to E_\infty$ weakly as Caccioppoli sets. Then, $\Gamma_j \to \Gamma_\infty$ in C^∞_{loc}.

Proposition 8.12 (Maximum principle for Jacobi fields). Let $\Gamma^n \hookrightarrow \mathbb{R}^{n+1}$ be a two-sided minimal hypersurface with unit normal ν, and $f \in C^\infty(\Gamma)$. If $L_\Gamma f \neq 0$ and $f \geq 0$, then either $f \equiv 0$ or $f > 0$.

Lemma 8.13. Let $\Sigma^{n-1} \hookrightarrow S^n$ be a two-sided minimal hypersurface with unit normal ν, and let $a \in \mathfrak{X}(\mathbb{R}^{n+1})$ be a parallel vector field (with respect to the Euclidean connection $\overline{\nabla}$). Then, $f = \langle \nu, a \rangle$ is an eigenfunction of the Jacobi operator of Σ:

$$L_\Sigma f - (n-1)f = 0.$$

Proof. 1. Denote by $\nabla, \mathring{\nabla}$, and $\overline{\nabla}$ the connections on Σ, S^n, and $\mathbb{R}^{n+1}$, respectively. At $p \in \Sigma$, pick $E_1, \dots, E_{n-1}$ a local orthonormal frame that is ∇-parallel at p. Note that this means that $E_1|_p, \dots, E_{n-1}|_p, \nu_p, \partial_r|_p \in T_p\mathbb{R}^{n+1}$ is an orthonormal basis. Pick η to be a $\overline{\nabla}$-parallel extension of $\partial_r|_p$ (note that $\eta \neq \partial_r$ except at p; this will help simplify our computations). In these coordinates, $\nabla_{E_i} E_j = 0$, so we have that

$$\Delta_\Sigma f = \sum_{i=1}^{n-1} \nabla_{E_i} \nabla_{E_i} \langle \nu, a \rangle .$$

2. Denote by A_{S^n} the scalar second fundamental form of $S^n \hookrightarrow \mathbb{R}^{n+1}$. By Proposition 2.32, we have that $A_{S^n} = g_{S^n}$. By the Gauss formula, we conclude that since $\overline{\nabla} a = 0$, we have for any $X \in \mathfrak{X}(S^n)$,

$$0 = \overline{\nabla} a(X) = \overline{\nabla}_X a = \mathring{\nabla}_X a + A(X, a)\partial_r,$$

and hence at p we can write

$$\mathring{\nabla}_X a = -\langle X, a \rangle \partial_r = -\langle X, a \rangle \eta. \tag{8.4}$$

3. We now compute $\nabla_{E_i} \langle \nu, a \rangle$. Using the same trick(s) as above, we get that

$$\nabla_{E_i} \langle \nu, a \rangle = E_i \langle \nu, a \rangle = \mathring{\nabla}_{E_i} \langle \nu, a \rangle = \left\langle \mathring{\nabla}_{E_i} \nu, a \right\rangle + \left\langle \nu, \mathring{\nabla}_{E_i} a \right\rangle . \tag{8.5}$$

We analyze the two terms separately. Define $(-)^\top : T_p\mathbb{R}^{n+1} \to T_p\Sigma$ to be the projection map, so that

$$a = a^\top + \langle a, \nu \rangle \nu + \langle a, \eta \rangle \eta.$$

Then, the first term $\left\langle \mathring{\nabla}_{E_i} \nu, a \right\rangle$ is

$$\left\langle \mathring{\nabla}_{E_i} \nu, a \right\rangle = \left\langle \mathring{\nabla}_{E_i} \nu, a^\top \right\rangle + \langle a, \nu \rangle \left\langle \mathring{\nabla}_{E_i} \nu, \nu \right\rangle + \langle a, \eta \rangle \left\langle \mathring{\nabla}_{E_i} \nu, \eta \right\rangle .$$

The second term vanishes because $\left\langle \mathring{\nabla}_{E_i} \nu, \nu \right\rangle = \frac{1}{2}\mathring{\nabla}_{E_i} \langle \nu, \nu \rangle = 0$ and the third term vanishes because $\mathring{\nabla}_{E_i} \nu \in T_p S^n$ and $\eta \in (T_p S^n)^\perp$. Hence, we get that

$$\left\langle \mathring{\nabla}_{E_i} \nu, a \right\rangle = A_\Sigma(E_i, a^\top).$$

For the second term in (8.5), we get by (8.4) that

$$\left\langle \nu, \mathring{\nabla}_{E_i} a \right\rangle = -\langle E_i, a\rangle \langle \nu, \eta\rangle = 0.$$

Therefore, we conclude that

$$\nabla_{E_i} \langle \nu, a\rangle = A_\Sigma(E_i, a^\top). \tag{8.6}$$

4. We now compute the second derivative term. By (8.6),

$$\nabla_{E_i}\nabla_{E_i} \langle \nu, a\rangle = \nabla_{E_i}(A_\Sigma(E_i, a^\top)) = (\nabla A_\Sigma)(E_i, a^\top, E_i) + A_\Sigma(E_i, \nabla_{E_i} a^\top),$$

since $\nabla_{E_i} E_i = 0$. By the Codazzi equation (Lemma 2.29), we get by the same computation in the Euclidean case that

$$\sum_{i=1}^{n-1} (\nabla A_\Sigma)(E_i, a^\top, E_i) = \sum_{i=1}^{n-1} (\nabla A_\Sigma)(E_i, E_i, a^\top) = \sum_{i=1}^{n-1} \nabla_{a^\top} A_\Sigma(E_i, E_i) - 2A_\Sigma(\nabla_{a^\top} E_i, E_i) = 0$$

by minimality. Therefore,

$$\sum_{i=1}^{n-1} \nabla_{E_i}\nabla_{E_i} \langle \nu, a\rangle = \sum_{i=1}^{n-1} A_\Sigma(E_i, \nabla_{E_i} a^\top). \tag{8.7}$$

5. We now compute the $\nabla_{E_i} a^\top$ term. Since $a^\top = a - \langle a, \nu\rangle \nu - \langle a, \eta\rangle \eta$ and $\overline{\nabla} a = 0$,

$$\overline{\nabla}_{E_i} a^\top = -\overline{\nabla}_{E_i} \langle a, \nu\rangle \nu - \overline{\nabla}_{E_i} \langle a, \eta\rangle \eta = -\left((E_i \langle a, \nu\rangle)\nu + (E_i \langle a, \eta\rangle)\eta + \langle a, \nu\rangle \overline{\nabla}_{E_i} \nu + \langle a, \eta\rangle \overline{\nabla}_{E_i} \eta \right).$$

Since η was chosen to be $\overline{\nabla}$-parallel, $\overline{\nabla}_{E_i} \eta = 0$. Then, by Gauss's formula applied twice (and noting that $E_i, a^\top$ are both vector fields parallel to Σ and hence also to S^n), we have that

$$\overline{\nabla}_{E_i} a^\top = \mathring{\nabla}_{E_i} a^\top + \left\langle E_i, a^\top \right\rangle \eta = \nabla_{E_i} a^\top + A_\Sigma(E_i, a^\top)\nu + \left\langle E_i, a^\top \right\rangle \eta$$

and hence $\nabla_{E_i} a^\top = (\mathring{\nabla}_{E_i} a^\top)^\top = (\overline{\nabla}_{E_i} a^\top)^\top$. So,

$$\nabla_{E_i} a^\top = \left(\overline{\nabla}_{E_i} a^\top\right)^\top = -\langle a, \nu\rangle \left(\overline{\nabla}_{E_i} \nu\right)^\top = -\langle a, \nu\rangle \left(\mathring{\nabla}_{E_i} \nu\right)^\top.$$

Finally, expanding $\left(\mathring{\nabla}_{E_i} \nu\right)^\top$, we conclude that

$$\nabla_{E_i} a^\top = -\langle \nu, a\rangle \sum_{j=1}^{n-1} A_\Sigma(E_i, E_j) E_j. \tag{8.8}$$

6. Combining (8.7) and (8.8), we get that

$$\Delta_\Sigma \langle \nu, a\rangle = -\langle \nu, a\rangle \sum_{i,j=1}^{n-1} A_\Sigma(E_i, E_j)^2 = -\langle \nu, a\rangle |A_\Sigma|^2,$$

and hence

$$L_\Sigma \langle \nu, a\rangle = \Delta_\Sigma \langle \nu, a\rangle + |A_\Sigma|^2 \langle \nu, a\rangle + (n-1)\langle \nu, a\rangle = (n-1)\langle \nu, a\rangle.$$

□

Proposition 8.14. Let $\Sigma^{n-1} \hookrightarrow S^n$ be a two-sided minimal hypersurface with unit normal ν. Then, if Σ^{n-1} is not a totally geodesic S^{n-1}, $\dim \ker L_\Sigma^\circ > 1$.

Proof. 1. Assume Σ is not totally geodesic. Pick $p_1 \in \Sigma$, and define $a_1(p_1) = \nu(p_1) \in T_{p_1}\mathbb{R}^{n+1}$, then extend a_1 to a parallel vector field in $\mathbb{R}^{n+1}$ (with respect to the Euclidean connection $\overline{\nabla}$). By construction, $f_1(p_1) = \langle \nu(p_1), a_1(p_1) \rangle = 1$.
2. Now suppose to the contrary that $f_1 \equiv 1$. Then, we would have $\nu = a_1$ everywhere on Σ, so Σ would be an equatorial S^{n-1}, a contradiction.
3. Hence, there exists a point $p_2 \in \Sigma$ such that $f_1(p_2) < 1$. Then, define $a_2(p_2) = \nu(p_2) \in T_{p_2}\mathbb{R}^{n+1}$, and similarly extend to a $\overline{\nabla}$-parallel vector field on $\mathbb{R}^{n+1}$. Again by construction, $f_2(p_2) = 1$. Moreover, since both vector fields are parallel,

$$f_2(p_1) = \langle \nu(p_1), a_2(p_1) \rangle = \langle a_2(p_1), a_1(p_1) \rangle = \langle a_2(p_2), a_1(p_2) \rangle = \langle \nu(p_2), a_1(p_2) \rangle = f_1(p_2) < 1.$$

4. It follows that f_1, f_2 are both in $\ker L_\Sigma^\circ$ but are not linearly dependent: if they were, there would exist a constant $c \in \mathbb{R}$ such that $f_1 = cf_2$. Evaluating at p_1 shows that we need $c > 1$, but evaluating at p_2 shows that we need $c < 1$, a contradiction. Hence $\dim \ker L_\Sigma^\circ \geq 2$. $\square$

We are finally ready to prove the non-smooth blowdown theorem, or the easy version of de Giorgi's cone splitting theorem.

Proof of Theorem 8.9. 1. By the monotonicity formula (see Proposition 8.10), C is not flat. Assume, for the sake of contradiction, that C is a *smooth* cone at infinity to Γ; that is, $\operatorname{sing} C = \{0\}$. Denote by ν_j and ν_C the unit normals to Γ_j and C, respectively. By construction, $\left\langle \nu_j, e_{n+1} \right\rangle > 0$ for each j. By Proposition 8.11, we have $\Gamma_j \to C$ in C^∞_{loc}. Hence, $\left\langle \nu_j, e_{n+1} \right\rangle \to \langle \nu_C, e_{n+1} \rangle$, so $\langle \nu_C, e_{n+1} \rangle \geq 0$.
2. By Lemma 3.13, $f = \langle \nu_C, e_{n+1} \rangle$ is a Jacobi field: $L_C f = 0$. By the maximum principle (Proposition 8.12), either $f \equiv 0$ or $f > 0$; we consider these cases in turn.
3. We first consider the case where $f \equiv 0$. It follows that e_{n+1} is tangent to C (i.e., $e_{n+1} \in T_pC$ for all $p \in C$). Therefore, C is invariant under the flow of e_{n+1}, which is translation in the e_{n+1} direction. Therefore C splits a line: there exists $C' \subseteq \mathbb{R}^n$ such that $C = C' \times \mathbb{R}$. But, this is a contradiction: by assumption, $\operatorname{sing} C = \{0\}$, but since C is invariant under translation in the e_{n+1} direction, so should $\operatorname{sing} C$.
4. We now consider the case where $f > 0$. Let $\Sigma^{n-1} \hookrightarrow S^n$ denote the link $\Sigma = C \cap S^n$; by assumption Σ is smooth. Let ν_Σ denote the unit normal to Σ. Since $\nu_\Sigma = \nu_C$ along Σ, $\langle \nu_\Sigma, e_{n+1} \rangle > 0$. By Lemma 8.13, we get that $L_\Sigma^\circ \langle \nu_\Sigma, e_{n+1} \rangle = 0$. Since $\langle \nu_\Sigma, e_{n+1} \rangle > 0$, by Theorem 3.9 it is the first eigenfunction, and hence the dimension of the 0-eigenspace is equal to 1. However, this is a contradiction: by Proposition 8.14, since C is nonflat we must have that the dimension of the 0-eigenspace is at least 2. $\square$

Chapter III

A return to the smooth world

The iterated tangent cone argument allows us to reduce our study of (potentially singular) minimizing cones to *smooth* minimizing cones. Then, the dimension reduction argument shows us that Bernstein's problem for graphs $\Gamma^n \hookrightarrow \mathbb{R}^{n+1}$ can be resolved if every *smooth* minimizing cone $C^k \hookrightarrow \mathbb{R}^{k+1}$ is flat, for $k \leq n-1$.
The aim of this section is to finish the proof of Bernstein's theorem for $n \leq 7$ by proving Simons's theorem on cones, originally proved in [Sim68]. Recall that a minimizing cone must be stable (considering the cone without the origin).

> Theorem (Simons). Let $2 \leq n \leq 6$, and suppose $\Sigma^{n-1} \hookrightarrow S^n$ is a smooth minimal submanifold with cone $C^n = C(\Sigma) \hookrightarrow \mathbb{R}^{n+1}$. If C is stable, then C is flat.

The proof strategy is as follows. Recall that for a two-sided minimal hypersurface $\Sigma^n \hookrightarrow (M^{n+1}, g)$, we write the *stability operator* L_Σ and the *Jacobi operator* L_Σ° as

$$L_\Sigma = \Delta_\Sigma + |A_\Sigma|^2 + \mathrm{Ric}_M(\nu,\nu) \qquad \text{and} \qquad L_\Sigma^\circ = \Delta_\Sigma + |A_\Sigma|^2 .$$

(a) Using a separation of variables argument, we relate the first eigenvalue of the Jacobi operator on the link $\lambda_1(L_\Sigma^\circ)$ with the first eigenvalue of the stability operator on the cone. Then, the assumption that C is stable gives a lower bound on $\lambda_1(L_\Sigma^\circ)$.

(b) Then, we prove the *Simons inequality* on Σ, which is an inequality in terms of the norm of the second fundamental form and its derivatives that we can use to gain an upper bound on $\lambda_1(L_\Sigma^\circ)$.

(c) Finally, we observe that the upper and lower bounds derive a contradiction unless the dimension is $2 \leq n \leq 6$.

Section 9

Separation of variables

For technical reasons which will become clear, it is useful for us to define the notion of a *weighted first eigenvalue*.

> Definition 9.1. Let $\Sigma^n \hookrightarrow (M^{n+1}, g)$ be a two-sided minimal hypersurface. For $w \in C^\infty(\Sigma)$ with $w > 0$, the

weighted first eigenvalue of the stability operator on $\Omega \Subset \Sigma$ is

$$\lambda_1(L_\Sigma, w) = \inf_{\varphi \in C^\infty(\Sigma)\setminus\{0\}} \frac{\int_\Sigma \varphi L_\Sigma^\circ \varphi}{\int_\Sigma w|\varphi|^2}.$$

Remark 9.2. Nevertheless, for the purposes of ascertaining stability, the weight is irrelevant: if $\lambda_1(L_\Sigma, w) \geq 0$ then the numerator in the Rayleigh quotient is always nonnegative, so normalizing by the usual L^2 norm keeps the quotient nonnegative. Hence Σ is stable if and only if $\lambda_1(L_\Sigma, w) \geq 0$.

Our setup is as follows. Let $\Sigma^{n-1} \hookrightarrow S^n$ be a two-sided minimal hypersurface, and let $C^n = C(\Sigma) \hookrightarrow \mathbb{R}^{n+1}$ denote the cone over Σ. Let ν denote the unit normal on Σ; note that it extends radially to a unit normal on C. Note that the stability and Jacobi operators on the cone are identical, but on the link they differ by a factor of $n-1$ corresponding to the $\mathrm{Ric}_{S^n}(\nu,\nu)$ term.

The aim of this section is to prove the following theorem:

Theorem 9.3 (Separation of variables). Let $\Sigma^{n-1} \hookrightarrow S^n$ be a two-sided smooth minimal hypersurface and let $C^n = C(\Sigma)$ be the (punctured) cone over Σ. Then,

$$\lambda_1(L_C, r^{-2}) = \lambda_1(L_\Sigma^\circ) + \frac{(n-2)^2}{4}.$$

The crux of the proof lies in analyzing the stability operator of the cone L_C with respect to spherical coordinates; that is, computing the numerator of the Rayleigh quotient with respect to the conformal change of coordinates $\mathbb{R}^{n+1}\setminus\{0\} \xrightarrow{\sim} S^n \times \mathbb{R}$ given by $x \mapsto \left(\frac{x}{|x|}, \log|x|\right)$. For the sake of expository clarity, we split the proof of this proposition into several lemmas.

§9.1 Proof of separation of variables

In the following, we denote $(\theta, r) = \left(\frac{x}{|x|}, |x|\right) \in S^n \times (0,\infty)$ to be spherical coordinates on $\mathbb{R}^{n+1}\setminus\{0\}$ and we denote $(\theta, t) \in S^n \times \mathbb{R}$ to be the usual cylindrical coordinates.

Lemma 9.4 (Scaling formulas for link and cone). Let $\Sigma^{n-1} \hookrightarrow S^n$ be a two-sided smooth minimal hypersurface and let $C^n = C(\Sigma)$ be the (punctured) cone over Σ. Let $g^\Sigma = \iota_\Sigma^* g^{\mathbb{R}^{n+1}}$ and $g^C = \iota_C^* g^{\mathbb{R}^{n+1}}$ be the induced metrics under the respective embeddings, and let $g^{\Sigma\times\mathbb{R}} := g^\Sigma + dt^2$. Then, the following formulae hold.

(a) $r^{-2} g^C = g^{\Sigma\times\mathbb{R}}$,

(b) $r^2 |A_C|^2(\theta, r) = |A_{\Sigma\times\mathbb{R}}|^2(\theta, t) = |A_\Sigma|^2(\theta)$,

(c) $|\nabla_C \varphi|^2 = r^{-2}\left(|\nabla_\Sigma \varphi|^2 + |\partial_t \varphi|^2\right)$

Proof. 1. Note that since $\log r = t$, we have $r\partial_r = \partial_t$. Pick $p = (\theta, r) \in C$, and let $e_1, \dots, e_{n-1}$ be an orthonormal basis for $T_p(r\Sigma)$, so that $e_1, \dots, e_{n-1}, \nu, \partial_r$ is an orthonormal basis for $T_p\mathbb{R}^{n+1}$ with respect to $g^{\mathbb{R}^{n+1}}$. Under the change of coordinates, we get that $re_1, \dots, re_{n-1}, r\nu, r\partial_r = \partial_t$ is an orthonormal basis for $T_\theta\Sigma \times T_t\mathbb{R}$, so we automatically get that $g^C_{ij} = r^2 g^{\Sigma\times\mathbb{R}}_{ij}$.

2. Then, denoting $e_n = \nu$ and $e_{n+1} = \partial_r$, we have that

$$|A_C|^2(\theta, r) = \sum_{i,j\neq n} \left|A_C(e_i, e_j)\right|^2(\theta, r) = r^{-2} \sum_{i,j\neq n} \left|A_{\Sigma\times\mathbb{R}}(re_i, re_j)\right|^2(\theta, t) = r^{-2}|A_{\Sigma\times\mathbb{R}}|^2(\theta, t).$$

Finally, since $g^{\Sigma\times\mathbb{R}} = g^\Sigma + dt^2$ splits as a product metric and dt^2 is flat, we have that $|A_{\Sigma\times\mathbb{R}}|^2(\theta, t) = |A_\Sigma|^2(\theta)$.
3. The final computation is straightforward from the equation $r^{-2}g^C = g^{\Sigma\times\mathbb{R}}$:

$$|\nabla_C\varphi|^2 = r^{-2}|\nabla_{\Sigma\times\mathbb{R}}\varphi|^2 = r^{-2}\left(|\nabla_\Sigma\varphi|^2 + |\partial_t\varphi|^2\right).$$

□

We can now use this computation to phrase the first eigenvalue inequality for C in cylindrical coordinates on $\Sigma \times \mathbb{R}$.

Lemma 9.5. Let $\Sigma^{n-1} \hookrightarrow S^n$ be a two-sided smooth minimal hypersurface and let $C^n = C(\Sigma)$ be the (punctured) cone over Σ. Then, for all $\varphi \in C_c^\infty(\Sigma\times\mathbb{R})$,

$$\int_{\mathbb{R}}\int_\Sigma \left(\lambda_1(L_C, r^{-2}) + |A_\Sigma|^2\right)\varphi^2 e^{(n-2)t}\,\mathrm{dvol}_\Sigma\,dt \le \int_{\mathbb{R}}\int_\Sigma \left(|\nabla_\Sigma\varphi|^2 + |\partial_t\varphi|^2\right)e^{(n-2)t}\,\mathrm{dvol}_\Sigma\,dt$$

Proof. By definition of the first eigenvalue, for all $\varphi \in C_c^\infty(C)$, we have the inequality

$$\lambda_1(L_C, r^{-2})\int_C r^{-2}\varphi^2 \le \int_C |\nabla_C\varphi|^2 - |A_C|^2\varphi^2.$$

Expressing these integrals into spherical coordinates and using the computation for the scaling of the second fundamental form and gradient in Lemma 9.4, after rearranging we get

$$\int_0^\infty\int_\Sigma \left(\lambda_1(L_C, r^{-2}) + |A_\Sigma|^2\right)\varphi^2 r^{n-3}\,\mathrm{dvol}_\Sigma\,dr \le \int_0^\infty\int_\Sigma \left(|\nabla_\Sigma\varphi|^2 + |\partial_t\varphi|^2\right)r^{n-3}\,\mathrm{dvol}_\Sigma\,dr.$$

Converting to cylindrical coordinates (and noting that $dr = r\,dt$ and $r = e^t$), we get

$$\int_{\mathbb{R}}\int_\Sigma \left(\lambda_1(L_C, r^{-2}) + |A_\Sigma|^2\right)\varphi^2 e^{(n-2)t}\,\mathrm{dvol}_\Sigma\,dt \le \int_{\mathbb{R}}\int_\Sigma \left(|\nabla_\Sigma\varphi|^2 + |\partial_t\varphi|^2\right)e^{(n-2)t}\,\mathrm{dvol}_\Sigma\,dt,$$

as desired. □

Note that this almost looks like a first eigenvalue inequality on the product, but there is an additional factor of $e^{(n-2)t}$ preventing this. Our next idea is to absorb the $e^{(n-2)t}$ term into φ by replacing φ with $\psi = \varphi e^{\frac{n-2}{2}t}$. Note that the LHS and the $|\nabla_\Sigma\varphi|^2$ term absorb the exponential factor without any trouble (∇_Σ is linear with respect to $C^\infty(\mathbb{R})$), but that the $\partial_t\varphi$ term will have interesting cross-terms; this is precisely where the improvement for $\lambda_1(L_\Sigma^\circ)$ comes from. Moreover, note that since $e^{\frac{n-2}{2}t} > 0$, multiplication by this function is a bijection from $C_c^\infty(\Sigma\times\mathbb{R})$ to itself, meaning that nothing is lost by making this substitution.

Lemma 9.6. Let $\varphi \in C_c^\infty(\Sigma\times\mathbb{R})$ and let $\psi = \varphi e^{\frac{n-2}{2}t} \in C_c^\infty(\Sigma\times\mathbb{R})$. Then,

$$|\nabla_\Sigma\varphi|^2 e^{(n-2)t} = |\nabla_\Sigma\psi|^2 \quad\text{and}\quad |\partial_t\varphi|^2 e^{(n-2)t} = |\partial_t\psi|^2 - (n-2)\psi\partial_t\psi + \frac{(n-2)^2}{4}\psi^2.$$

Proof. 1. The first equality is straightforward: if $e_1, \dots, e_{n-1}$ is an orthonormal basis of $T_\theta\Sigma \subseteq T_\theta\Sigma \times T_t\mathbb{R}$, then

$$|\nabla_\Sigma\psi|^2 = \sum_{i=1}^{n-1}\left|\nabla_{e_i}(\varphi e^{\frac{n-2}{2}t})\right|^2 = e^{(n-2)t}\sum_{i=1}^{n-1}\left|\nabla_{e_i}\varphi\right|^2 = e^{(n-2)t}|\nabla_\Sigma\varphi|^2.$$

2. The second equality is also a straightforward computation:

$$\begin{aligned} e^{(n-2)t}\,|\,\partial_t\varphi\,|^2 &= e^{(n-2)t}\left(\partial_t(\psi e^{-\frac{n-2}{2}t})\right)^2 \\ &= e^{(n-2)t}\left((\partial_t\psi)e^{-\frac{n-2}{2}t} - \left(\frac{n-2}{2}\right)e^{-\frac{n-2}{2}t}\psi\right)^2 \\ &= \left(\partial_t\psi - \left(\frac{n-2}{2}\right)\psi\right)^2 \\ &= |\,\partial_t\psi\,|^2 - (n-2)\psi\partial_t\psi + \frac{(n-2)^2}{4}\psi^2. \end{aligned}$$

□

Inputting this computation into our previous lemma 9.5, we get the following.

Lemma 9.7. Let $\Sigma^{n-1} \hookrightarrow S^n$ be a two-sided smooth minimal hypersurface and let $C^n = C(\Sigma)$ be the (punctured) cone over Σ. Then,

$$\lambda_1(L_C, r^{-2}) - \frac{(n-2)^2}{4} = \inf_{\psi\in C_c^\infty(\Sigma\times\mathbb{R})\setminus\{0\}} \frac{\int_{\Sigma\times\mathbb{R}} |\nabla_\Sigma\psi|^2 - |A_\Sigma|^2\psi^2 + |\partial_t\psi|^2 - (n-2)\psi\partial_t\psi}{\int_{\Sigma\times\mathbb{R}}\psi^2}.$$

Proof. From Lemma 9.5 and 9.6, we get

$$\int_{\mathbb{R}}\int_\Sigma \left(\lambda_1(L_C, r^{-2}) + |A_\Sigma|^2\right)\psi^2\,\mathrm{dvol}_\Sigma\,dt \le \int_{\mathbb{R}}\int_\Sigma |\nabla_\Sigma\psi|^2 + |\partial_t\psi|^2 - (n-2)\psi\partial_t\psi + \frac{(n-2)^2}{4}\psi^2\,\mathrm{dvol}_\Sigma\,dt.$$

Rearranging,

$$\lambda_1(L_C, r^{-2}) - \frac{(n-2)^2}{4} \le \frac{\int_{\Sigma\times\mathbb{R}} |\nabla_\Sigma\psi|^2 - |A_\Sigma|^2\psi^2 + |\partial_t\psi|^2 - (n-2)\psi\partial_t\psi}{\int_{\Sigma\times\mathbb{R}}\psi^2}.$$

Moreover, since $\lambda_1(L_C, r^{-2})$ is defined as the greatest number making the inequality in its definition hold, we get the desired result. □

The second equation in Lemma 9.6, as well as the RHS in Lemma 9.7, suggests that we define a new (ordinary) differential operator

$$L_t = -\partial_t^2 - (n-2)\partial_t.$$

Note that in spherical coordinates, we can write

$$L_t = -\partial_t(r\partial_r) - (n-2)r\partial_r = -(\partial_t r)\partial_r - r^2\partial_r^2 - (n-2)r\partial_r = -r^2\partial_r^2 - (n-1)r\partial_r$$

since $\partial_t r = \partial_t e^t = e^t = r$. This is precisely Simons's original expression for the radial operator in [Sim68] (see Section 6, particularly Lemmas 6.1.3–4).

Lemma 9.8 (First eigenvalue of radial operator).

$$\lambda_1(L_t) := \lambda_1(L_t, \mathbb{R}) = \inf_{\rho\in C_c^\infty(\mathbb{R})\setminus\{0\}} \frac{\int_{\mathbb{R}}\rho L_t\rho\,dt}{\int_{\mathbb{R}}\rho^2\,dt} = \inf_{\rho\in C_c^\infty(\mathbb{R})\setminus\{0\}} \frac{\int_{\mathbb{R}}(\rho')^2 - (n-2)\rho\rho'\,dt}{\int_{\mathbb{R}}\rho^2\,dt} = 0.$$

Proof. First, note that for any compactly supported $\rho \in C_c^\infty(\mathbb{R})$,

$$\int_{\mathbb{R}}(n-2)\rho\rho' = \frac{n-2}{2}\int_{\mathbb{R}}(\rho^2)' = 0,$$

so

$$\lambda_1(L_t) = \inf_{\rho \in C_c^\infty(\mathbb{R}) \setminus \{0\}} \frac{\int_{\mathbb{R}} (\rho')^2 - (n-2)\rho\rho' \, dt}{\int_{\mathbb{R}} \rho^2 \, dt} = \inf_{\rho \in C_c^\infty(\mathbb{R}) \setminus \{0\}} \frac{\int_{\mathbb{R}} (\rho')^2}{\int_{\mathbb{R}} \rho^2}.$$

Let $\rho_R \in C_c^\infty(\mathbb{R})$ be a bump function with $\rho_R \equiv 1$ on $[-R, R]$, $\operatorname{supp} \rho_R \subseteq [-2R, 2R]$, and $\sup |\rho_R'| \leq 2R^{-1}$ (for instance, by taking the linear cutoff and smoothing it at the discontinuities). Then, we get that

$$\int_{\mathbb{R}} (\rho')^2 \leq (2R^{-1})(4R) = 8$$

but $\int_{\mathbb{R}} \rho^2 \geq 2R$, and hence

$$\frac{\int_{\mathbb{R}} (\rho')^2}{\int_{\mathbb{R}} \rho^2} \leq \frac{4}{R} \searrow 0$$

as $R \to \infty$, so $\lambda_1(L_t) \leq 0$. Finally, observe that we always have $\int_{\mathbb{R}} (\rho')^2 \geq 0$, so $\lambda_1(L_t) \geq 0$, and we are done. ☐

This lemma indicates that we should expect the contribution of the $\psi L_t \psi$ term to be zero, leaving us just with the $\frac{(n-2)^2}{4}\psi^2$ term. This allows us to prove one direction of Theorem 9.3.

Proof of Theorem 9.3. 1. We begin with the $\leq$ direction. Let $\eta \in C^\infty(\Sigma)$ be arbitrary, let $\rho \in C_c^\infty(\mathbb{R})$ with $\int_{\mathbb{R}} \rho^2 = 1$, and define $\psi \in C_c^\infty(\Sigma \times \mathbb{R})$ by $\psi(\theta, t) = \eta(\theta)\rho(t)$. By Lemma 9.6, we have

$$\int_\Sigma \left(\lambda_1(L_C, r^{-2}) + |A_\Sigma|^2\right) \eta^2 \, d\mathrm{vol}_\Sigma \leq \int_\Sigma |\nabla_\Sigma \eta|^2 + \frac{(n-2)^2}{4}\eta^2 \, d\mathrm{vol}_\Sigma + \left(\int_\Sigma \eta^2 \, d\mathrm{vol}_\Sigma\right) \left(\int_{\mathbb{R}} |\rho'|^2 - (n-2)\rho\rho' \, dt\right).$$

Since ρ could have been chosen arbitrarily and $\lambda_1(L_t) = 0$ by Lemma 9.8, we get

$$\int_\Sigma \left(\lambda_1(L_C, r^{-2}) + |A_\Sigma|^2\right) \eta^2 \, d\mathrm{vol}_\Sigma \leq \int_\Sigma |\nabla_\Sigma \eta|^2 + \frac{(n-2)^2}{4}\eta^2 \, d\mathrm{vol}_\Sigma.$$

Rearranging,

$$\lambda_1(L_C, r^{-2}) - \frac{(n-2)^2}{4} \leq \frac{\int_\Sigma |\nabla_\Sigma \eta|^2 - |A_\Sigma|^2 \eta^2 \, d\mathrm{vol}_\Sigma}{\int_\Sigma \eta^2 \, d\mathrm{vol}_\Sigma},$$

and hence taking the infimum over all $\eta \in C^\infty(\Sigma)$, we get that

$$\lambda_1(L_C, r^{-2}) - \frac{(n-2)^2}{4} \leq \lambda_1(L_\Sigma^\circ).$$

2. We prove the $\geq$ direction. Let $\psi \in C_c^\infty(\Sigma \times \mathbb{R})$ be arbitrary; define $\psi_t(\theta) = \psi(\theta, t)$. Then, by definition of $\lambda_1(L_\Sigma^\circ)$,

$$\int_\Sigma \left(\lambda_1(L_\Sigma^\circ) + |A_\Sigma|^2\right) \psi_t^2 \, d\mathrm{vol}_\Sigma \leq \int_\Sigma |\nabla_\Sigma \psi_t|^2 \, d\mathrm{vol}_\Sigma.$$

Integrating over $t \in \mathbb{R}$,

$$\int_{\mathbb{R}} \int_\Sigma \left(\lambda_1(L_\Sigma^\circ) + |A_\Sigma|^2\right) \psi^2 \, d\mathrm{vol}_\Sigma \, dt \leq \int_{\mathbb{R}} \int_\Sigma |\nabla_\Sigma \psi|^2 \, d\mathrm{vol}_\Sigma \, dt. \tag{9.1}$$

Now, define $\psi_\theta(t) = \psi(\theta, t)$. Since $\lambda_1(L_t) = 0$, we get

$$0 \leq \int_{\mathbb{R}} |\psi_\theta'|^2 - (n-2)\psi_\theta' \psi_\theta \, dt.$$

Integrating over $\theta \in \Sigma$, we get

$$0 \leq \int_{\mathbb{R}} \int_\Sigma |\partial_t \psi|^2 - (n-2)\psi \partial_t \psi \, d\mathrm{vol}_\Sigma \, dt. \tag{9.2}$$

Combining (9.1) and (9.2), we get

$$\int_{\mathbb{R}} \int_\Sigma \left(\lambda_1(L_\Sigma^\circ) + |A_\Sigma|^2\right) \psi^2 \, d\mathrm{vol}_\Sigma \, dt \leq \int_{\mathbb{R}} \int_\Sigma |\nabla_\Sigma \psi|^2 + |\partial_t \psi|^2 - (n-2)\psi \partial_t \psi \, d\mathrm{vol}_\Sigma \, dt.$$

Rearranging,

$$\lambda_1(L_\Sigma^\circ) \leq \frac{\int_{\Sigma\times\mathbb{R}} |\nabla_\Sigma \psi|^2 - |A_\Sigma|^2 \psi^2 + |\partial_t \psi|^2 - (n-2)\psi \partial_t \psi}{\int_{\Sigma\times\mathbb{R}} \psi^2}.$$

Taking the infimum over all $\psi \in C_c^\infty(\Sigma\times\mathbb{R}) \setminus \{0\}$,

$$\lambda_1(L_\Sigma^\circ) \leq \lambda_1(L_C, r^{-2}) - \frac{(n-2)^2}{4},$$

as desired.

□

Section 10

Simons equation

Let us recall the Euclidean Bochner formula.

Proposition 10.1 (Euclidean Bochner formula). Let $u \in C^\infty(\mathbb{R}^n)$ be a smooth function. Then,

$$\frac{1}{2}\Delta|\nabla u|^2 = \left|\nabla^2 u\right|^2 + \langle \nabla u, \nabla \Delta u\rangle.$$

Proof. The formula follows from a straightforward computation. In accordance with the notation for tensors, let us denote partial derivatives by $\partial_j u = u_{;j}$.

$$\frac{1}{2}\Delta|\nabla u|^2 = \frac{1}{2}\sum_{i,j=1}^n (u_{;i}^2)_{;jj} = \sum_{i,j=1}^n (u_{;i}u_{;ij})_{;j} = \sum_{i,j=1}^n u_{;ij}^2 + u_{;i}u_{;ijj} = \sum_{i,j=1}^n u_{;ij}^2 + u_{;i}u_{;jji} = \left|\nabla^2 u\right|^2 + \langle \nabla u, \nabla \Delta u\rangle.$$

□

Note that the final step relies on the fact that mixed partials commute (in particular, that $u_{;ijj} - u_{;jji} = 0$), but otherwise all of the other computations are purely local. Hence, we can expect for there to be a Bochner formula on an arbitrary Riemannian manifold involving a curvature term. Indeed, this is true: for $u \in C^\infty(M)$ for a Riemannian manifold M,

$$\frac{1}{2}\Delta|\nabla u|^2 = \left|\nabla^2 u\right|^2 + \langle \nabla u, \nabla \Delta u\rangle + \mathrm{Ric}(\nabla u, \nabla u).$$

Also, recall that for a two-sided hypersurface $\Sigma^n \hookrightarrow (M^{n+1}, g)$, the unit normal ν and the second fundamental form A_Σ are related by

$$A_\Sigma(X,Y) = -\langle \nabla_X \nu, Y\rangle = -\langle \nabla \nu(X), Y\rangle,$$

and hence (up to raising an index and sign), $\nabla \nu = A_\Sigma$. Hence, we might expect to be able to find a Bochner-type formula for $\Delta|A_\Sigma|^2$ in terms of the second fundamental form, its derivatives, and various curvature terms.

In this section, we will find such a formula, the *Simons identity*, in the case where M has constant sectional curvature. Then, we derive the *Simons inequality*, which will be a crucial ingredient in the proof of Simons's cone theorem.

§10.1 Simons identity

In this section, we prove a Bochner-style formula for $\Delta_\Sigma |A_\Sigma|^2$, the *Simons identity*. For this section, we fix a Riemannian manifold (M^{n+1}, g) with constant sectional curvature κ and a two-sided hypersurface $\Sigma^n \hookrightarrow M$ with unit normal ν.

Remark 10.2. For the application we have in mind, we will be taking $M = S^n$ (note the dimension shift), but this same computation works in a Euclidean ambient space, in which then it is possible to prove Simons's theorem by working on the cone.

We denote by $\nabla, \mathrm{Rm}, R, \mathrm{Ric}$ the Levi–Civita connection, covariant Riemann curvature tensor, (1,3) Riemann curvature tensor, Ricci curvature tensor, respectively on Σ, and $\widetilde{\nabla}, \widetilde{\mathrm{Rm}}, \widetilde{R}, \widetilde{\mathrm{Ric}}$ the corresponding objects on M. We denote the components of the scalar second fundamental form in coordinates (x^i) by $(A_\Sigma)(\partial_i, \partial_j) = A_{ij}$.

Proposition 10.3 (Simons identity). Let (M^{n+1}, g) be a Riemannian manifold with constant sectional curvature κ, and let $\Sigma^n \hookrightarrow (M^{n+1}, g)$ be a two-sided hypersurface. Then, the scalar second fundamental form A_Σ satisfies the equation

$$\frac{1}{2}\Delta |A_\Sigma|^2 = |\nabla A_\Sigma|^2 + \left\langle A_\Sigma, \nabla^2 H_\Sigma \right\rangle - |A_\Sigma|^4 + H_\Sigma \operatorname{tr}_g A_\Sigma^3 - \kappa H_\Sigma^2 + \kappa n |A_\Sigma|^2,$$

where $H_\Sigma = \operatorname{tr}_g A_\Sigma$ is the scalar mean curvature and A_Σ^3 is the symmetric covariant 2-tensor with components $(A_\Sigma)_{ij} = A_{ik}A^{kl}A_{lj}$. In particular, if $H_\Sigma = 0$, we get that

$$|A_\Sigma| \Delta |A_\Sigma| + |\nabla |A_\Sigma||^2 = \frac{1}{2}\Delta |A_\Sigma|^2 = |\nabla A_\Sigma|^2 - |A_\Sigma|^4 + \kappa n |A_\Sigma|^2$$

and in particular

$$|A_\Sigma| L_\Sigma^\circ |A_\Sigma| = |\nabla A_\Sigma|^2 - |\nabla |A_\Sigma||^2 + \kappa n |A_\Sigma|^2.$$

Before we prove the Simons identity, we prove the following lemma.

Lemma 10.4 (Symmetries of $\nabla^2 T$). Let (M, g) be a Riemannian manifold, and let $T \in \Gamma(T^{(0,2)}TM)$ be a covariant 2-tensor. Then, for any four vector fields $W, X, Y, Z \in \mathfrak{X}(M)$,

$$(\nabla^2 T)(W, X, Y, Z) - (\nabla^2 T)(W, X, Z, Y) = T(R(Y, Z)W, X) + T(W, R(Y, Z)X).$$

In coordinates,

$$T_{ij;kl} - T_{ij;lk} = \mathrm{Rm}_{klim}\, T^m{}_j + \mathrm{Rm}_{kljm}\, T_i{}^m.$$

Proof. 1. Assume W, X, Y, Z are parallel at a point $p \in M$. Expanding $(\nabla^2 T)(W, X, Y, Z)$,

$$\begin{aligned}
(\nabla^2 T)(W, X, Y, Z) &= \nabla_Z(\nabla T)(W, X, Y) - (\nabla T)(\nabla_Z W, X, Y) - (\nabla T)(W, \nabla_Z X, Y) - (\nabla T)(W, X, \nabla_Z Y)\\
&= \nabla_Z \nabla_Y T(W, X) - \nabla_Z T(\nabla_Y W, X) - \nabla_Z T(W, \nabla_Y X)\\
&\quad - \nabla_Y T(\nabla_Z W, X) + T(\nabla_Y \nabla_Z W, X) + T(\nabla_Z W, \nabla_Y X)\\
&\quad - \nabla_Y T(\nabla_Y W, \nabla_Z X) + T(\nabla_Y W, \nabla_Z X) + T(W, \nabla_Y \nabla_Z X)\\
&\quad - \nabla_{\nabla_Z Y} T(W, X) + T(\nabla_{\nabla_Z Y} W, X) + T(W, \nabla_{\nabla_Z Y} X).
\end{aligned}$$

Let us label these terms (1)–(12). Since we assumed W, X, Y, Z to be parallel at p, the terms (2)–(4), (6)–(8) vanish. (Note that (10), (11), and (12) also vanish, but it is convenient to include them here.) Let us denote by (i') the term

(i) but with Y, Z swapped. Then, the commutator is given by

$$\begin{aligned}(\nabla^2 T)(W,X,Y,Z) - (\nabla^2 T)(W,X,Z,Y) &= [(1) - (10) - (1') + (10')] \\ &\quad + [(5) + (11) - (5') - (11')] \\ &\quad + [(9) + (12) - (9') - (12')] \\ &= \nabla^2_{Z,Y} T(W,X) - \nabla^2_{Y,Z} T(W,X) \\ &\quad + T(\nabla_Y \nabla_Z W - \nabla_Z \nabla_Y W - \nabla_{[Y,Z]} W, X) \\ &\quad + T(W, \nabla_Y \nabla_Z X - \nabla_Z \nabla_Y X - \nabla_{[Y,Z]} X) \\ &= T(R(Y,Z)W, X) + T(W, R(Y,Z)X).\end{aligned}$$

Note that since $T(W,X)$ is a function, the Hessian $\nabla^2_{Z,Y} T(W,X)$ is symmetric in Z and Y, so the $(1) - (10)$ term vanishes.

2. Pick coordinates (x^i) with coordinate vector fields (∂_i). Then, we get

$$T_{ij;kl} - T_{ij;lk} = \mathrm{Rm}_{kli}{}^{m} T_{mj} + T_{im} \mathrm{Rm}_{klj}{}^{i}.$$

Lowering and raising an index, we get the desired formula.

□

Using these lemmas, we can prove the Simons identity.

Proof of Proposition 10.3 (Simons identity). 1. Let (x^i) be a regular coordinate patch at $p \in \Sigma$, such that (∂_i) is an orthonormal frame parallel at p. In these coordinates, $\Delta_\Sigma |A_\Sigma|^2 = \sum_{i,j,k} (A^2_{ij})_{;kk}$. Expanding the summand,

$$\frac{1}{2}(A^2_{ij})_{;kk} = (A_{ij} A_{ij;k})_{;k} = A_{ij} A_{ij;kk} + A^2_{ij;k}.$$

Tracing the second term with respect to i, j, k, we simply get

$$A_{ij;k} A^{ij;k} = |\nabla A_\Sigma|^2.$$

For the first term, we apply Lemma 10.4 to commute $A_{ij;kk}$ for $A_{kk;ij}$:

$$\begin{aligned}A_{ij;kk} - A_{kk;ij} &= A_{ik;jk} - A_{ik;kj} \\ &= \mathrm{Rm}_{jkim} A^m{}_k + \mathrm{Rm}_{jkkm} A^m{}_i \\ &= (A_{jm} A_{ki} - A_{ji} A_{km} + \kappa(g_{jm} g_{ki} - g_{ji} g_{km})) A^m{}_k \\ &\quad + (A_{jm} A_{kk} - A_{jk} A_{km} + \kappa(g_{jm} g_{kk} - g_{jk} g_{km})) A^m{}_i\end{aligned}$$

Thus tracing the first term with respect to i, j, k, we get

$$\begin{aligned}A^{ij} A_{ij;k}{}^{;k} &= A^{ij} A^k{}_{k;ij} + A^{ij} A_{jm} A_{ki} A^{mk} - A^{ij} A_{ji} A_{km} A^{mk} + A^{ij} A_{jm} A_k{}^k A^m{}_i - A^{ij} A_{jk} A_{km} A^m{}_i \\ &\quad + \kappa A^{ij} g_{jm} g_{ki} A^{mk} - \kappa A^{ij} g_{ji} g_{km} A^{mk} + \kappa A^{ij} g_{jm} g_k{}^k A^m{}_i - \kappa A^{ij} g_{jk} g^k{}_m A^m{}_i\end{aligned} \tag{10.3}$$

2. Let us label the terms in (10.3) by (0)–(8) and compute them individually. Recall that $(A^3_\Sigma)_{ij} = A_{ik} A^{km} A_{kj}$. Let us similarly define $(A^4_\Sigma)_{ij} = A_{ik} A^{km} A_m{}^l A_{lj}$.

$$\begin{aligned}(0) &= A^{ij} A^k{}_{k;ij} = A^{ij} H_{;ij} = \langle A_\Sigma, \nabla^2 H_\Sigma \rangle \\ (1) &= A^{ij} A_{jm} A_{ki} A^{mk} = A^i{}_j A^j{}_m A^m{}_k A^k{}_i = \mathrm{tr}_g\, A^4_\Sigma \\ (2) &= -A^{ij} A_{ji} A_{km} A^{mk} = -A^i{}_j A^j{}_i A^k{}_m A^m{}_k = -|A_\Sigma|^4\end{aligned}$$

$$\begin{aligned}
(3) &= A^{ij}A_{jm}A_k{}^k A^m{}_i = A^k{}_k A^i{}_j A^j{}_m A^m{}_i = H_\Sigma \operatorname{tr}_g A_\Sigma^3 \\
(4) &= -A^{ij}A_{jk}A_{km}A^m{}_i = -A^i{}_j A^j{}_k A^k{}_m A^m{}_i = -\operatorname{tr}_g A_\Sigma^4 \\
(5) &= \kappa A^{ij} g_{jm} g_{ki} A^{mk} = \kappa A^i{}_m A^m{}_i = \kappa \,|\, A_\Sigma \,|^2 \\
(6) &= -\kappa A^{ij} g_{ji} g_{km} A^{mk} = -\kappa A^i{}_i A^k{}_k = -\kappa H_\Sigma^2 \\
(7) &= \kappa A^{ij} g_{jm} g_k{}^k A^m{}_i = \kappa g^k{}_k A^i{}_m A^m{}_i = \kappa n \,|\, A_\Sigma \,|^2 \\
(8) &= -\kappa A^{ij} g_{jk} g^k{}_m A^m{}_i = -\kappa A^i{}_k A^k{}_i = -\kappa \,|\, A_\Sigma \,|^2 .
\end{aligned}$$

Note that we are summing over $\partial_1, \dots, \partial_n \in T_p\Sigma$, so $g^k{}_k = n$. Therefore, we have that

$$A^{ij}A_{ij;k}{}^{;k} = \left\langle A_\Sigma, \nabla^2 H_\Sigma \right\rangle - |\, A_\Sigma \,|^4 + H_\Sigma \operatorname{tr}_g A_\Sigma^3 - \kappa H_\Sigma^2 + \kappa n \,|\, A_\Sigma \,|^2 \tag{10.4}$$

3. Adding these two parts together, we get that

$$\frac{1}{2}\Delta \,|\, A_\Sigma \,|^2 = |\, \nabla A_\Sigma \,|^2 + \left\langle A_\Sigma, \nabla^2 H_\Sigma \right\rangle - |\, A_\Sigma \,|^4 + H_\Sigma \operatorname{tr}_g A_\Sigma^3 - \kappa H_\Sigma^2 + \kappa n \,|\, A \,|_\Sigma^2 . \tag{10.5}$$

Hence, in the particular case where $H_\Sigma \equiv 0$, we get

$$\frac{1}{2}\Delta \,|\, A_\Sigma \,|^2 = |\, \nabla A_\Sigma \,|^2 - |\, A_\Sigma \,|^4 + \kappa n \,|\, A_\Sigma \,|^2 . \tag{10.6}$$

Note that for functions, $\frac{1}{2}\nabla u^2 = u \nabla u$ and therefore $\frac{1}{2}\Delta u^2 = |\, \nabla u \,|^2 + u \Delta u$. Taking $u = |\, A_\Sigma \,|$, we have

$$\frac{1}{2}\Delta \,|\, A_\Sigma \,|^2 = |\, A_\Sigma \,|\, \Delta \,|\, A_\Sigma \,| + |\, \nabla \,|\, A_\Sigma \,|\,|^2 . \tag{10.7}$$

Note that since M has constant sectional curvature, $\widetilde{\operatorname{Ric}} = \kappa n g$, so $\widetilde{\operatorname{Ric}}(\nu, \nu) = \kappa n$ and hence $L_\Sigma^\circ = \Delta + |\, A_\Sigma \,|^2$. From (10.6) and (10.7), we get

$$\begin{aligned}
|\, A_\Sigma \,|\, L_\Sigma^\circ \,|\, A_\Sigma \,| &= |\, A_\Sigma \,|\, \Delta \,|\, A_\Sigma \,| + |\, A_\Sigma \,|^4 \\
&= |\, \nabla A_\Sigma \,|^2 - |\, \nabla \,|\, A_\Sigma \,|\,|^2 + \kappa n \,|\, A_\Sigma \,|^2 .
\end{aligned} \tag{10.8}$$

□

§10.2 Simons inequality

The Simons identity contains the term $|\, \nabla A_\Sigma \,|^2 - |\, \nabla \,|\, A_\Sigma \,|\,|^2$; at the moment it is unclear whether this term is positive or negative. In this section we prove that it is positive, and hence we can turn the Simons identity into an inequality to put into the stability inequality.

Lemma 10.5 (Kato inequality). Let (M^n, g) be a Riemannian manifold and let $T \in \operatorname{Sym}^2(T^*M)$ be a symmetric covariant 2-tensor. Then, it satisfies the inequality

$$|\, \nabla \,|\, T \,|\,|^2 \leq |\, \nabla T \,|^2 .$$

If in addition $\operatorname{tr}_g T = 0$ and ∇T is symmetric, then it satisfies the *improved Kato inequality*

$$\left(1 + \frac{2}{n}\right) |\, \nabla \,|\, T \,|\,|^2 \leq |\, \nabla T \,|^2 .$$

Proof. 1. Since T is symmetric, at a point p we can pick normal coordinates (x^i) such that (∂_i) diagonalizes T: $T_{ij} = \kappa_i \delta_{ij}$. Moreover, we will drop Einstein notation and write sums explicitly; all sums range from $1, \dots, n$. We can expand $\left|\nabla|T|^2\right|^2$ as

$$\left|\nabla|T|^2\right|^2 = \sum_{i,j,k,l,m}(T_{ij}^2)_{;k}(T_{lm}^2)_{;k} = \sum_{i,j,k,l,m} 4T_{ij}T_{lm}T_{ij;k}T_{lm;k} = 4\sum_{i,k,l}(\kappa_i\kappa_l)(T_{ii;k}T_{ll;k}) = 4\sum_k\left(\sum_{i,l}(\kappa_i\kappa_l)\left(T_{ii;k}T_{ll;k}\right)\right).$$

By the Cauchy–Schwarz inequality (here applied in the form $\sum_{i,l}\alpha_{i,l}\beta_{i,l} \le \left(\sum_{i,l}\alpha_{i,l}^2\right)^{1/2}\left(\sum_{i,l}\beta_{i,l}^2\right)^{1/2}$ with $\alpha_{i,l} = \kappa_i\kappa_l$ and $\beta_{i,l} = T_{ii;k}T_{ll;k}$ for each k), we get

$$\begin{aligned} 4\sum_k\left(\sum_{i,l}(\kappa_i\kappa_l)\left(T_{ii;k}T_{ll;k}\right)\right) &\le 4\sum_k\left(\sum_{i,l}\kappa_i^2\kappa_l^2\right)^{1/2}\left(\sum_{i,l}\left(T_{ii;k}T_{ll;k}\right)^2\right)^{1/2} \\ &= 4|T|^2\sum_{i,k}T_{ii;k}^2 \end{aligned}$$

Since $\nabla|T|^2 = 2|T|\nabla|T|$, we have $\left|\nabla|T|^2\right|^2 = 4|T|^2|\nabla|T||^2$. Therefore,

$$|\nabla|T||^2 \le \sum_{i,k}T_{ii;k}^2. \tag{10.9}$$

Since $\sum_{i,k}T_{ii;k}^2 \le \sum_{i,j,k}T_{ij;k}^2 = |\nabla T|^2$, we have proved the first part of the lemma.

2. Since $\operatorname{tr}_g T = 0$, we have that $\sum_i T_{ii;k} = 0$ for any k; in particular, $T_{ii;k}^2 = \left(\sum_{i\ne j}T_{jj;k}\right)^2$. Using this and the Cauchy–Schwarz inequality,

$$\begin{aligned} \sum_{i,k}T_{ii;k}^2 &= \sum_{i\ne k}T_{ii;k}^2 + \sum_i T_{ii;i^2} \\ &= \sum_{i\ne k}T_{ii;k}^2 + \sum_i\left(\sum_{j\ne i}T_{jj;i}\cdot 1\right)^2 \\ &\le \sum_{i\ne k}T_{ii;k}^2 + (n-1)\sum_i\sum_{j\ne i}T_{jj;i}^2 \\ &= \sum_{i\ne k}T_{ii;k}^2 + (n-1)\sum_{j\ne i}T_{jj;i}^2 \\ &= n\sum_{i\ne k}T_{ii;k}^2. \end{aligned}$$

3. Since ∇T is symmetric, we can write

$$\sum_{i,k}T_{ii;k}^2 \le n\sum_{i\ne k}T_{ii;k}^2 = \frac{n}{2}\sum_{i\ne k}T_{ik;i} + T_{ki;i}.$$

Hence,

$$\left(1+\frac{2}{n}\right)|\nabla|T||^2 \le \sum_{i\ne k}T_{ii;k} + T_{ik;i} + T_{ki;i} \le \sum_{i,j,k}T_{ij;k}^2 = |\nabla T|^2. \tag{10.10}$$

□

Combining this lemma with the Simons identity (Proposition 10.3), we get the following inequality.

> Corollary 10.6 (Simons inequality). Let (M^{n+1}, g) be a Riemannian manifold with constant sectional curvature κ and let $\Sigma^n \hookrightarrow (M^{n+1}, g)$ be a two-sided minimal hypersurface. Then,
>
> $$|A_\Sigma|L_\Sigma^\circ|A_\Sigma| \ge \frac{2}{n}|\nabla|A_\Sigma||^2 + \kappa n|A_\Sigma|^2 \ge \kappa n|A_\Sigma|^2.$$

Section 11

Simons cone theorem and Bernstein's theorem

We are now, more or less, done. Using the Simons inequality (Corollary 10.6), we get the following estimate for the first eigenvalue of the link:

Proposition 11.1 (Upper bound for first eigenvalue of the link). Let $\Sigma^{n-1} \hookrightarrow S^n$ be a minimal two-sided hypersurface. Then, either Σ is totally geodesic or

$$\lambda_1(L_\Sigma^\circ) \leq -(n-1).$$

Proof. Integrating the Simons inequality (Corollary 10.6), we get

$$-\int_\Sigma |A_\Sigma| L_\Sigma^\circ |A_\Sigma| \leq -(n-1)\int_\Sigma |A_\Sigma|^2 .$$

Assume Σ is not totally geodesic, so $\|A_\Sigma\|_{L^2} > 0$. Dividing on both sides and taking the infimum, we get

$$\lambda_1(L_\Sigma^\circ) \leq -(n-1),$$

as desired. □

Theorem 11.2 (Simons). Let $2 \leq n \leq 6$, and suppose $\Sigma^{n-1} \hookrightarrow S^n$ is a smooth minimal submanifold with cone $C^n = C(\Sigma) \hookrightarrow \mathbb{R}^{n+1}$. If C is stable, then C is flat.

Proof. Since C is stable, by the separation of variables theorem (Theorem 9.3), we have the inequality

$$\lambda_1(L_\Sigma^\circ) \geq -\frac{(n-2)^2}{4}.$$

On the other hand, suppose Σ is not totally geodesic (i.e., C is not flat). Then, we have the inequality

$$-\frac{(n-2)^2}{4} \leq \lambda_1(L_\Sigma^\circ) \leq -(n-1).$$

This inequality is only possible when

$$0 \leq -(n-1) + \frac{(n-2)^2}{4} = \frac{n^2}{4} - 2n + 2.$$

The polynomial on the right has roots $4-\sqrt{8} \approx 1.17157$ and $4+\sqrt{8} \approx 6.82842$ and has positive leading term. Hence, if $2 \leq n \leq 6$, this is a contradiction. □

As a remark, there exist stable but not minimizing cones for $n = 1$; consider the cross $\{|x| = |y|\} \subseteq \mathbb{R}^2$. However, this cone is not minimizing; the only smooth minimizing cones in $\mathbb{R}^2$ are flat. Hence, the Simons theorem has the following corollary:

Corollary 11.3. Let $1 \leq n \leq 6$, and suppose $\Sigma^{n-1} \hookrightarrow S^n$ is a smooth minimal submanifold with cone $C^n = C(\Sigma) \hookrightarrow \mathbb{R}^{n+1}$. If C is minimizing, then C is flat.

Finally, we can prove Bernstein's theorem for $n \leq 7$.

Theorem 11.4 (Bernstein's theorem, $n \leq 7$). Let $u : \mathbb{R}^n \to \mathbb{R}$ be an entire solution to the MSE. If $n \leq 7$, then $\Gamma = \operatorname{graph} u$ is flat.

Proof. Let Γ be a nonflat minimal graph. Let C be a tangent cone at infinity to Γ, which exists because Γ is minimizing and hence has area bounds on compact sets, which ensure convergence as a Caccioppoli set. By the non-smooth blowdown theorem, C is not smooth, and hence by the iterated tangent cone argument there exists a smooth minimizing cone $C' \subseteq \mathbb{R}^{k+1}$ for $k + 1 < n + 1 \leq 8$. But, this means that $k \leq 6$, so C' must be flat by Simons's theorem. But, since nonflat graphs must blow up to nonflat cones, this is a contradiction. □

Appendix

The coarea formula is a generalization of Fubini's theorem which holds in greater generality. Let $\mathcal{H}^k$ denote the Hausdorff k-dimensional measure on $\mathbb{R}^n$ (see [CM11] Equation (1.59) for this version, see [Sim14] Chapter 1.2 for a version in arbitrary codimension, and see [Giu84] Theorem 1.23 for a version involving Caccioppoli sets).

Theorem A.1 (Co-area formula). Let $\Sigma^k \hookrightarrow \mathbb{R}^n$ be a C^1 smooth submanifold with $f \in C^{0,1}(\Sigma)$ Lipschitz with $f^{-1}((-\infty, t])$ compact for all $t \in \mathbb{R}$. Then, for any function $\varphi \in L^1_{\mathrm{loc}}(\Sigma)$,

$$\int_{\{f \le t\}} |\nabla f| \, \varphi \, d\mathcal{H}^k = \int_{-\infty}^{t} \int_{\{f=\tau\}} \varphi \, d\mathcal{H}^{k-1} \, d\tau.$$

We state the parametric transversality theorem, which we use as a technical tool to ensure that things are transverse; see [LeeSM] Theorem 6.35.

Theorem A.2 (Parametric transversality theorem). Let X, Y, S, Z be smooth manifolds with $Z \subseteq Y$ embedded, and suppose $F : X \times S \to Y$ is transverse to Z. Then, for almost all $s \in S$, the map $F_s : X \to Y$ defined by $F_s : x \mapsto F(x, s)$ is transverse to Z.

See [Eva10] Section 5.7 Theorem 1.

Theorem A.3 (Rellich compactness). Let $\Omega \subseteq \mathbb{R}^n$ be a bounded open set with C^1 boundary. Then, for $p \in [1, n]$ and $q \in [1, p^*)$,

$$W^{1,p}(\Omega) \Subset L^q(\Omega).$$

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